

LEE–YANG PRESERVING QUANTUM MARKOV SEMIGROUPS AND CHANNELS

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ABSTRACT. We classify continuous quantum Markov semigroups that preserve the Lee–Yang property of every density matrix in the class. On two or more qubits, ordinary preservation forces tensor-product dynamics, with each local generator characterized by two explicit semidefinite conditions; it is then equivalent to preservation with arbitrary inert reference systems. For arbitrary finite spins with at least one spin greater than one half, preservation of all pure Lee–Yang inputs already excludes intersite terms and forces isotropic diffusion with axial drift at every higher-spin site. We also classify invertible static qubit maps with stable full Choi tensors, give the complete one-qubit full-Choi classification including singular maps, and obtain a root parametrization for independently covariant maps between finite spins. Finally, correlated inputs yield an exact first-order defect and dimension-independent quantitative rigidity for diagonal Hamiltonians, whereas product tests do not provide uniform robustness. The proofs combine physical positivity with polynomial root collisions and coherent tensor contraction.

CONTENTS

1. Introduction	2
2. Conventions and zero-free kernels	3
3. Imported polynomial generator results	6
4. Physical quantum Markov dynamics	8
5. Physical dynamics at arbitrary finite spins	23
6. Static channels and stable encodings	30
7. The complete one-qubit static classification	47
8. Independently covariant channels between finite spins	52
9. Quantitative rigidity and correlated tests	56
Disclosure of artificial intelligence use	63
References	63

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1. INTRODUCTION

The Lee–Yang property associates to a quantum density matrix a polynomial that is zero-free when its independent row and column variables lie in the unit disk. It extends the use of stable polynomial tensors from equilibrium operators to quantum channels. A channel can preserve this property for a particular state, for every state in the class on its given system, or even in the presence of an arbitrary inert reference system. These requirements lead to different static problems. Our main result is that continuous physical dynamics impose a much more rigid structure.

On two or more qubits, ordinary preservation of all Lee–Yang density matrices forces a completely positive, trace-preserving (CPTP) semigroup to be a tensor product of one-qubit semigroups. Each local generator is specified by an axial Hamiltonian and two explicit positive semidefinite matrix conditions (Theorem 4.1). No locality, unitality, or restriction on the initial GKLS noise operators is assumed. A finite family of equatorial product states already detects every intersite term. Further two-site inputs prevent compensation between local defects and recover the exact local cone. Ordinary and complete preservation consequently coincide for these semigroups. On one qubit, the ordinary cone is larger, and we give its separate characterization in Proposition 4.14.

For arbitrary finite spins, the classification becomes stricter. If at least one site has spin at least one, preservation of all pure Lee–Yang inputs for sufficiently small times already forces a sum of local generators (Theorem 5.1). Every site of spin at least one carries isotropic spin diffusion and an axial Hamiltonian; qubit sites retain the preceding semidefinite cone. The proof first reduces arbitrary noise operators to linear spin operators by repeated-root contacts of physical pure states. Positivity makes the coefficients of forbidden high-order terms sums of squares, so they cannot cancel. This is where polynomial root geometry enters the physical classification.

The static problem retains features that a flow from the identity cannot exhibit. We classify all invertible Hermiticity-preserving (HP), trace-preserving (TP) qubit maps with stable full Choi tensors as local maps followed by a permutation of the physical wires (Theorem 6.6). We then give the complete one-qubit full-Choi classification, including singular maps, and structural results for singular multisite channels and stable isometries. The separate covariant problem at arbitrary input and output spins has an exact root parametrization (Theorem 8.1). Under independent local $SU(2)$ covariance and trace preservation, stable coherent Choi kernels force tensor factorization and complete positivity, even when positivity was not imposed initially.

The last section measures a limitation of product tests. For diagonal Hamiltonians, correlated two-block inputs recover an exact first-order trace-norm defect. This defect controls distance to the local diagonal commutator space with a dimension-independent constant (Theorem 9.2). In contrast, product-input tests alone admit families with vanishing defects while the distance to the local space stays bounded below. The comparison constant comes from the Kalton–Roberts approximate-additivity theorem; the polynomial contribution is the exact root-contact identity connecting that theorem to physical distance.

Relation to earlier work. Our use of disk-stable tensors and contraction follows the classical Asano and stability-preserver theories [3, 5, 6]. Wong, Bravyi, Gosset, and Liu [19] formulate Lee–Yang quantum tensors and provide useful preserving

channels. Here the main question is necessity: which continuous physical dynamics preserve the entire class, when neither locality nor a prescribed family of jumps is given? The finite-dimensional GKLS representation [10, 15] supplies the general physical generator; the zero-free constraints supply the additional classification.

Isotropic spin diffusion and its multipole spectrum are established objects [8, 9]. Our higher-spin result derives that form from Lee–Yang preservation without assuming rotation covariance. Likewise, the representation theory of covariant channels is inherited from [1, 4]; the root restrictions and forced factorization in Theorem 8.1 are additional consequences of zero-freeness. The short application to rotation convolution semigroups is stated with the Gaussian recognition results of Voit [16] in view, rather than as a separate characterization of Brownian motion.

The finite-box polynomial generator classification and its two precise local tools are imported from the companion manuscript [17]. Section 3 states them with all degree conventions: the real and complex generator theorems, the order-two box decomposition, and the lower-half-plane root-cluster lemma. In particular, the necessity proof at higher spin uses the latter two statements separately. The physical and static proofs retained here are given in full, with the relevant imported statement cited at each use.

Organization. Section 2 fixes the polynomial and coherent normalizations, and Section 3 records the imported results. Sections 4 and 5 prove the qubit and finite-spin dynamical classifications. Sections 6 and 7 treat static qubit maps. Section 8 gives the independently covariant finite-spin classification, and Section 9 proves the quantitative rigidity results.

2. CONVENTIONS AND ZERO-FREE KERNELS

2.1. Regions, degrees, and preservation. Write $\mathbb{H} = \{z \in \mathbb{C} : \Im z > 0\}$ and $\mathbb{D}_r = \{z \in \mathbb{C} : |z| < r\}$, with $\mathbb{D} = \mathbb{D}_1$.

Definition 2.1 (Stability and preservation). A nonzero polynomial is Ω -stable if it has no zero when all its variables lie in Ω . For a product of different regions the corresponding meaning is used. A polynomial is *real stable* if its coefficients are real and it is $\mathbb{H}$ -stable. In one variable this is equivalent to real-rootedness. An operator *preserves* one of these classes when it sends every member to another member or to zero.

Let $\mathbb{N} = \{0, 1, 2, \dots\}$. For $\kappa \in \mathbb{N}^n$ put

$$V_\kappa = \{f \in \mathbb{C}[z_1, \dots, z_n] : \deg_{z_i} f \leq \kappa_i\}, \quad \binom{\kappa}{\alpha} = \prod_i \binom{\kappa_i}{\alpha_i}.$$

Variables of capacity zero can be deleted. A degree bound is not an assertion that the polynomial has that degree. All finite-box statements include degree drops and constants. For a real operator, its action on complex polynomials always means its complex-linear extension. Every operator on V_κ has a unique canonical expression

$$A = \sum_{\alpha \leq \kappa} P_\alpha(z) \partial^\alpha, \quad (1)$$

as an operator on V_κ . Uniqueness follows by applying the expression successively to monomials, starting with 1: the equation for z^α determines $\alpha!P_\alpha$ after all lower terms are known. The phrase *differential order* refers to this expression.

A finite-dimensional generator A preserves a class when e^{tA} preserves it for all $t \geq 0$. The polynomial and quantum operator classifications below take place in finite-dimensional spaces.

Definition 2.2 (Complete stability preservation). An operator is *completely stable* if its tensor product with the identity in arbitrarily many extra polynomial variables preserves the same complex stable class, for every finite choice of the extra degree bounds.

This is a statement about polynomial stability, distinct from complete positivity of a quantum channel.

2.2. Polarization and the disk kernel. The normalized polarization of z_i^a into κ_i binary slots is

$$z_i^a \mapsto \binom{\kappa_i}{a}^{-1} e_a(z_{i1}, \dots, z_{i\kappa_i}).$$

It is symmetric in the slots and specializes to z_i^a on the diagonal. Polarization preserves stability in the convex circular regions used here, including open disks and half-planes; diagonalization preserves it directly. We use the Grace–Walsh–Szegő coincidence theorem in this form; see [5, 6, 7]. The binomial normalization is part of the definition, not an optional choice of basis.

For $T : V_\kappa \rightarrow V_\lambda$ define its disk kernel by

$$K_T(z, w) = T_z \prod_i (1 + z_i w_i)^{\kappa_i} = \sum_{\alpha \leq \kappa} \binom{\kappa}{\alpha} (Tz^\alpha)(z) w^\alpha. \quad (2)$$

The notation z_i in the first expression denotes the input variables on which T acts; its result is a polynomial in the output variables. The kernel has output degree bound λ and input degree bound κ . In particular $K_{\text{Id}}(z, w) = \prod_i (1 + z_i w_i)^{\kappa_i}$.

For two polynomials of degree at most κ in paired variables define their bilinear contraction by

$$[f(u), g(v)]_\kappa = \sum_{\alpha \leq \kappa} \binom{\kappa}{\alpha}^{-1} f_\alpha g_\alpha, \quad f(u) = \sum_\alpha f_\alpha u^\alpha, \quad g(v) = \sum_\alpha g_\alpha v^\alpha. \quad (3)$$

There is no complex conjugation in this formula. Matrix composition is exactly contraction of the corresponding kernels:

$$K_{ST}(z, w) = [K_S(z, u), K_T(v, w)]_\kappa,$$

where κ is the intermediate capacity.

Lemma 2.3 (Disk contraction). *Suppose $f(z, u)$ and $g(v, w)$ are stable in products of disks and $\deg_{u_i} f, \deg_{v_i} g \leq \kappa_i$. If their paired radii r_i, s_i satisfy $r_i s_i \geq 1$, then their contraction (3) is stable in the remaining disks or is identically zero. If all $r_i s_i > 1$, contraction at any point in the remaining disks is nonzero.*

Proof. First consider a multiaffine polynomial $F(u, v) = a + bu + cv + duv$ that is nonzero on $\mathbb{D}_r \times \mathbb{D}_s$. Then $a \neq 0$ and $F(rt, st)$ is nonzero for $|t| < 1$. If $d \neq 0$, the product of its two roots has modulus at least one, so $|d|rs \leq |a|$; if $d = 0$, this inequality is immediate. Thus $a + d \neq 0$ when $rs > 1$. The same argument applies after fixing any uncontracted variables in their respective disks. Consequently the operation $a + bu + cv + duv \mapsto a + d$ preserves stability under strict separation of

the paired radii. It may be iterated on a general multiaffine polynomial, without a factorization assumption.

Polarize the paired variables of f and g into κ_i slots for coordinate i , and apply this operation to each pair of corresponding slots in their product. For a fixed multi-index α , the matching slot subsets have cardinality $\binom{\kappa}{\alpha}$, while the two normalized polarizations contribute the factor $\binom{\kappa}{\alpha}^{-2}$. The result is therefore exactly (3). This proves strict nonvanishing, including degree deficiencies. It is the weighted form of circular-domain contraction; see [3, 5, 6]. For equality of paired radii, replace every paired variable in each input polynomial by $(1 - \varepsilon)$ times that variable, apply the strict statement, and let $\varepsilon \downarrow 0$. Hurwitz's theorem gives the stable-or-zero conclusion. The argument is valid after fixing any values of the unpaired variables in their respective disks. $\square$

The finite-box symbol theorem [5, 6] identifies stability preservers of rank at least two by the appropriate stable kernel; the low-rank alternatives are retained whenever they arise. In particular an invertible disk-stability preserver has a stable kernel (2).

The Cayley isomorphism is

$$(\mathcal{C}_\kappa f)(x) = \prod_i (x_i + i)^{\kappa_i} f\left(\frac{x_1 - i}{x_1 + i}, \dots, \frac{x_n - i}{x_n + i}\right). \quad (4)$$

It is a linear isomorphism between the degree boxes, and f is $\mathbb{D}$ -stable if and only if $\mathcal{C}_\kappa f$ is $\mathbb{H}$ -stable. Thus the generator classifications in half-plane coordinates and the physical classifications in disk coordinates concern conjugate finite-dimensional semigroups.

2.3. Coherent coordinates and physical tensors. Equip V_κ with the inner product for which $\binom{\kappa}{\alpha}^{1/2} z^\alpha$ is an orthonormal basis. Equivalently, identify the standard basis $|\alpha\rangle$ of $\bigotimes_i \mathbb{C}^{\kappa_i+1}$ with $\binom{\kappa}{\alpha}^{1/2} z^\alpha$. The polynomial of a vector ψ is

$$f_\psi(z) = \sum_{\alpha \leq \kappa} \binom{\kappa}{\alpha}^{1/2} \psi_\alpha z^\alpha.$$

An operator M on this Hilbert space has coherent tensor

$$F_M(z, w) = \sum_{\alpha, \beta \leq \kappa} \binom{\kappa}{\alpha}^{1/2} \binom{\kappa}{\beta}^{1/2} M_{\alpha\beta} z^\alpha w^\beta. \quad (5)$$

This is precisely (2) for its action in coherent coordinates. Bra variables are independent complex variables; they are not conjugated when testing zero-freeness. For a positive semidefinite density matrix ρ , the assertion that F_ρ is $\mathbb{D}$ -stable is called its *Lee–Yang property*.

For a linear map Φ between matrix algebras the full coherent Choi tensor is defined by replacing $M_{\alpha\beta}$ in (5) with the coefficients $\Phi(|\gamma\rangle\langle\delta|)_{\alpha\beta}$ and adding input monomials $u^\gamma v^\delta$ with the corresponding square-root binomial weights. Its stability is called *full-Choi Lee–Yang preservation*. For qubits all capacities are one, so these weights are one. They cannot be omitted for higher spin. Ordinary physical preservation tests only positive semidefinite Lee–Yang inputs; it is a weaker condition for isolated maps.

We use the Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus $E_i = z_i \partial_i$ represents $|1\rangle\langle 1|_i$ and $Z_i = I - 2E_i$ on a qubit. For general capacity $E_i = z_i \partial_i$ has eigenvalues $0, \dots, \kappa_i$.

3. IMPORTED POLYNOMIAL GENERATOR RESULTS

We state the exact polynomial results needed in the physical proofs. All four statements in this section are imported from [17]; the physical classifications and their proofs are developed below.

Let $\kappa = (\kappa_1, \dots, \kappa_d)$ have finite positive integer coordinates, and write

$$V_\kappa^{\mathbb{F}} = \mathbb{F}[z_1, \dots, z_d]_{\deg_{z_i} \leq \kappa_i}, \quad \mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}.$$

Coordinates of capacity zero are omitted. If no coordinate remains, the space consists of constants and every scalar generator has the requisite preservation property. Stability and complete preservation have the meanings of Definitions 2.1 and 2.2; in this section the domain of stability is $\mathbb{H}^d$. A real operator acts on $V_\kappa^{\mathbb{C}}$ by complexification.

For a polynomial q in one variable of degree at most two, define

$$J_{i, \kappa_i}(q) = q(z_i) \partial_i - \frac{\kappa_i}{2} q'(z_i). \quad (6)$$

These are endomorphisms of the box. The three choices $q = 1, z_i, z_i^2$ give

$$J_i^- = \partial_i, \quad J_i^0 = z_i \partial_i - \frac{\kappa_i}{2}, \quad J_i^+ = z_i^2 \partial_i - \kappa_i z_i.$$

Their span is the sitewise $\mathfrak{sl}_2$ algebra.

Every endomorphism has a unique canonical normal representation

$$A = \sum_{\alpha \leq \kappa} P_\alpha(z) \partial^\alpha. \quad (7)$$

Here the coefficient polynomials are not assumed to belong to the original box. The order of A means the largest $|\alpha|$ for which its canonical coefficient P_α is nonzero. The uniqueness of this representation follows by triangular recursion on the monomial basis and is important at small capacities: in particular ∂_i^2 is absent when $\kappa_i = 1$.

Theorem 3.1 (Real generators; [17]). *Let $A \in \text{End}_{\mathbb{R}}(V_\kappa^{\mathbb{R}})$. The following conditions are equivalent.*

- (1) *For every $t \geq 0$, e^{tA} preserves real stability.*
- (2) *For every $t \geq 0$, its complexification preserves complex stability, also after adjoining arbitrary inert polynomial variables.*
- (3) *The canonical representation of A has order at most two, and its second-order coefficients satisfy*

$$P_{2e_i}(s) \leq 0 \quad \text{for all } s \in \mathbb{R}, \quad \kappa_i \geq 2, \quad (8)$$

$$P_{e_i+e_j}(s, t) \leq 0 \quad \text{for all } (s, t) \in \mathbb{R}^2, \quad i < j. \quad (9)$$

In the third condition, box preservation forces P_{2e_i} to depend only on z_i , with degree at most four, and $P_{e_i+e_j}$ to depend only on z_i, z_j , with bidegree at most $(2, 2)$. There are no further inequalities on the zeroth- and first-order coefficients beyond the identities already forced by $A(V_\kappa^\mathbb{R}) \subseteq V_\kappa^\mathbb{R}$.

The coefficient $P_{e_i+e_j}$ in this statement is the full coefficient of $\partial_i \partial_j$, with $i < j$. Thus a convention writing the second-order part as $\sum_{i,j} Q_{ij} \partial_i \partial_j$, with Q symmetric, has $P_{e_i+e_j} = 2Q_{ij}$. We use the full-coefficient convention throughout this section.

Theorem 3.2 (Complex generators; [17]). *Let $A \in \text{End}_\mathbb{C}(V_\kappa^\mathbb{C})$. The following conditions are equivalent.*

- (1) *For every $t \geq 0$, e^{tA} preserves complex stability.*
- (2) *For every $t \geq 0$, e^{tA} preserves complex stability after adjoining arbitrary inert polynomial variables.*
- (3) *There are a real generator R satisfying Theorem 3.1, a real scalar c , and real polynomials q_i of degree at most two, nonnegative on $\mathbb{R}$, such that*

$$A = R + i \left(c \text{Id} + \sum_{i=1}^d J_{i,\kappa_i}(q_i) \right). \quad (10)$$

Equivalently, the canonical representation has order at most two, all its second-order coefficients are real and satisfy (8)–(9), and every $\Im P_{e_i}$ is nonnegative on $\mathbb{R}^d$. In this formulation box preservation forces $\Im P_{e_i}(z) = q_i(z_i)$ with $\deg q_i \leq 2$, and forces the corresponding imaginary zeroth-order term in (10).

The two theorems identify ordinary and complete polynomial preservation for semigroups starting at the identity. Their proofs are in the companion paper [17]; they are used here as imported results.

Proposition 3.3 (Box decomposition; [17]). *Every box endomorphism of canonical order at most two is a linear combination of Id , the operators J_i^-, J_i^0, J_i^+ , and their products of length two, including products at the same site. Every endomorphism of order at most one has a unique expression*

$$c \text{Id} + \sum_{i=1}^d J_{i,\kappa_i}(q_i), \quad c \in \mathbb{F}, \quad q_i \in \mathbb{F}[z_i]_{\leq 2}. \quad (11)$$

Consequently the second-order coefficients have the variable dependence and degree bounds stated in Theorem 3.1. The vector space of endomorphisms of order at most two has dimension

$$1 + 3d + 5\#\{i : \kappa_i \geq 2\} + 9 \binom{d}{2}. \quad (12)$$

Lemma 3.4 (Lower-half-plane root clusters; [17]). *Let $p_t(s)$, $t \geq 0$, be a curve of complex polynomials of uniformly bounded degree whose coefficients are right differentiable at zero. Suppose*

$$p_0(s) = s^m g(s), \quad m \geq 1, \quad g(0) \neq 0,$$

and that p_t has no zero in $\mathbb{H}$ for all sufficiently small t . Let C_t be the monic factor consisting of the m roots converging to zero. Its coefficients are right differentiable, and

$$\left. \frac{d}{dt} C_t(s) \right|_{t=0} = c_1 s^{m-1} + c_2 s^{m-2}, \quad \Im c_1 \geq 0, \quad c_2 \in \mathbb{R}, \quad c_2 \leq 0. \quad (13)$$

When $m = 1$, the c_2 term is absent. If every p_t is real rooted, then c_1 is real as well.

4. PHYSICAL QUANTUM MARKOV DYNAMICS

We classify continuous physical evolutions that preserve the Lee–Yang property of every density matrix in the class. The physical tensor decomposition and the computational basis are fixed throughout this section.

For a matrix M on n qubits, put

$$F_M(z, w) = \sum_{a, b \in \{0, 1\}^n} M_{ab} z^a w^b. \quad (14)$$

A density matrix is called *Lee–Yang* if this polynomial does not vanish on $\mathbb{D}^{2n}$. Thus the row and column variables are independent. For a pure state, $F_{|\psi\rangle\langle\psi|}(z, w) = f_\psi(z) \overline{f_\psi(w)}$, where $f_\psi(z) = \sum_a \psi_a z^a$. A channel is *ordinarily preserving* if it preserves these density matrices on its given system, and is *completely preserving* if it does so after adjoining any number of inert qubits. Complete positivity of a channel and complete preservation in this sense are distinct conditions.

Write $\sigma_x = X$, $\sigma_y = Y$, and $\sigma_z = Z$, with $Z = \text{diag}(1, -1)$. The Pauli matrices in all generator formulas below are unnormalized. We use the standard finite-dimensional GKLS representation [10, 15]. If $\mathcal{P}_n$ denotes the nonidentity Hermitian Pauli strings, every generator of a continuous CPTP semigroup has a representation

$$\mathcal{L}(\rho) = -i[H, \rho] + \sum_{P, Q \in \mathcal{P}_n} K_{PQ} (P\rho Q - \tfrac{1}{2}\{QP, \rho\}), \quad H = H^*, \quad K \succeq 0. \quad (15)$$

Identity components in the noise have been absorbed into the Hamiltonian.

Theorem 4.1 (Ordinary and complete preservation). *Let $n \geq 2$, and let $\Phi_t = e^{t\mathcal{L}}$ be a continuous CPTP semigroup on n qubits. The following conditions are equivalent.*

- (1) Φ_t ordinarily preserves Lee–Yang density matrices for every $t \geq 0$.
- (2) The same preservation holds for all t in some interval $[0, \varepsilon)$.
- (3) Φ_t completely preserves Lee–Yang density matrices for every $t \geq 0$.
- (4) The full Choi tensor of Φ_t is Lee–Yang for every $t \geq 0$.
- (5) The generator is a sum of one-qubit generators,

$$\mathcal{L} = \sum_{j=1}^n \mathcal{L}_j \otimes \text{Id}_{\hat{j}}, \quad \mathcal{L}_j(\rho) = -i[h_j Z, \rho] + \sum_{\alpha, \beta = x, y, z} (K_j)_{\alpha\beta} (\sigma_\alpha \rho \sigma_\beta - \tfrac{1}{2}\{\sigma_\beta \sigma_\alpha, \rho\}), \quad (16)$$

where $h_j \in \mathbb{R}$ and, for a real symmetric 2×2 matrix B_j , a vector $d_j \in \mathbb{R}^2$, and $c_j \in \mathbb{R}$,

$$K_j = \begin{pmatrix} B_j & id_j \\ -id_j^\top & c_j \end{pmatrix} \succeq 0, \quad \begin{pmatrix} c_j I_2 - \text{adj } B_j & d_j \\ d_j^\top & \text{Tr } B_j \end{pmatrix} \succeq 0. \quad (17)$$

In this case $\Phi_t = \bigotimes_j e^{t\mathcal{L}_j}$. The equivalence between ordinary and complete preservation is false on one qubit.

We prove the theorem in four steps. A fixed finite family of product inputs first excludes every interaction. Two-site pure states then fix the reality of the remaining local coefficients. Finally, positive definite separable states recover the individual

local inequalities even though both sites evolve. The resulting inequalities are reduced to (17) without dividing by a possibly vanishing trace.

4.1. A finite family detecting every interaction.

Lemma 4.2 (A one-sided root collision). *Let g_t be a differentiable curve in a finite-dimensional polynomial space, with $g_0 \not\equiv 0$. Suppose 0 is a root of g_0 of multiplicity m , and g_t is upper-half-plane stable along a sequence $t \downarrow 0$ of positive times. If $m \geq 3$, then $\dot{g}_0(0) = 0$. If $m = 2$ and $g_0(x) = cx^2 + O(x^3)$, then $\dot{g}_0(0)/c$ is real and nonpositive.*

Proof. Choose a small circle about zero containing no other root of g_0 . The nearby cluster has a monic factor $b_t(x) = x^m + a_1(t)x^{m-1} + \cdots + a_m(t)$ with $b_0 = x^m$. Its coefficients are differentiable: the contour integrals of $x^k g'_t(x)/g_t(x)$ give the cluster power sums, and Newton identities give its elementary symmetric functions. This construction also applies when the ambient degree changes. The complementary factor is nonzero at zero.

At each of the specified stable times, write the cluster roots as $\lambda_j = x_j + iy_j$, where $y_j \leq 0$. Differentiability of the first power sum gives $\sum_j |y_j| = O(t)$. The second power sum is also $O(t)$, so its real part gives $\sum_j x_j^2 = O(t)$, since $\sum_j y_j^2 = O(t^2)$. Thus every $\lambda_j = O(\sqrt{t})$ and $\sum_j \lambda_j^k = o(t)$ for $k \geq 3$. Newton identities imply

$$a_k(t) = o(t) \quad (k \geq 3), \quad a_2(t) = -\frac{1}{2} \sum_j x_j^2 + o(t).$$

For the second equality, the imaginary part of the second power sum is $O(t^{3/2})$ and $a_1(t)^2 = O(t^2)$. Consequently $a'_2(0)$ is real and nonpositive, while $a'_k(0) = 0$ for $k \geq 3$. Multiplication by the complementary factor proves the assertions. All derivatives are fixed by the differentiable curve, so estimates along the given sequence suffice. $\square$

Theorem 4.3 (Finite equatorial tests). *Let $\Phi_t = e^{t\mathcal{L}}$ be a continuous CPTP semi-group on n qubits. Set $\mathcal{E} = \{0, \sqrt{3}, -\sqrt{3}\}$ and*

$$\psi_a = \frac{(1+ia)|0\rangle + (1-ia)|1\rangle}{\sqrt{2(1+a^2)}}, \quad \rho_a = \bigotimes_{j=1}^n |\psi_{a_j}\rangle\langle\psi_{a_j}|, \quad \mathbf{a} \in \mathcal{E}^n.$$

Suppose that, for each of these 3^n densities, there is a sequence of positive times tending to zero along which its image is Lee-Yang. Then $\mathcal{L}$ is a sum of one-qubit GKLS generators. The sequences may depend on the input.

Proof. The coefficient Cayley transform is

$$C = \frac{1}{\sqrt{2}} \begin{pmatrix} i & -i \\ 1 & 1 \end{pmatrix}, \quad f(z) \mapsto \frac{s+i}{\sqrt{2}} f\left(\frac{s-i}{s+i}\right). \quad (18)$$

It takes disk stability to upper-half-plane stability. Applied to all row and column legs of a density tensor, it is a coefficient transformation, not a unitary conjugation of that density matrix. Direct multiplication gives $C\psi_a = (-a, 1)^T/\sqrt{1+a^2}$ and $C\overline{\psi_a} = (a, 1)^T/\sqrt{1+a^2}$. The transformed input is therefore a nonzero scalar multiple of

$$p_{\mathbf{a}}(z, w) = \prod_j (z_j - a_j)(w_j + a_j). \quad (19)$$

For a binary coefficient matrix $M = (M_{ab})$, its polynomial action has the first-order form

$$M_{00} + M_{10}s + [M_{01} + (M_{11} - M_{00})s - M_{10}s^2]\partial_s.$$

The identities $CXC^* = -Z$, $CYC^* = X$, $CZC^* = -Y$ consequently give the derivative coefficients

$$v(s) = (2s, 1 - s^2, i(1 + s^2)) \quad (20)$$

for left multiplication. Right multiplication uses transposed Paulis; its derivative vector at a general column coordinate t is $\tilde{v}(t) = (2t, -(1 - t^2), i(1 + t^2))$. In particular $\tilde{v}(-a) = -\overline{v(a)}$.

Use (15). A zero diagonal entry of a positive semidefinite matrix has a zero row and column. Suppose that $k \geq 2$ is the greatest weight of a Pauli string with $K_{PP} > 0$, and choose such a string with support S , $|S| = k$. All noise rows of larger weight vanish. Put both coordinates of each site in S at their roots in (19), keeping every other coordinate away from its root, and restrict to a line with every coordinate slope positive. The initial polynomial has a root of multiplicity $2k \geq 4$. Lemma 4.2 makes its first variation zero at this point.

Hamiltonian and anticommutator terms act only on row legs or only on column legs, so leave selected roots untouched. A sandwich $P\rho Q$ can survive only if both supports contain S . Maximality makes both supports exactly S . Dividing out the nonzero complementary product and the input scalar therefore gives

$$0 = (-1)^k \sum_{\text{supp } P = \text{supp } Q = S} K_{PQ} V_P(\mathbf{a}) \overline{V_Q(\mathbf{a})}, \quad V_P(\mathbf{a}) = \prod_{j \in S} v_{P_j}(a_j). \quad (21)$$

The sum, without its sign, is a positive semidefinite quadratic form. Thus the exact-support block K_{SS} kills $\overline{V(\mathbf{a})}$ for every $\mathbf{a} \in \mathcal{E}^S$. The three components in (20) form a basis of the quadratic polynomials. Evaluation at the three distinct points of $\mathcal{E}$ is invertible. The resulting tensor products span $\mathbb{C}^{3^k}$, forcing $K_{SS} = 0$, a contradiction. Every surviving noise index consequently has weight one. This still permits cross-site noise blocks; we eliminate them below.

The anticommutator operator $\sum K_{PQ} QP$ now has support at most two. If a Hamiltonian term has maximal support S with $|S| = k \geq 3$, put just the row coordinates of S at their roots. All column coordinates and remaining row coordinates stay away from their roots. The k -fold contact again has zero first variation. No noise term, anticommutator, or column Hamiltonian removes all these row roots. The surviving row Hamiltonian equals a nonzero common factor times

$$-i \sum_{\text{supp } P = S} h_P \prod_{j \in S} v_{P_j}(a_j).$$

The same finite tensor evaluation basis forces every displayed h_P to vanish. Descending in the maximal support leaves only Hamiltonian terms of weight at most two. The transformed superoperator now acts on at most two individual row or column legs at a time.

Fix distinct physical sites i, j . A row-row double contact with coordinates $s, t \in \mathcal{E}$ uses input parameters $a_i = s, a_j = t$. A row-column contact uses $a_i = s, a_j = -t$, which is available because $\mathcal{E} = -\mathcal{E}$. Keep all other coordinates off their roots. Only the corresponding mixed principal coefficient survives. Dividing by the initial quadratic coefficient divides that coefficient by the positive product

of the two selected slopes. Lemma 4.2 therefore makes both mixed coefficients real and nonpositive on this nine-point grid.

Let K^{ij} be the 3×3 cross-site noise block, and write the Hamiltonian pair as $\sum_{\alpha,\beta} h_{\alpha\beta} \sigma_{i,\alpha} \sigma_{j,\beta}$ with h real. The row-column coefficient matrix is K^{ij} . Since distinct-site Paulis commute, the two conjugate noise blocks in the anticommutator combine to give the row-row coefficient matrix

$$-\Re K^{ij} - ih. \quad (22)$$

Here the real part is taken entrywise; there is no transpose in this formula.

Put

$$u(s) = \frac{(2s, 1 - s^2)}{1 + s^2}, \quad R = \text{diag}(1, -1).$$

The normalized row and column derivative vectors are $(u(s), i)$ and $(Ru(t), i)$. Uniform averaging over $\mathcal{E}$ gives

$$\mathbb{E}u = 0, \quad \mathbb{E}uu^\top = \frac{1}{2}I_2. \quad (23)$$

Initially write $K^{ij} = \begin{pmatrix} A & B \\ D & e \end{pmatrix}$ with complex blocks. The imaginary part of its normalized row-column coefficient is the imaginary part of $u^\top ARv + iu^\top B + iDRv - e$. Its nine values vanish. Averaging these values, and then their products with u , Rv , and $u(Rv)^\top$, proves $\Im e = 0$, $\Re B = 0$, $\Re D = 0$, and $\Im A = 0$. Hence $B = ib$, $D = id^\top$ with A, b, d, e real, and

$$u^\top ARv - b^\top u - d^\top Rv - e \leq 0. \quad (24)$$

The imaginary part obtained from (22) is $-u^\top h_{TT}v + h_{zz}$, where $T = \{x, y\}$. The same moments give $h_{TT} = 0 = h_{zz}$. Write $k = h_{Tz}$ and $\ell^\top = h_{zT}$. The real row-row inequality is

$$e - u^\top Av + k^\top u + \ell^\top v \leq 0. \quad (25)$$

Both inequalities have already been divided by the positive factor $(1 + s^2)(1 + t^2)$. Their unweighted averages are respectively $-e$ and e , so $e = 0$. Each of the nine nonpositive values in each inequality has zero average, and thus each value is zero. Applying (23) once more gives $A = b = d = k = \ell = 0$. Consequently $K^{ij} = 0$ and the pair Hamiltonian vanishes.

Every remaining block K^{ii} is positive semidefinite. The remaining terms of (15) are therefore genuine one-qubit GKLS generators. They commute on distinct tensor factors. The proof compared only derivatives at zero, so the different stable time sequences present no compatibility issue. $\square$

Corollary 4.4. *If a CPTP generator is not a sum of one-qubit generators, one of the fixed inputs in Theorem 4.3 has a non-Lee-Yang output at every sufficiently small positive time.*

Proof. Otherwise each input would have Lee-Yang output at positive times arbitrarily close to zero, providing the sequences in that theorem. $\square$

The theorem makes no assertion that its number of tests is minimal or that testing the output polynomial is computationally efficient. Nor can a single isolated positive time replace the sequences: the interaction $H = Z_1 Z_2$ generates the identity channel at $t = \pi$.

4.2. The local reality forced by a second physical site. We first record the elementary one-qubit state geometry.

Lemma 4.5. *A qubit density matrix $\rho = (I + r \cdot \sigma)/2$ is Lee–Yang if and only if $r_z \geq 0$.*

Proof. Write $\rho = \begin{pmatrix} a & b \\ \bar{b} & d \end{pmatrix}$, so $a + d = 1$, $|b|^2 \leq ad$, and $F_\rho(z, w) = g(w) + zh(w)$ with $g = a + bw$, $h = \bar{b} + dw$. On $|w| = 1$,

$$|g(w)|^2 - |h(w)|^2 = (a - d)[a + d + 2\Re(bw)]. \quad (26)$$

If $a > d$, then $a > |b|$, so g is zero-free on the closed disk. The right side is nonnegative. The maximum principle applied to h/g proves $|h/g| \leq 1$ on the disk and hence the desired zero-freeness. If $a = d$, add a positive multiple of $|0\rangle\langle 0|$ and take its nonzero coefficient limit; Hurwitz’s theorem gives the assertion. If $a < d$, positivity gives $d > |b|$, and (26) is strictly negative on the circle. Thus a nearby interior point has $|h| > |g|$ and $h \neq 0$; choosing $z = -g/h$ gives an interior zero. This proves both directions. $\square$

Proposition 4.6. *Suppose $n \geq 2$ and a CPTP semigroup ordinarily preserves Lee–Yang densities on a neighborhood of time zero. In the local representation given by Theorem 4.3, each noise matrix and Hamiltonian have the form*

$$K = \begin{pmatrix} B & id \\ -id^\top & c \end{pmatrix}, \quad B = B^\top \in \mathbb{R}^{2 \times 2}, \quad d \in \mathbb{R}^2, \quad H = hZ. \quad (27)$$

Writing $T = \text{Tr } B$ and $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, the local Bloch equation is

$$\dot{r}_\perp = Mr_\perp + k, \quad \dot{r}_z = -2Tr_z, \quad M = 2[B - (T + c)I_2] + 2hJ, \quad k = 4Jd. \quad (28)$$

In particular the finite-time map has the form

$$(r_\perp, r_z) \mapsto (D_t r_\perp + \tau_t, \lambda_t r_z), \quad D_t \in GL_2(\mathbb{R}), \quad \lambda_t = e^{-2Tt} > 0. \quad (29)$$

Proof. After locality is known, let $q_j(s, w)$ denote the coefficient of $\partial_s \partial_w$ in the Cayley-transformed generator at site j . For a general Hermitian noise matrix write

$$K = \begin{pmatrix} \alpha & \delta + i\eta & r + i\mu \\ \delta - i\eta & \beta & t + i\nu \\ r - i\mu & t - i\nu & \gamma \end{pmatrix}.$$

Multiplying the row and column derivative vectors in (20) gives

$$\Im q(s, w) = 2(s + w)[- \eta(1 - sw) + r(1 + sw) + t(w - s)]. \quad (30)$$

The Hamiltonian and anticommutator act on single legs and do not enter this coefficient.

Fix two sites. For real a, b and $c_0 > 0$, the polynomial $f(z_1, z_2) = (z_1 - a)(z_2 - b) - c_0$ is upper-half-plane stable: solving for z_2 when $z_1 \in \mathbb{H}$ gives a point in the lower half-plane. Its inverse Cayley transform defines a pure physical Lee–Yang input. The transformed conjugate factor is $g(w_1, w_2) = (w_1 + a)(w_2 + b) - c_0$, since $\bar{C} = \text{diag}(-1, 1)C$ and there are two sites. For arbitrary nonzero real u, v , choose the real contact

$$z_1 = a + u, \quad z_2 = b + c_0/u, \quad w_1 = -a + v, \quad w_2 = -b + c_0/v.$$

Here $f = g = 0$, $\nabla f = (c_0/u, u)$ and $\nabla g = (c_0/v, v)$. Their directional derivatives along any strictly positive direction are nonzero. The initial product has a double

root with nonzero real leading coefficient. Lower-order generator terms vanish at the contact, and its first variation is

$$h = \frac{c_0^2}{uv} q_1(a+u, -a+v) + uv q_2(b+c_0/u, -b+c_0/v). \quad (31)$$

Lemma 4.2 makes h real. Additional sites may be padded by equatorial factors kept away from their roots; their generators contribute zero at this contact.

Formula (30) gives $\Im q_2(b, -b) = 0$. The coefficient of c_0 in the imaginary part of (31) is therefore

$$v \partial_s \Im q_2(b, -b) + u \partial_w \Im q_2(b, -b) = 0.$$

Independent variation of u, v makes both derivatives vanish. By (30), this says $-\eta(1+b^2) + r(1-b^2) - 2tb = 0$ for every real b . Its three coefficients give $\eta = r = t = 0$. Interchanging the sites proves the same conclusion at the other site, and every site has a partner. This proves the stated shape of K .

At an equatorial Bloch vector the dissipator with this shape has zero longitudinal velocity. A Hamiltonian $h_x X + h_y Y + h_z Z$ contributes $\dot{r}_z = 2(h_x r_y - h_y r_x)$. Product inputs and Lemma 4.5 require this quantity to be nonnegative at every equatorial point. Replacing that point by its negative gives $h_x = h_y = 0$. Finally, multiplication of the Pauli matrices in the generator gives (28); in particular $\mathcal{L}(I) = -4d_y X + 4d_x Y$, fixing the displacement sign. The linear differential equation gives $D_t = e^{tM}$ and $\lambda_t = e^{-2Tt}$, as claimed. $\square$

4.3. Separable conditional blocks prevent compensation.

Lemma 4.7 (Conditional-block cancellation). *Let a qubit channel Ψ have the affine form $(r_\perp, r_z) \mapsto (Dr_\perp + \tau, \lambda r_z)$, with $\lambda > 0$. If $\alpha, \beta \in \mathbb{C}$ satisfy $|\beta| > |\alpha|$, the polynomial of $\alpha\Psi(|0\rangle\langle 0|) + \beta\Psi(|1\rangle\langle 1|)$ has a zero in $\mathbb{D}^2$.*

Proof. The scalar $\alpha + \beta$ is nonzero. Dividing the matrix by $(\alpha + \beta)/2$ gives $I + \tau \cdot \sigma_\perp + qZ$, where

$$q = \lambda \frac{\alpha - \beta}{\alpha + \beta}, \quad \Re q = \lambda \frac{|\alpha|^2 - |\beta|^2}{|\alpha + \beta|^2} < 0.$$

Its polynomial is $(1+q) + (\tau_x + i\tau_y)z + (\tau_x - i\tau_y)w + (1-q)zw$. Choose a phase θ making $(\tau_x + i\tau_y)e^{i\theta} = |\tau|$ and substitute $z = e^{i\theta}u$, $w = e^{-i\theta}u$; if $\tau = 0$ any phase works. The quadratic has nonzero leading coefficient $1-q$ and root product $(1+q)/(1-q)$ of modulus less than one. At least one root has modulus less than one, giving the asserted pair of disk coordinates. $\square$

The lemma applies to the actual evolving channel at a second site. It does not require that this channel fix the reference state. In particular, if positive matrices A, B give a Lee–Yang input proportional to $A \otimes |0\rangle\langle 0| + B \otimes |1\rangle\langle 1|$, ordinary preservation by $\Phi \otimes \Psi$ forces

$$|F_{\Phi(B)}(z, w)| \leq |F_{\Phi(A)}(z, w)| \quad (z, w \in \mathbb{D}). \quad (32)$$

Indeed a strict reverse inequality supplies α, β for Lemma 4.7. This also covers $\alpha = 0$.

Proposition 4.8 (Recovery of each local inequality). *Under the assumptions of Proposition 4.6, each local matrix in (27) satisfies*

$$r^\top Bv + d \cdot (v - r) \leq c \quad (r, v \in \mathbb{R}^2, |r| = |v| = 1). \quad (33)$$

Necessity follows from positive definite separable inputs classical on one of two selected sites.

Proof. Fix an active site and a distinct reference site. For $0 < \kappa < 1/2$ and $e \in \mathbb{R}^2$ with $|e| = 1$, put

$$U = I + \frac{1}{2}Z, \quad V = \frac{1}{8}I + \frac{\sqrt{\kappa(1-\kappa)}}{4}\sigma_e + \frac{\kappa}{4}Z, \quad \sigma_e = e_x X + e_y Y,$$

and define

$$A = \frac{1}{2}(U + V), \quad B_0 = \frac{1}{2}(U - V), \quad \rho_{\kappa,e} = \frac{1}{2}[A \otimes |0\rangle\langle 0| + B_0 \otimes |1\rangle\langle 1|]. \quad (34)$$

Both blocks are positive definite, because $\lambda_{\min}(U) = 1/2$ and $\|V\| = 1/8 + \sqrt{\kappa}/4 < 1/2$. Their traces sum to 2, so this is a normalized positive definite separable density matrix.

For a Hermitian matrix $Q = q_0 I + q_\perp \cdot \sigma_\perp + q_z Z$, parameterize the distinguished boundary by $z = e^{i(\gamma+\delta)}$, $w = e^{i(\gamma-\delta)}$, and set $x = \cos \gamma$, $y = \sin \gamma$, $n = (\cos \delta, -\sin \delta)$. Expansion of (14) gives

$$\frac{1}{2}e^{-i\gamma}F_Q(z, w) = q_0 x + q_\perp \cdot n - iq_z y. \quad (35)$$

Write $u = 1/2$, $v_0 = 1/8$, $w_0 = 2v_0\sqrt{\kappa(1-\kappa)}$ and $\zeta = \kappa v_0/u$, so $U = I + uZ$ and $V = v_0 I + w_0 \sigma_e + \zeta Z$. Formula (35) yields

$$\begin{aligned} \mathcal{D}_0(x, n) &:= \frac{1}{4}(|F_A|^2 - |F_{B_0}|^2) \\ &= v_0[(1-\kappa)x^2 + 2\sqrt{\kappa(1-\kappa)}x(e \cdot n) + \kappa] \\ &\geq v_0(\sqrt{1-\kappa}|x| - \sqrt{\kappa})^2 \geq 0. \end{aligned} \quad (36)$$

The matrix A is positive definite with strictly positive Z coefficient. Consequently its diagonal entries obey $a > d > 0$, and (26) is strictly positive on the circle. The proof of Lemma 4.5 then gives zero-freeness of F_A on the closed bidisk. Applying the maximum principle separately in z, w to F_{B_0}/F_A extends (36) to that bidisk. Hence $F_A + z_2 w_2 F_{B_0}$ is nonzero on $\mathbb{D}^4$, proving that (34) is Lee–Yang. Other sites may be padded by maximally mixed states.

For the active map write D_t, τ_t, λ_t as in (29). At time zero their derivatives are $M, k, -2T$ in (28). The reference map has $\lambda_t > 0$ at every finite time. By (32), ordinary preservation forces the active output block difference to be nonnegative on the disk and, by continuity, on its boundary. Its exact normalized expression is

$$\mathcal{D}_t(x, n) = (x + \tau_t \cdot n)[v_0(x + \tau_t \cdot n) + w_0 n^\top D_t e] + \lambda_t^2 u \zeta (1 - x^2). \quad (37)$$

Choose $n = -e$ and $x = \rho := \sqrt{\kappa/(1-\kappa)} \in (0, 1)$. The input difference is zero. Its derivative must therefore be nonnegative. Differentiating (37) gives

$$\begin{aligned} \frac{\mathcal{D}'_0}{2v_0\kappa} &= \rho k \cdot n - n^\top M n - 2T \frac{1-2\kappa}{1-\kappa} \\ &= 2(c - n^\top B n) + 2T\rho^2 + \rho k \cdot n \geq 0. \end{aligned} \quad (38)$$

For explicit verification, the three derivative contributions before division are $(k \cdot n)(2v_0\rho - w_0)$, $-\rho w_0 n^\top M n$, and $-4T u \zeta (1 - \rho^2)$. The Hamiltonian term drops out because $n^\top J n = 0$.

Set $n = Jm$, $|m| = 1$. Since $k = 4Jd$ and $(Jm)^\top B(Jm) = T - m^\top B m$, we obtain

$$c - T + m^\top B m + T\rho^2 + 2\rho m \cdot d \geq 0 \quad (|m| = 1, 0 \leq \rho \leq 1). \quad (39)$$

The endpoints follow by continuity. For arbitrary unit r, v , set $(v - r)/2 = \rho m$. Their mean is perpendicular to m and has squared length $1 - \rho^2$; when $\rho = 0$, choose either perpendicular direction for m . In two dimensions this gives

$$r^\top Bv + d \cdot (v - r) = T - m^\top Bm - T\rho^2 + 2\rho m \cdot d.$$

Apply (39) with $-m$ to prove (33). $\square$

4.4. The exact local semidefinite cone.

Lemma 4.9 (Reduction of the circle inequality). *Suppose $B = B^\top \in \mathbb{R}^{2 \times 2}$ and $\begin{pmatrix} B & id \\ -id^\top & c \end{pmatrix} \succeq 0$. Then (33) is equivalent to*

$$\begin{pmatrix} cI_2 - \text{adj } B & d \\ d^\top & \text{Tr } B \end{pmatrix} \succeq 0. \quad (40)$$

If $T = \text{Tr } B > 0$, the equivalent scalar condition is $c \geq \lambda_{\max}(\text{adj } B + dd^\top/T)$. If $T = 0$, positivity forces $B = d = 0$, and every $c \geq 0$ is allowed.

Proof. Unitary congruence, followed if necessary by changing the sign of the last coordinate, gives $\begin{pmatrix} B & d \\ d^\top & c \end{pmatrix} \succeq 0$. In particular $B \succeq 0$ and $dd^\top \preceq cB$. The assertion for $T = 0$ follows immediately. Suppose $T > 0$. Writing $(v - r)/2 = \rho m$ as above, the expression to maximize is

$$T - m^\top Bm - T\rho^2 + 2\rho d \cdot m, \quad |m| = 1, \quad 0 \leq \rho \leq 1.$$

If $c < T$, then $|d|^2 \leq c\lambda_{\max}(B) < T^2$. Choose the sign of m so $d \cdot m \geq 0$. The unconstrained maximum in ρ is attained at $\rho = |d \cdot m|/T < 1$. The maximum is consequently

$$\lambda_{\max} \left(TI_2 - B + \frac{dd^\top}{T} \right) = \lambda_{\max} \left(\text{adj } B + \frac{dd^\top}{T} \right).$$

This proves equivalence in this case.

If $c \geq T$, put $b_m = m^\top Bm$ and use $|d \cdot m| \leq \sqrt{cb_m}$. The difference between c and the expression above is bounded below by

$$c - T + b_m + T\rho^2 - 2\sqrt{cb_m}\rho = (\sqrt{b_m} - \sqrt{c}\rho)^2 + (c - T)(1 - \rho^2) \geq 0.$$

The scalar matrix condition also holds automatically, because

$$dd^\top \preceq cB \preceq T[B + (c - T)I_2],$$

the second inequality being equivalent to $(c - T)(TI_2 - B) \succeq 0$. Finally, the Schur complement with positive last entry T proves equivalence of the scalar condition and (40). The separate $c \geq T$ argument avoids an invalid unconstrained radial maximization when $|d| > T$. $\square$

Lemma 4.10 (Choi tensors and complete preservation). *For a CPTP map Φ on n qubits, the following are equivalent: its full Choi tensor is Lee–Yang; it completely preserves physical Lee–Yang states; and its extension by n inert qubits takes the maximally entangled state to a Lee–Yang state. For any invertible complex-linear map, without a positivity or trace assumption, its full Choi tensor is Lee–Yang if and only if its coefficient action preserves all complex disk-stable polynomials in the $2n$ row and column variables.*

Proof. With $|\Omega\rangle = \sum_a |a\rangle|a\rangle$, the Choi entries are $J_{ai,bj} = [\Phi(|i\rangle\langle j|)]_{ab}$. The normalized maximally entangled input is Lee–Yang, since its polynomial is proportional to $\prod_k (1 + z_k z'_k)(1 + w_k w'_k)$. Its output is the normalized Choi matrix. Conversely, for a joint input ρ , the output coefficients are

$$[(\Phi \otimes \text{Id})(\rho)]_{ar,bs} = \sum_{i,j} J_{ai,bj} \rho_{ir,js}.$$

This is bilinear tensor contraction with no conjugation of entries. Lemma 2.3 gives a Lee–Yang polynomial or zero; trace preservation excludes zero. This proves the physical equivalences.

Let M be the superoperator matrix in coefficient coordinates. Its full tensor differs from the Choi tensor only by a permutation of indices. Contraction with an arbitrary stable coefficient vector gives stability or zero, and invertibility excludes zero. In the other direction, if M preserves all stable polynomials, apply it for each fixed $w \in \mathbb{D}^{2n}$ to $\prod_j (1 + z_j w_j)$. The resulting joint polynomial is exactly the full tensor of M and is nonzero for every $(z, w) \in \mathbb{D}^{4n}$. $\square$

Proposition 4.11 (One-qubit complete generators). *A one-qubit CPTP semigroup is completely preserving if and only if its generator has the form (27) and satisfies the two positive semidefinite conditions in (17).*

Proof. For necessity, extend the semigroup by one inert qubit. Propositions 4.6 and 4.8 give the real local shape and (33) for the active site. Lemma 4.9 gives the two matrix conditions.

For sufficiency, transform both coefficient legs by (18). Write $B = \begin{pmatrix} a & u \\ u & b \end{pmatrix}$ and $d = (x, y)^\top$. The transformed generator is real. This follows directly from the Pauli identities: the mixed terms pair the purely imaginary transverse–longitudinal noise entries with the purely imaginary Z derivative; the anticommutator is $-(a + b + c)I + y(Z_1 + Z_2) + x(X_1 - X_2)$; and the longitudinal Hamiltonian is real after the transform. Its full coefficient of $\partial_s \partial_t$ is

$$\begin{aligned} q(s, t) = & 4ast - b(1 - s^2)(1 - t^2) - c(1 + s^2)(1 + t^2) \\ & + 2u[t(1 - s^2) - s(1 - t^2)] \\ & + 2x[t(1 + s^2) - s(1 + t^2)] - y[(1 - s^2)(1 + t^2) + (1 + s^2)(1 - t^2)]. \end{aligned} \quad (41)$$

There is no extra factor of two in this convention. With

$$r(s) = \frac{(2s, 1 - s^2)}{1 + s^2}, \quad v(t) = \frac{(2t, t^2 - 1)}{1 + t^2},$$

we have

$$\frac{q(s, t)}{(1 + s^2)(1 + t^2)} = r(s)^\top B v(t) + d \cdot (v(t) - r(s)) - c \leq 0. \quad (42)$$

The real two-variable specialization of the finite-box generator theorem (Theorem 3.2) says that a real endomorphism of the binary polynomial box generates upper-half-plane stability preservers precisely when this mixed coefficient is nonpositive on $\mathbb{R}^2$. It applies here because the transformed operator is an endomorphism of that box. Thus its exponential preserves complex stability. Undoing the Cayley similarity and applying Lemma 4.10, with invertibility of the exponential, proves complete preservation. $\square$

Proof of Theorem 4.1. Assume preservation on a neighborhood of zero. Theorem 4.3 gives locality. Propositions 4.6, 4.8, and Lemma 4.9 give (16)–(17). Conversely those conditions give complete preservation of every local semigroup by Proposition 4.11. Their tensor product completely preserves by successive application to the individual sites. Lemma 4.10 equates this with the full Choi condition. Complete preservation implies ordinary preservation, which implies its small-time version. The one-qubit exception is proved next. $\square$

4.5. The Hermiticity-preserving branch without positivity. For the full Choi condition, locality does not require complete positivity. The local semidefinite reduction does require it, and must be replaced by the circle inequality when positivity is absent.

Theorem 4.12 (Hermiticity- and trace-preserving semigroups). *Let $T_t = e^{t\mathcal{L}}$ be a continuous semigroup of complex-linear, Hermiticity-preserving, trace-preserving maps on n qubits. Its full Choi tensor is Lee–Yang for every $t \geq 0$ if and only if $\mathcal{L} = \sum_j \mathcal{L}_j \otimes \text{Id}_j$, where each $\mathcal{L}_j$ has the algebraic form (16), with*

$$H_j = h_j Z, \quad K_j = \begin{pmatrix} B_j & id_j \\ -id_j^\top & c_j \end{pmatrix}, \quad B_j = B_j^\top \in \mathbb{R}^{2 \times 2}, \quad d_j \in \mathbb{R}^2, \quad c_j, h_j \in \mathbb{R},$$

and satisfies

$$c_j \geq \max_{r, v \in S^1} [r^\top B_j v + d_j \cdot (v - r)]. \quad (43)$$

Here no positive semidefiniteness of K_j is imposed. Requiring the Choi property only on a neighborhood of zero is equivalent. If complete positivity is also imposed, the local conditions reduce exactly to (17).

Proof. We first justify the algebraic representation used in the proof. A Hermiticity-preserving, trace-annihilating generator has a representation (15) with $H = H^*$ and $K = K^*$, without the condition $K \succeq 0$. Indeed, its Choi matrix is Hermitian. In the orthogonal vectorized Pauli basis, its compression to the orthogonal complement of $|I\rangle\rangle$ uniquely specifies the Hermitian coefficients K_{PQ} of the sandwich terms. Subtract the corresponding dissipator, which is Hermiticity-preserving and trace-annihilating. The remaining Hermitian Choi matrix has zero compression to that complement, hence is of the form $|B\rangle\rangle\langle\langle I| + |I\rangle\rangle\langle\langle B|$ for some matrix B . It represents $\rho \mapsto B\rho + \rho B^*$. Trace annihilation gives $B + B^* = 0$, so $B = -iH$ with H Hermitian. This proves the claimed representation.

The last assertion of Lemma 4.10 and the Cayley transform show that the full Choi hypothesis places the coefficient generator in Theorem 3.2 on $2n$ binary legs. It consequently has differential order at most two. This implies tensor support at most two on those individual legs: expand in the local endomorphism basis $I, \partial_s, s\partial_s - 1/2, s^2\partial_s - s$. At a nonzero support of maximal size, its full derivative coefficient is a linear combination of independent products of $1, s, s^2$. If the support size exceeds two, the order bound forces all its coefficients to vanish. Descending eliminates every larger support. Local Cayley similarity preserves the identity and traceless subspaces, so this conclusion also holds before the transform.

For nonidentity Pauli strings P, Q , the tensor coefficient supported on their respective row and column legs is exactly the sandwich coefficient K_{PQ} , up to the harmless sign of a transposed Y . Neither a Hamiltonian nor an anticommutator

term can cancel a coefficient involving both row and column legs. Therefore

$$\text{wt}(P) + \text{wt}(Q) > 2 \implies K_{PQ} = 0.$$

In particular every row indexed by a string of weight at least two vanishes, directly and without a PSD argument. The anticommutator now has support at most two, so the same support bound excludes every Hamiltonian term of higher weight.

For distinct physical sites, the cross-site matrices and their normalized mixed coefficients are precisely (22)–(25). The general generator theorem makes these coefficients real and nonpositive at every real pair. Restricting to the trine grid, the discrete moment calculation (23) eliminates every cross-site block and pair Hamiltonian exactly as in Theorem 4.3. That calculation uses Hermiticity and trace annihilation, but never PSD of K . Thus the generator is a sum of local Hermiticity-preserving, trace-annihilating generators.

At one site, the nine products of the two Pauli derivative triples are independent. Reality of their mixed coefficient forces K_{xy} to be real and K_{xz}, K_{yz} to be purely imaginary, giving the displayed local shape. Its dissipator is then real after the Cayley transform. For a Hamiltonian $h_x X + h_y Y + h_z Z$, the remaining imaginary first-order coefficients are

$$h_y s^2 - 2h_x s - h_y, \quad h_y t^2 + 2h_x t - h_y.$$

The complex generator criterion requires each quadratic to be nonnegative on the real line. Comparing leading and constant terms forces $h_y = 0$, and then the linear term forces $h_x = 0$. The real mixed coefficient is (41), whose normalization (42) is nonpositive exactly under (43). This proves necessity without any positivity assumption.

Conversely, the displayed local algebraic generators are Hermiticity-preserving and trace-annihilating. Their Cayley transforms are real binary-box endomorphisms with nonpositive mixed coefficient. Theorem 3.2 and Lemma 4.10 prove full Choi stability of their exponentials and their tensor product. Small-time preservation extends to all times by the semigroup identity and Lemma 2.3, with invertibility excluding zero. Finally, complete positivity imposes $K_j \succeq 0$ in the canonical representation, and Lemma 4.9 gives exactly the second matrix condition in (17). $\square$

4.6. Cancellation at a fixed positive time. The mechanism used above has a finite-time version. It applies to different local channels and does not assume that they belong to a semigroup.

Theorem 4.13 (Static tensor cancellation). *Let $n \geq 2$. For each j , let Ψ_j be an invertible CPTP qubit map with Bloch action*

$$(r_\perp, r_z) \mapsto (D_j r_\perp + \tau_j, \lambda_j r_z), \quad D_j \in GL_2(\mathbb{R}), \quad \lambda_j > 0.$$

Then $\bigotimes_j \Psi_j$ ordinarily preserves all physical Lee–Yang densities if and only if every Ψ_j has a Lee–Yang Choi tensor. If one factor fails this condition, a positive definite separable input, classical on a second selected site and maximally mixed on the other sites, detects the failure, whatever the parameters of the second factor.

Proof. The outputs of the two computational basis states give $|\tau|^2 + \lambda^2 \leq 1$, so $|\tau| < 1$. The static one-qubit criterion in Theorem 7.1 says that the Choi condition for the active factor Ψ is equivalent to

$$\|D^\top n\|^2 \leq (x - \tau \cdot n)^2 + \lambda^2(1 - x^2) \quad (n \in S^1, -1 \leq x \leq 1). \quad (44)$$

Changing x to $-x$ gives the alternative sign convention for its transverse displacement. Suppose the condition fails. By strictness and continuity choose $x \in (-1, 1)$ and $n \in S^1$ such that, with $v = D^\top n$, $V = |v|$ and $y = x - \tau \cdot n$,

$$V^2 > y^2 + \lambda^2(1 - x^2). \quad (45)$$

In particular $V > 0$. Set $u = yv/V^2$, so $|u| < 1$ and $u \cdot v = y$. The matrix $W = I_2 - uu^\top$ is positive definite and $v^\top W v = V^2 - y^2 > \lambda^2(1 - x^2)$. Choose $0 < \theta < 1$ sufficiently close to one and put

$$d_0 = \theta \frac{Wv}{\sqrt{v^\top W v}}.$$

Then $W - d_0 d_0^\top \succ 0$, as follows by congruence with $W^{-1/2}$, and

$$d_0 \cdot v > \lambda \sqrt{1 - x^2}. \quad (46)$$

Choose $\beta > 0$ with $\beta(1 - |u|) > |d_0|$ and define

$$A_R = R(I + u \cdot \sigma_\perp) + Z, \quad B_0 = \beta I + (\beta u + d_0) \cdot \sigma_\perp.$$

The second matrix is positive definite. The first is positive definite once $R^2(1 - |u|^2) > 1$.

We claim that, for all sufficiently large finite R , F_{A_R} is zero-free on the closed bidisk and

$$|F_{B_0}(s, w)| < |F_{A_R}(s, w)| \quad (|s|, |w| \leq 1). \quad (47)$$

On the distinguished boundary use (35) with $X = \cos \gamma$, $m \in S^1$, and $\ell = X + u \cdot m$. The squared-modulus difference divided by four is

$$G_R(X, m) = (R^2 - \beta^2)\ell^2 - 2\beta\ell(d_0 \cdot m) + 1 - X^2 - (d_0 \cdot m)^2.$$

On $\ell = 0$, its last two terms are at least $\delta := \lambda_{\min}(I_2 - uu^\top - d_0 d_0^\top) > 0$. If $|\ell| \leq \epsilon$, then $|X^2 - (u \cdot m)^2| \leq 2\epsilon$. Choose $\epsilon > 0$ so $2\epsilon + 2\beta\epsilon|d_0| < \delta/2$. For $R \geq \beta$, this gives $G_R > 0$ on that set. On $|\ell| \geq \epsilon$, using $|\ell| \leq 2$ gives

$$G_R \geq (R^2 - \beta^2)\epsilon^2 - 4\beta|d_0| - |d_0|^2 > 0$$

for sufficiently large R . This proves strict torus domination.

The diagonal entries of A_R are $R + 1, R - 1$ and its off-diagonal magnitude is $R|u|$. The constant coefficient in the representation $F_{A_R} = g(w) + sh(w)$ is nonzero on the closed disk because $R + 1 > R|u|$. On the circle,

$$|g(w)|^2 - |h(w)|^2 = 4R + 4\Re((A_R)_{01}w) > 0.$$

The maximum principle gives closed-bidisk zero-freeness of F_{A_R} . Apply it separately in both coordinates to F_{B_0}/F_{A_R} to obtain (47). Consequently the density proportional to $A_R \otimes |0\rangle\langle 0| + B_0 \otimes |1\rangle\langle 1|$ is positive definite, separable, and Lee–Yang.

At the output torus point in (35) with $X = -x$, $m = n$ and $Y = \sqrt{1 - x^2}$, the affine action gives

$$\frac{1}{2}e^{-i\gamma}F_{\Psi(A_R)} = R[-x + \tau \cdot n + u \cdot v] - i\lambda Y = -i\lambda Y,$$

and

$$\frac{1}{2}e^{-i\gamma}F_{\Psi(B_0)} = \beta[-x + \tau \cdot n + u \cdot v] + d_0 \cdot v = d_0 \cdot v.$$

The first value is nonzero, while (46) makes the second strictly larger in modulus. The strict inequality persists when the active coordinates move slightly into the disk. Lemma 4.7 then gives a zero of the two-site output after any allowed partner channel. Padding by maximally mixed states proves necessity for every $n \geq 2$. Conversely,

if all factors have Lee–Yang Choi tensors, Lemma 4.10 gives complete preservation at each site and therefore ordinary preservation of their tensor product. $\square$

The theorem assumes the displayed local affine form. It is not a factorization theorem for arbitrary isolated channels. In particular, the continuous-time locality theorem does not by itself imply such a static conclusion.

4.7. The one-qubit exception and dynamical consequences.

Proposition 4.14 (The ordinary one-qubit cone). *Let a one-qubit CPTP semigroup have Bloch equation $\dot{r} = Ar + b$, with real A, b . It ordinarily preserves Lee–Yang states if and only if*

$$b_z \geq \sqrt{A_{zx}^2 + A_{zy}^2}. \quad (48)$$

Proof. By Lemma 4.5, the additional condition is invariance of the northern Bloch hemisphere. At $r_z = 0$, its minimum outward-coordinate velocity is $b_z - \sqrt{A_{zx}^2 + A_{zy}^2}$, since the transverse vector ranges over the closed unit disk. A negative value gives immediate failure. Conversely, complete positivity keeps $|r(t)| \leq 1$. Under (48), $\dot{r}_z \geq A_{zz}r_z$ at every time. Multiplication by $e^{-A_{zz}t}$ and integration give $r_z(t) \geq e^{A_{zz}t}r_z(0) \geq 0$. $\square$

Example 4.15 (Amplitude damping). The amplitude-damping channel toward $|0\rangle$ has Kraus operators $K_0 = \text{diag}(1, \sqrt{\eta})$ and $K_1 = \sqrt{1-\eta}|0\rangle\langle 1|$, where $0 \leq \eta \leq 1$. The semigroup $\eta = e^{-\gamma t}$ satisfies $\dot{r}_z = \gamma(1 - r_z)$ and therefore ordinarily preserves one-qubit Lee–Yang states. It is not completely preserving for $0 < \eta < 1$. Indeed, its action on one half of a Bell state gives, up to a positive scalar, the polynomial

$$(1 + \sqrt{\eta}z_1z_2)(1 + \sqrt{\eta}w_1w_2) + (1 - \eta)z_2w_2.$$

At $z_1 = z_2 = w_1 = w_2 = ir$ its value is $1 - (2\sqrt{\eta} + 1 - \eta)r^2 + \eta r^4$. It is positive at $r = 0$ and equals $2\eta - 2\sqrt{\eta} < 0$ at $r = 1$. An interior root $0 < r < 1$ proves the claim.

Remark 4.16. For diagonal $K = \text{diag}(a, b, c)$ the complete criterion is $c \geq \max(a, b)$, recovering the Pauli-semigroup portion of the known sufficient channel family [19, Theorem 3]. The full cone also permits nonunital noise: $\sqrt{\gamma}(X + iZ)/2$ is an allowed jump and damps toward an equatorial pure state. Within the completely preserving cone, only rotations around the fixed Z axis are allowed as coherent motion. More generally, ordinary one-qubit dynamics with Hamiltonian $(\omega_x X + \omega_y Y + \omega_z Z)/2$, downward and upward jump rates $\gamma_\downarrow, \gamma_\uparrow$, and arbitrary Z dephasing satisfies (48) exactly when $\gamma_\downarrow - \gamma_\uparrow \geq \sqrt{\omega_x^2 + \omega_y^2}$.

Corollary 4.17. *An ordinarily preserving CPTP semigroup on $n \geq 2$ qubits cannot take a product input to a nonproduct state, either at a finite time or in a limit as $t \rightarrow \infty$. In particular, a semigroup converging from every input to a fixed nonproduct density matrix cannot ordinarily preserve the full Lee–Yang class.*

Proof. Theorem 4.1 keeps every product input a tensor product. The set of product densities is closed, being the continuous image of a compact product of one-site state spaces. $\square$

This conclusion concerns universal preservation and preparation from product inputs. It does not exclude a correlated stationary state of a nonergodic product semigroup, nor preservation along one specially chosen trajectory. The identity semigroup illustrates the first distinction.

4.8. Explicit boundaries of restricted tests. The following examples distinguish preservation of selected inputs or classical populations from preservation of the whole physical class. Their zeros provide direct certificates independent of the classification.

Lemma 4.18 (Pair defects of real hyperbola contacts). *Let $\mathcal{L}$ be a sum of local GKLS generators of the real form (27), on $n \geq 2$ sites, and let q_j be its local mixed coefficient (41). Define*

$$\Delta_j = \sup_{s, w \in \mathbb{R}, s+w \neq 0} \frac{q_j(s, w)}{(s+w)^2}.$$

All the real double-contact inequalities from the two-site pure hyperbola family (31) hold if and only if every Δ_j is finite and

$$\Delta_i + \Delta_j \leq 0 \quad (i \neq j). \quad (49)$$

In particular, these are necessary conditions for ordinary preservation. They are not asserted to be sufficient for ordinary preservation.

Proof. Fix a pair of sites and number them one and two. The double-root leading coefficient of the product of the two hyperbola factors in the proof of Proposition 4.6 has sign uv . Thus its contact inequality is

$$\frac{c_0^2}{u^2 v^2} q_1(a_0 + u, -a_0 + v) + q_2(b_0 + c_0/u, -b_0 + c_0/v) \leq 0.$$

Writing $S = u + v$ and $R = c_0 S/(uv)$, when $S \neq 0$ this is the sum of the two ratios defining Δ_1, Δ_2 . Every pair of real points with nonzero sums S, R is attained: choose a nonzero real P with the sign of S/R , taking $0 < P < S^2/4$ when it is positive; let u, v be the roots of $X^2 - SX + P$, put $c_0 = PR/S > 0$, and choose a_0, b_0 to fit the first coordinate in each pair. Thus the contact inequalities with nonzero sums say that every value of the first ratio plus every value of the second is nonpositive. Fixing a finite value of either ratio bounds the other ratio above. Each ratio also has at least one finite value, so both suprema are real and their sum is nonpositive. The converse follows directly from the definition of the suprema.

When $S = 0$, also $R = 0$. The paired Pauli derivative vectors obey $\tilde{v}(-s) = -\overline{v(s)}$ in (20), so positivity of the local noise matrix gives

$$q_j(s, -s) = -v(s)^T K_j \overline{v(s)} \leq 0.$$

These contacts therefore impose no additional condition. For $n > 2$, pad the input by pure local factors that are nonzero at the selected contact. Local terms on the other sites then vanish to second order at that contact, leaving exactly the displayed two-site inequality. This proves the claim for every pair and the necessity statement. $\square$

Proposition 4.19 (Real hyperbola contacts are insufficient). *For $a > 0$ and $c \geq 0$, the two-qubit generator*

$$\mathcal{L}(\rho) = a(X_1 \rho X_1 - \rho) + c(Z_2 \rho Z_2 - \rho)$$

does not ordinarily preserve Lee–Yang densities. Nevertheless, when $c \geq a$, it satisfies every real double-contact inequality obtained from the pure input family used in (31).

Proof. Here $q_1(s, w) = 4asw$ and $q_2(s, w) = -c(1 + s^2)(1 + w^2)$. The identities $4sw \leq (s + w)^2$ and $(1 + s^2)(1 + w^2) - (s + w)^2 = (sw - 1)^2$ give $\Delta_1 = a$, $\Delta_2 = -c$, with equality attained in each supremum. Lemma 4.18 proves the assertion about all real hyperbola contacts.

For actual failure, use the positive input

$$\rho = \frac{2}{7} \left[\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \otimes |0\rangle\langle 0| + \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \otimes |1\rangle\langle 1| \right].$$

It has trace one. Up to its positive scalar its polynomial is $2 + sw + \frac{1}{4}(1 + s)(1 + w)zv$, which is zero-free on $\mathbb{D}^4$ because $|2 + sw| > 1$ and $|\frac{1}{4}(1 + s)(1 + w)zv| < 1$ there. The second-site dephasing vanishes on this input and its entire first-site trajectory. At $t = (\log 2)/a$, the first diagonal block is $\text{diag}(13/8, 11/8)$, whereas the second block is fixed. At $s = w = 19i/20$ the output polynomial becomes

$$\frac{1229}{3200} + \frac{39 + 760i}{1600}zv.$$

Choose $zv = -1229/[2(39 + 760i)]$. Its modulus is $1229/1522 < 1$, since $39^2 + 760^2 = 761^2$. Taking both z, v to be the same square root puts all variables in the open disk and makes the output zero. $\square$

Example 4.20 (Random interchange and incoherent exclusion). Let S interchange two qubits. For $0 < p < 1$, the channel $\rho \mapsto p\rho + (1 - p)S\rho S$ fails ordinary preservation. The Lee–Yang input $|+0\rangle\langle +0|$ gives an output polynomial proportional to $p(1 + z_1)(1 + w_1) + (1 - p)(1 + z_2)(1 + w_2)$. It vanishes at

$$z_1 = w_1 = -1 + \sqrt{1 - p}e^{i\pi/4}, \quad z_2 = w_2 = -1 + \sqrt{p}e^{-i\pi/4}.$$

Every coordinate is in the open disk: for $0 < x < 1$, $|-1 + xe^{\pm i\pi/4}|^2 = 1 + x^2 - \sqrt{2}x < 1$. In particular the random-interchange semigroup fails at each positive time, since its mixture parameter is $(1 + e^{-2\gamma t})/2$.

The two jumps $\sqrt{\gamma}|01\rangle\langle 10|$ and $\sqrt{\gamma}|10\rangle\langle 01|$, with $\gamma > 0$, also fail at all sufficiently small positive times. On the same input, let $u = e^{-\gamma t/2}$. The 00–10 coherence decays by u , and the two excited populations exchange at rate 2γ , giving

$$2F_t = 1 + u(z_1 + w_1) + \frac{1+u^4}{2}z_1w_1 + \frac{1-u^4}{2}z_2w_2.$$

At $z_1 = w_1 = -r$, $z_2 = w_2 = ir$, this is $1 - 2ur + u^4r^2$. Its value at $r = 0$ is positive. At $r = 1$ it is $u^4 - 2u + 1$, negative for $u < 1$ sufficiently close to one, because this function vanishes at one and has derivative two there. The intermediate value theorem supplies an interior zero. Thus this natural quantum extension of population exchange does not preserve the entire physical class.

Example 4.21 (Equal local damping does not restore preservation). Apply the same amplitude-damping channel to both halves of a Bell state. Its output polynomial, up to the positive factor $1/2$, is

$$1 + (1 - \eta)^2 + \eta(1 - \eta)(z_1w_1 + z_2w_2) + \eta^2z_1z_2w_1w_2 + \eta(z_1z_2 + w_1w_2).$$

At all four variables ir , the polynomial is $1 + (1 - \eta)^2 - 2\eta(2 - \eta)r^2 + \eta^2r^4$. It is positive at zero and equals $2(1 - \eta)(1 - 2\eta) < 0$ at one for $1/2 < \eta < 1$. Hence equal

local damping fails ordinary joint preservation at every sufficiently small positive time. For instance, $\eta = 3/4$ has the exact interior root $r^2 = (5 - 2\sqrt{2})/3$. Vacuum padding extends the example to any larger system.

5. PHYSICAL DYNAMICS AT ARBITRARY FINITE SPINS

The physical classification becomes more restrictive when a site has spin at least one. We keep the physical basis fixed and use the coherent normalization of Section 2. For positive integers $\kappa_i = 2j_i$, put

$$\mathcal{H} = \bigotimes_{i=1}^n \mathbb{C}^{\kappa_i+1}, \quad b_a = \prod_i \binom{\kappa_i}{a_i}, \quad f_\psi(z) = \sum_{a \leq \kappa} \sqrt{b_a} \psi_a z^a,$$

and define

$$F_\rho(z, w) = \sum_{a, b \leq \kappa} \sqrt{b_a b_b} \rho_{ab} z^a w^b. \quad (50)$$

A density matrix is coherently Lee–Yang when this polynomial is nonzero on $\mathbb{D}^{2n}$, with independent row and column variables. Ordinary preservation refers to this class on $\mathcal{H}$. Complete preservation means preservation after adjoining any finite collection of inert finite spins, with the same encoding. A pure input means a rank-one density matrix with disk-stable vector polynomial; the vector need not be a spin-coherent state. Adjoints are always taken in the physical orthonormal basis, equivalently in the polynomial inner product $\langle z^a, z^b \rangle = \delta_{ab}/b_a$.

Theorem 5.1 (Finite-spin physical generators). *Suppose that $\kappa_i \geq 2$ for at least one site, and let $\Phi_t = e^{t\mathcal{L}}$ be a continuous CPTP semigroup on $\mathcal{H}$. The following conditions are equivalent.*

- (1) Φ_t ordinarily preserves coherently Lee–Yang densities for every $t \geq 0$.
- (2) For some $\varepsilon > 0$, Φ_t preserves every pure coherently Lee–Yang input for $0 \leq t < \varepsilon$.
- (3) Φ_t completely preserves coherently Lee–Yang densities for every $t \geq 0$.
- (4) The generator is a sum of local generators,

$$\mathcal{L} = \sum_i \mathcal{L}_i \otimes \text{Id}_{\hat{\kappa}_i}. \quad (51)$$

At a site with $j_i \geq 1$, its local term is

$$\mathcal{L}_i(\rho) = -i[h_i S_i^z, \rho] - \frac{\gamma_i}{2} \sum_{\alpha=x,y,z} [S_i^\alpha, [S_i^\alpha, \rho]], \quad h_i \in \mathbb{R}, \quad \gamma_i \geq 0. \quad (52)$$

At a qubit site, its local term is

$$\mathcal{L}_i(\rho) = -i[h_i Z, \rho] + \sum_{\alpha, \beta=x,y,z} (K_i)_{\alpha\beta} (\sigma_\alpha \rho \sigma_\beta - \frac{1}{2} \{\sigma_\beta \sigma_\alpha, \rho\}), \quad h_i \in \mathbb{R},$$

where $B_i = B_i^\top \in \mathbb{R}^{2 \times 2}$, $d_i \in \mathbb{R}^2$, and

$$K_i = \begin{pmatrix} B_i & id_i \\ -id_i^\top & c_i \end{pmatrix} \succeq 0, \quad \begin{pmatrix} c_i I_2 - \text{adj } B_i & d_i \\ d_i^\top & \text{Tr } B_i \end{pmatrix} \succeq 0. \quad (53)$$

In particular, $\Phi_t = \bigotimes_i e^{t\mathcal{L}_i}$.

The dissipator in (52) is the familiar isotropic spin diffusion, with Casimir multipole rates [8, 9]. The assertion here is its necessity from physical zero-freeness, without an assumption of covariance, linearity of the noise, locality, or unitality. The axial Hamiltonian commutes with the isotropic dissipator; a nonzero axial drift need not make the full semigroup rotation covariant.

We first establish the reduction from arbitrary GKLS operators using only physical pure inputs. The spin realization is

$$S_i^x = \frac{1}{2}[(1 - z_i^2)\partial_i + \kappa_i z_i], \quad S_i^y = \frac{1}{2i}[(1 + z_i^2)\partial_i - \kappa_i z_i], \quad S_i^z = \kappa_i/2 - z_i\partial_i. \quad (54)$$

The capacity-dependent Cayley transform is

$$(Cf)(s) = 2^{-|\kappa|/2} \prod_i (s_i + i)^{\kappa_i} f\left(\frac{s_1 - i}{s_1 + i}, \dots, \frac{s_n - i}{s_n + i}\right). \quad (55)$$

It is the symmetric-power realization of (18), and is unitary in the weighted inner product. Applied separately to both coefficient legs of a pure density, it gives

$$p(z)\mathcal{R}p(w), \quad \mathcal{R}p(w) = (-1)^{|\kappa|} \overline{p(-\bar{w})}, \quad p = Cf_\psi. \quad (56)$$

Indeed this follows in capacity one from $\bar{C} = \text{diag}(-1, 1)C$, and then by symmetric powers and tensor products. This two-leg transformation is not unitary conjugation of the density matrix. Every stable nonzero p in the capacity box is obtained from a physical pure input after normalization. Its degree may be strictly below the capacity; the inverse of (55) still belongs to the prescribed box.

Lemma 5.2 (Physical order reduction). *Under condition (2) of Theorem 5.1, every jump in any finite GKLS representation has, after removal of its scalar part, the form*

$$V_\nu \in \text{span}_{\mathbb{C}}\{S_i^x, S_i^y, S_i^z : 1 \leq i \leq n\}.$$

The Hamiltonian has spin degree at most two: it is a sum of a scalar, linear spin terms, on-site symmetric quadratics, and two-site bilinears. At qubit sites the on-site quadratics reduce to a scalar and linear terms.

Proof. Write a finite-dimensional GKLS representation

$$\mathcal{L}(\rho) = -i[H, \rho] + \sum_\nu (V_\nu \rho V_\nu^* - \frac{1}{2}\{V_\nu^* V_\nu, \rho\}).$$

In the vector-polynomial coordinates, let

$$Q_\nu = CV_\nu C^{-1} = \sum_{\alpha \leq \kappa} P_{\nu, \alpha}(z) \partial^\alpha$$

be its canonical representation. Let R be the largest order among all these operators. Suppose $R \geq 2$, and fix an index $|\alpha| = R$. For arbitrary $a \in \mathbb{R}^n$ take the stable pure input $p_a(z) = \prod_i (z_i - a_i)^{\kappa_i}$. In local coordinates $u = z - a$, $v = w + a$, its doubled polynomial is exactly $u^\kappa v^\kappa$, up to a positive normalization constant. Put $M = 2|\kappa|$.

Restrict the transformed output to $(u, v) = \tau\xi$ for any $\xi \in \mathbb{R}_{>0}^{2n}$. The initial polynomial is a nonzero multiple of τ^M , and the output remains upper-half-plane stable near time zero. Lemma 3.4 makes every coefficient of its first variation below degree $M - 2$ vanish. Varying ξ over the positive orthant shows, by polynomial identity, that each multivariate homogeneous part below that degree vanishes as well.

Consider the coefficient of $u^{\kappa-\alpha}v^{\kappa-\alpha}$. Hamiltonian and anticommutator terms act on one leg only, leaving the opposite full root factor untouched, and do not contribute. The sandwich term from V_ν is $(Q_\nu p_a)(z)\mathcal{R}(Q_\nu p_a)(w)$. To obtain $u^{\kappa-\alpha}$ from a canonical term with derivative index β , even allowing Taylor terms of its coefficient polynomial at a , one must have $\beta \geq \alpha$. Since $|\beta| \leq R = |\alpha|$, this requires $\beta = \alpha$ and the constant Taylor coefficient. Reflection therefore gives the total coefficient

$$(-1)^R(\kappa)_\alpha^2 \sum_\nu |P_{\nu,\alpha}(a)|^2, \quad (\kappa)_\alpha = \prod_i \frac{\kappa_i!}{(\kappa_i - \alpha_i)!}. \quad (57)$$

Its degree is $M - 2R < M - 2$, so it is zero. Every square vanishes for every real a . Thus every coefficient of maximal order vanishes identically, contradicting the definition of R .

All jumps have order at most one. Proposition 3.3 identifies this space as a scalar plus the local $\mathfrak{sl}_2$ actions. Formula (54) and the Cayley rotation identify it with the stated spin span. Absorb scalar jump parts into the Hamiltonian. The dissipative superoperator now has differential order at most two on the doubled box.

If CHC^{-1} had maximal canonical order $R \geq 3$, use the same input and choose $|\alpha| = R$. The coefficient of $u^{\kappa-\alpha}v^\kappa$ in the first variation can come only from the row Hamiltonian and its constant maximal-order coefficient at a . The column Hamiltonian retains u^κ , and the dissipator cannot lower total degree by R . Its degree $M - R < M - 2$ forces that Hamiltonian coefficient to vanish for every a , again a contradiction. Proposition 3.3 identifies the resulting order-two space with products of at most two local spin operators. Hermiticity gives real coefficients in the Hermitian basis consisting of scalars, local spins, local symmetric quadrupoles, and products on distinct sites. The antisymmetric local products are spin commutators, and the scalar symmetric component is the Casimir. At capacity one the remaining quadrupoles vanish. This proves the assertion. $\square$

Lemma 5.3 (Locality at finite spins). *Under the same hypothesis, the generator is a sum of local GKLS generators as in (51).*

Proof. Put $T_i^\alpha = 2S_i^\alpha$. Lemma 5.2 gives a PSD Gram matrix K in the basis of these local spin operators. Its (i, j) block is denoted K^{ij} . The transformed left and right actions have respective derivative vectors

$$\mathbf{v}(s) = (2s, 1 - s^2, i(1 + s^2)), \quad \tilde{\mathbf{v}}(t) = (2t, t^2 - 1, i(1 + t^2)). \quad (58)$$

These vectors are independent of the capacity. Only their zeroth-order coefficients depend on κ_i .

Fix distinct sites i, j . The physical pure product with transformed vector $\prod_k (z_k - a_k)$ allows independent row-row contacts at $z_i = s, z_j = t$ and row-column contacts at $z_i = s, w_j = t$, by choosing $a_i = s$ and respectively $a_j = t$ or $a_j = -t$. Keep all other legs off their roots and restrict to a line with all slopes positive. The initial root has multiplicity two. Only the chosen mixed principal coefficient survives at the contact; after division by the initial leading coefficient its multiplier is the reciprocal of a positive product of slopes. Lemma 3.4 makes it real and nonpositive for every real s, t . On-site quadrupoles and other terms leave at least one selected root untouched.

Write the pair Hamiltonian as $\sum_{\alpha,\beta} h_{\alpha\beta} T_i^\alpha T_j^\beta$, with h real. The row-column coefficient is $\mathbf{v}(s)^\top K^{ij} \tilde{\mathbf{v}}(t)$. The row-row coefficient is $\mathbf{v}(s)^\top (-\Re K^{ij} - ih) \mathbf{v}(t)$: the

conjugate Gram blocks combine in the anticommutator, since the two sites commute. Set

$$u(s) = \frac{(2s, 1 - s^2)}{1 + s^2}, \quad R_0 = \text{diag}(1, -1).$$

Its range is dense in the unit circle. The normalized derivative vectors are $(u(s), i)$ and $(R_0 u(t), i)$.

Reality of the row-column coefficient, by comparison of its constant, linear, and bilinear circle components, gives

$$K^{ij} = \begin{pmatrix} A & ib \\ id^\top & e \end{pmatrix}, \quad A, b, d, e \text{ real.}$$

Its sign condition is

$$u^\top A R_0 v - b \cdot u - d \cdot R_0 v - e \leq 0 \quad (u, v \in S^1). \quad (59)$$

For completeness, the comparison follows by averaging the imaginary part first alone, then against u , $R_0 v$, and $u(R_0 v)^\top$; the circle moments are $\mathbb{E}u = 0$ and $\mathbb{E}uu^\top = I_2/2$. Reality of the row-row coefficient similarly gives $h_{\perp\perp} = 0$ and $h_{zz} = 0$. With $k = h_{\perp z}$ and $\ell^\top = h_{z\perp}$, its remaining sign condition is

$$e - u^\top A v + k \cdot u + \ell \cdot v \leq 0. \quad (60)$$

Independent circle averages of (59) and (60) are $-e$ and e , so $e = 0$. Each continuous nonpositive function then has zero average and vanishes everywhere. The same circle moments give $A = b = d = k = \ell = 0$. Thus all cross Gram blocks and pair Hamiltonians vanish. The diagonal Gram blocks are PSD and define local GKLS generators, proving locality. $\square$

Proof of Theorem 5.1. We prove (2) $\Rightarrow$ (4) first. The preceding lemmas give locality. Each local semigroup preserves its pure Lee–Yang inputs near zero: pad such an input with arbitrary pure Lee–Yang states at the other sites. Its evolved global polynomial is a product of the local output polynomials, each nonzero since its density has trace one. Stability of that product implies stability of each factor.

Fix a site of capacity $\kappa \geq 2$, retaining the normalization $T = 2S$. Let $q(s, t)$ denote its row-column principal coefficient. Use the pure vector polynomial $p(z) = (z - s)(z + t)$, with $s + t \neq 0$. At row coordinate s and column coordinate t both factors in (56) have simple roots. A positive-slope line has a double root, and only $q(s, t)$ contributes at the contact. The common reflection sign cancels on dividing by the initial leading coefficient. Thus $q(s, t)$ is real and nonpositive; continuity adds the omitted line $s + t = 0$.

For a general Hermitian Gram matrix write

$$K = \begin{pmatrix} a & u + i\eta & r + i\mu \\ u - i\eta & b & v + i\nu \\ r - i\mu & v - i\nu & c \end{pmatrix}.$$

Multiplication of (58) gives

$$\Im q(s, t) = 2(s + t)[- \eta(1 - st) + r(1 + st) + v(t - s)].$$

Hence $\eta = r = v = 0$, and

$$K = \begin{pmatrix} B & id \\ -id^\top & c \end{pmatrix}, \quad B = B^\top \in \mathbb{R}^{2 \times 2}, \quad d \in \mathbb{R}^2. \quad (61)$$

With

$$r(s) = \frac{(2s, 1 - s^2)}{1 + s^2}, \quad v(t) = \frac{(2t, t^2 - 1)}{1 + t^2},$$

the mixed coefficient satisfies

$$\frac{q(s, t)}{(1 + s^2)(1 + t^2)} = r(s)^\top B v(t) + d \cdot (v(t) - r(s)) - c \leq 0. \quad (62)$$

Both circle variables range densely and independently. Taking them equal gives $B \preceq cI_2$.

There is a second, independent contact available at this capacity. Take $p(z) = (z - s)^2$ and a column coordinate away from its reflected root. Its pure-row principal coefficient must be real and nonpositive. This follows either by real specialization of the column, or by a positive-slope line through the contact; the nonzero complementary factor cancels from the root-cluster coefficient. Write the quadratic Hamiltonian as $H_2 = T^\top Q T$, with Q real symmetric. Put $u = r(s)$. The pure-row principal coefficient, divided by $(1 + s^2)^2$, is

$$\frac{1}{2}(c - u^\top B u) + 2Q_{\perp z} \cdot u - i(u^\top Q_{\perp \perp} u - Q_{zz}). \quad (63)$$

Indeed its dissipative part is $-\mathbf{v}(s)^\top (\Re K) \mathbf{v}(s)/2$, and its Hamiltonian part is $-i\mathbf{v}(s)^\top Q \mathbf{v}(s)$, using the three-component vector in (58). Reality gives $Q_{\perp \perp} = Q_{zz}I_2$. Remove the corresponding Casimir scalar and set $k = Q_{\perp z}$. Nonpositivity at u and $-u$ now gives $c - u^\top B u \leq 0$. Together with $B \preceq cI_2$, this yields $B = cI_2$; the remaining inequality $2k \cdot u \leq 0$ gives $k = 0$. The quadratic Hamiltonian is consequently scalar.

Equation (62) reduces to

$$c(r \cdot v - 1) + d \cdot (v - r) \leq 0 \quad (r, v \in S^1).$$

If $d \neq 0$, take $r \perp d$ and $v = (\cos \theta)r + (\sin \theta)d/|d|$ for small positive θ . The left side is $c(\cos \theta - 1) + |d| \sin \theta > 0$, a contradiction. Thus $d = 0$, $K = cI_3$, and positivity gives $c \geq 0$.

The transformed isotropic dissipator has the explicit form

$$\begin{aligned} \tilde{\mathcal{D}} = & -2c\kappa(\kappa zw + 1) + 2c\kappa z(zw - 1)\partial_z + 2c\kappa w(zw - 1)\partial_w \\ & - 2c(zw - 1)^2\partial_z\partial_w. \end{aligned} \quad (64)$$

To verify it directly, the left actions of (T_x, T_y, T_z) are

$$2z\partial_z - \kappa, \quad (1 - z^2)\partial_z + \kappa z, \quad i[(1 + z^2)\partial_z - \kappa z],$$

and the right actions are the same formulas in w except that the middle action has its sign reversed. Substitute them into $c \sum_\alpha (L_\alpha R_\alpha - (L_\alpha^2 + R_\alpha^2)/2)$. In particular this is a real operator and its pure-row and pure-column second-order coefficients vanish. For the remaining Hamiltonian $H = \hat{h}_x T_x + \hat{h}_y T_y + \hat{h}_z T_z$, use $p(z) = z - s$ and keep the reflected column off its root. The imaginary simple-root velocity comes only from the row Hamiltonian and is

$$\hat{h}_y s^2 - 2\hat{h}_x s - \hat{h}_y \geq 0 \quad (s \in \mathbb{R}),$$

by Lemma 3.4. Hence $\hat{h}_x = \hat{h}_y = 0$. Returning to $S = T/2$ gives (52), with $h = 2\hat{h}_z$ and $\gamma = 4c$.

It remains to recover the full qubit condition in a mixed-spin system. Fix a qubit and a high-spin partner whose local form has just been proved. Their mixed

principal coefficients will be denoted by $q_q(s, t)$ and $q_h(u, v) = -2c(uv - 1)^2$. For arbitrary real s, t with $s + t \neq 0$, choose

$$\begin{aligned} x &= (s + t)/2, & a &= (s - t)/2, & c_0 &> |x|, \\ b &= \sqrt{(c_0/x)^2 - 1}, & u_* &= b + c_0/x, & v_* &= -b + c_0/x. \end{aligned} \quad (65)$$

Then $u_*v_* = 1$. The polynomial

$$p(z, u) = (z - a)(u - b) - c_0 \quad (66)$$

is upper-half-plane stable, since for $z \in \mathbb{H}$ its root in u is $b + c_0/(z - a)$, which lies in the lower half-plane. It is a valid physical vector in capacities $(1, \kappa)$, including when its degree is below κ . Apart from the common parity sign, its reflected factor is $(w + a)(v + b) - c_0$.

At $(z, u, w, v) = (s, u_*, t, v_*)$ both factors vanish. Along the line with all four slopes one, the initial quadratic coefficient, apart from the same common sign, is $(c_0/x + x)^2 > 0$. Locality makes its first variation at the contact equal, with that sign, to

$$\frac{c_0^2}{x^2}q_q(s, t) + x^2q_h(u_*, v_*) = \frac{c_0^2}{x^2}q_q(s, t). \quad (67)$$

All lower-order terms vanish because one input factor remains zero; the high-spin pure-row and pure-column coefficients are zero by (64). Other sites can be padded by pure stable factors kept nonzero at the contact. The root-cluster condition thus makes $q_q(s, t)$ real and nonpositive. Continuity again adds $s + t = 0$. Consequently (61) and (62) hold at every qubit as well. For this Gram shape the dissipator is real after Cayley transformation, as in the proof of Proposition 4.11. The same simple-root test excludes its transverse Hamiltonian. Lemma 4.9 gives precisely (53), including $\text{Tr } B = 0$. This proves (2) $\Rightarrow$ (4), without treating the evolving partner as inert.

Conversely assume (4). Formula (64) is a real endomorphism of the doubled capacity box whose only second-order coefficient is $-2c(zw - 1)^2 \leq 0$. The axial Hamiltonian contributes only a real first-order endomorphism. At qubit sites, the proof of Proposition 4.11 gives a real box endomorphism with its only mixed coefficient nonpositive by Lemma 4.9. The sum therefore satisfies Theorem 3.1 on the full doubled capacity box. Its exponential preserves all complex stable polynomials after arbitrary inert-variable extensions. Undoing the coefficient Cayley transforms gives the asserted physical complete preservation, in particular for arbitrary inert finite spins. No convexity of the stable class is used. This proves (4) $\Rightarrow$ (3). Finally (3) $\Rightarrow$ (1) and (1) $\Rightarrow$ (2) are immediate. $\square$

The all-qubit ordinary classification is Theorem 4.1; it is not asserted there or here to follow from pure inputs alone. On a single qubit the ordinary cone is instead Proposition 4.14, and ordinary preservation need not be complete. Theorem 5.1 also makes no classification claim for isolated noncovariant high-spin channels: its semigroup hypothesis is essential to the argument.

5.1. An application to rotation convolution semigroups. Gaussian processes on compact groups can already be recognized through a faithful finite-dimensional representation and its tensor products; see [16, Theorem 2.8], which restates the earlier compact-group characterization. For the real defining representation of $SO(3)$, the spin-one conjugation action is its tensor square and contains the defining representation as its antisymmetric part. The following application combines that

recognition mechanism with the physical classification above. We give the short character proof in this case using Hunt's formula [11] and its twirling realization [2, Theorem 5.1].

Corollary 5.4 (Lee–Yang rotation dynamics). *Let $(\mu_t)_{t \geq 0}$ be a weakly continuous probability convolution semigroup on $SO(3)$ with $\mu_0 = \delta_e$. Let U be its spin-one representation and set*

$$\Phi_t(\rho) = \int_{SO(3)} U(g)\rho U(g)^* \mu_t(dg).$$

Then Φ_t preserves every pure coherently Lee–Yang input in a neighborhood of zero if and only if the group generator is

$$\mathcal{A} = hX_z + \frac{\gamma}{2}(X_x^2 + X_y^2 + X_z^2), \quad h \in \mathbb{R}, \quad \gamma \geq 0. \quad (68)$$

Here X_α are the left-invariant fields normalized by $U(\exp(tE_\alpha)) = e^{-itS_\alpha}$ and $X_\alpha f(g) = \frac{d}{dt}f(g \exp(tE_\alpha))|_{t=0}$. In particular the Lévy measure is zero, without a prior assumption of continuous sample paths or rotation invariance of μ_t .

Proof. The twirling maps form a continuous CPTP semigroup. Under the stated preservation hypothesis Theorem 5.1 gives (52). The spin-one operator space decomposes as the angular-momentum representations $0 \oplus 1 \oplus 2$. On its ℓ component the double-commutator Casimir is $\ell(\ell+1)I$; the axial commutator has trace zero. Thus the generator restrictions satisfy

$$\mathrm{Tr} \mathcal{L}_1 = -3\gamma, \quad \mathrm{Tr} \mathcal{L}_2 = -15\gamma.$$

Let χ_1, χ_2 be the characters of the three- and five-dimensional representations. For a rotation of angle θ , put

$$F(g) = 2 - \chi_1(g) + \frac{1}{5}\chi_2(g) = \frac{4}{5}(1 - \cos \theta)^2. \quad (69)$$

This smooth function is strictly positive off the identity and has zero jet through order two at the identity. Since $\int \chi_\ell d\mu_t = \mathrm{Tr}(e^{t\mathcal{L}_\ell})$, the displayed traces give $\mathcal{A}F(e) = 3\gamma - 15\gamma/5 = 0$.

Write Hunt's generator with drift, diffusion matrix $a \succeq 0$, and Lévy measure ν , taking its second-order part to be $\frac{1}{2} \sum_{\alpha, \beta} a_{\alpha\beta} X_\alpha X_\beta$. At the identity all local terms and the linear jump compensation vanish on F . Therefore

$$0 = \mathcal{A}F(e) = \int_{SO(3) \setminus \{e\}} F(g) \nu(dg).$$

The integral is finite: near the identity $F = O(\mathrm{dist}(g, e)^4)$, while a Lévy measure has the requisite local second-moment integrability; away from the identity compactness applies. Positivity of F off e implies $\nu = 0$, including for an initially infinite-activity Lévy measure.

On the dipole space the symmetric part of the remaining generator is $\frac{1}{2}(a - (\mathrm{Tr} a)I_3)$. Comparing with $-\gamma I_3$, first taking traces and then substituting back, gives $a = \gamma I_3$. Its antisymmetric part identifies the drift as hX_z . This proves (68). Conversely, that group generator induces (52) under twirling, so Theorem 5.1 gives complete preservation. $\square$

The group and the tested spin matter. Conjugation cannot detect jumps in the center $\{\pm I\}$ of $SU(2)$, so the zero-Lévy-measure conclusion is stated on $SO(3)$. Moreover, Poisson jumps of rate λ by angle π about the z axis induce on a qubit

$$\Phi_t(\rho) = \frac{1}{2}(1 + e^{-2\lambda t})\rho + \frac{1}{2}(1 - e^{-2\lambda t})Z\rho Z.$$

This is an allowed dephasing semigroup, but its rotation process has nonzero jumps. Thus spin one cannot be replaced uniformly by spin one-half in Corollary 5.4.

6. STATIC CHANNELS AND STABLE ENCODINGS

This section classifies invertible Hermiticity-preserving, trace-preserving maps with Lee–Yang full Choi tensors, and gives structural results for singular channels. The variables associated with different physical qubits are always distinct. In particular, permutation of physical wires is allowed, whereas an arbitrary change of the computational basis is not.

6.1. Coefficient tensors and complete preservation. For a matrix $M : (\mathbb{C}^2)^{\otimes m} \rightarrow (\mathbb{C}^2)^{\otimes n}$, put

$$G_M(z, w) = \sum_{a \in \{0,1\}^n} \sum_{b \in \{0,1\}^m} M_{ab} z^a w^b. \quad (70)$$

We call M *tensor-LY* if this polynomial is nonzero whenever all its variables belong to the open unit disk $\mathbb{D}$. Zero is excluded from this definition. For a complex-linear map $\Phi : M_{2^m}(\mathbb{C}) \rightarrow M_{2^n}(\mathbb{C})$, our Choi convention is

$$J(\Phi) = \sum_{c,d} \Phi(|c\rangle\langle d|) \otimes |c\rangle\langle d|. \quad (71)$$

Thus full Choi–LY means stability in all $2(m+n)$ variables of $G_{J(\Phi)}$. Hermiticity-preserving and trace-preserving maps are abbreviated HP and TP, respectively.

We use the disk tensor-contraction theorem in the following form: contracting two binary indices of a tensor-LY polynomial with unit radii produces a tensor-LY polynomial or zero. Tensor products, permutation of indices, and coefficientwise conjugation preserve tensor-LY. These facts follow from the classical Asano contraction theorem; the tensor and channel formulation is described in [19, Section 2]. The possible zero result at unit radius will be excluded explicitly where necessary.

Lemma 6.1 (The Choi bridge). *Let $\Phi : M_{2^m}(\mathbb{C}) \rightarrow M_{2^n}(\mathbb{C})$ be complex linear. If $J(\Phi)$ is tensor-LY, then Φ , and each of its extensions by identity maps, sends tensor-LY matrices to tensor-LY matrices or zero. Its Hilbert–Schmidt adjoint $\Phi^\dagger$ is also full Choi–LY. For a CPTP map, full Choi–LY is equivalent to preservation of tensor-LY density matrices with every finite qubit reference system left inert.*

Proof. The superoperator and Choi arrays satisfy

$$S(\Phi)_{(a,b),(c,d)} = \Phi(|c\rangle\langle d|)_{ab}, \quad J(\Phi)_{(a,c),(b,d)} = S(\Phi)_{(a,b),(c,d)}.$$

They therefore differ only by a permutation of tensor indices. Applying Φ is contraction of this array with the input coefficient array. Input transposes occurring in the matrix convention also only permute indices. The contraction theorem proves the first assertion, including inert indices. The superoperator of $\Phi^\dagger$ is obtained by conjugating entries and interchanging the input and output index pairs, which proves the adjoint assertion.

For a CPTP map the output of a density matrix has trace one, so cannot be the zero alternative. Conversely, take the normalized maximally entangled vector between the input and an equally large reference:

$$|\Omega_m\rangle = 2^{-m/2} \sum_c |c\rangle|c\rangle.$$

Its amplitude polynomial is $2^{-m/2} \prod_i (1 + x_i u_i)$, and its density-matrix polynomial is the product of this polynomial and its coefficientwise conjugate in independent variables. Both are stable. Applying Φ to one side produces $2^{-m} J(\Phi)$, proving the converse. $\square$

Ordinary multiplication of tensor-LY matrices is likewise a tensor contraction, with the same zero alternative. In particular, multiplication on either side by an invertible tensor-LY matrix whose inverse is tensor-LY preserves the class of tensor-LY matrices.

6.2. An elementary matrix edge and hyperbolic pencils. Write

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Subscripts denote physical sites, with identity factors suppressed. The real vector spaces needed below are

$$\mathfrak{g} = \text{span}_{\mathbb{R}}\{X, Y, iZ\}, \quad \mathcal{E} = \mathbb{R}I + \mathfrak{g}, \quad \mathfrak{e}_n = \mathbb{C}I + \bigoplus_{j=1}^n \mathfrak{g}_j. \quad (72)$$

Lemma 6.2 (Two-sided tangency). *Suppose $M(t)$ is a matrix-valued curve defined for real t near zero, differentiable at zero, with $M(0) = I$ and $M'(0) = B$. If $M(t)$ is tensor-LY for both signs of all sufficiently small t , then $B \in \mathfrak{e}_n$. Conversely, $B \in \mathfrak{e}_n$ implies that e^{tB} is tensor-LY for every real t .*

Proof. Fix a site j and $|\zeta| = 1$. Specializing $z_j = \zeta$ is justified by first using $r\zeta$, $r < 1$, and then applying Hurwitz's theorem. The specialization is stable or zero. At $t = 0$ its value with all remaining variables zero is one, so it is not the zero polynomial for sufficiently small $|t|$. Fix interior values $u = (z_k, w_k)_{k \neq j}$, and put $D(u) = \prod_{k \neq j} (1 + z_k w_k)$. The resulting affine polynomial in w_j is

$$D(u)(1 + \zeta w_j) + tG_B(\zeta, w_j, u) + o(t).$$

Its root $r(t)$ is differentiable at zero, with $r(0) = -\zeta^{-1}$, and stability implies $|r(t)| \geq 1$ for both signs of small t . Differentiating the root equation gives

$$\frac{r'(0)}{r(0)} = \frac{G_B(\zeta, -\zeta^{-1}, u)}{D(u)}.$$

The derivative of $|r(t)|^2$ at its two-sided minimum is zero. Consequently the holomorphic function on the remaining polydisk

$$u \mapsto \frac{G_B(\zeta, -\zeta^{-1}, u)}{D(u)} \quad (73)$$

has purely imaginary values, and is therefore a purely imaginary constant.

Expand B in the Pauli tensor basis, separating site j : $B = \sum_{\sigma, \tau} b_{\sigma, \tau} \sigma_j \otimes \tau$. At $(z_j, w_j) = (\zeta, -\zeta^{-1})$, the one-site polynomials are

$$G_I = 0, \quad G_X = \zeta - \zeta^{-1}, \quad G_Y = i(\zeta + \zeta^{-1}), \quad G_Z = 2.$$

Since the Pauli polynomials on the other sites are linearly independent, constancy of (73) gives, for every nonidentity word τ ,

$$b_{X,\tau}(\zeta - \zeta^{-1}) + ib_{Y,\tau}(\zeta + \zeta^{-1}) + 2b_{Z,\tau} = 0.$$

Taking $\zeta = 1, -1, i$ forces all three coefficients to vanish. As j was arbitrary, every Pauli coefficient of weight at least two is zero. Thus $B = cI + \sum_j (x_j X_j + y_j Y_j + d_j Z_j)$. The remaining scalar contact expression is

$$x_j(\zeta - \zeta^{-1}) + iy_j(\zeta + \zeta^{-1}) + 2d_j.$$

Its real part is zero. The same three values of ζ imply $x_j, y_j \in \mathbb{R}$ and $d_j \in i\mathbb{R}$, as asserted. The scalar c is unrestricted.

For the converse, if $N = aI + xX + yY + izZ \in \mathcal{E}$, direct multiplication gives

$$NZN^* = \delta Z, \quad \delta = a^2 + z^2 - x^2 - y^2 = \det N. \quad (74)$$

When $\delta > 0$, write $G_N(s, w) = A(s) + C(s)w$. Equation (74) implies

$$|A(s)|^2 - |C(s)|^2 = \delta(1 - |s|^2) > 0 \quad (|s| < 1),$$

so N is tensor-LY. Its inverse has the same property. For $b = xX + yY + izZ$, one has $b^2 = (x^2 + y^2 - z^2)I$. The exponential e^{tb} belongs to $\mathcal{E}$ for real t and has determinant one. Therefore it is tensor-LY. Finally, for $B = cI + \sum_j b_j$, $e^{tB} = e^{tc} \otimes_j e^{tb_j}$, which proves the assertion. $\square$

The space $\mathcal{E}$ is closed under multiplication. Its members sufficiently close to I , and their inverses, are tensor-LY by (74). We call these matrices local stability units. A traceless element $b = xX + yY + izZ \in \mathfrak{g}$ is *hyperbolic* if $x^2 + y^2 - z^2 > 0$. A diagonal phase conjugation followed by conjugation with e^{rY} , where $\tanh(2r) = z/\sqrt{x^2 + y^2}$, carries it to $\sqrt{x^2 + y^2 - z^2}X$. The conjugating matrices and their inverses are local stability units.

Lemma 6.3 (Hyperbolic affine pencil). *Let $B = cI + \sum_j b_j \in \mathfrak{e}_n$, and suppose one b_k is hyperbolic. If $I + tB$ is tensor-LY for some nonzero real t , then $b_j = 0$ for all $j \neq k$ and $c \in \mathbb{R}$.*

Proof. Conjugating by a local stability unit, assume $b_k = aX_k$, $a > 0$. For a one-site matrix b define $q_b(z, w) = G_b(z, w)/(1 + zw)$. The function

$$q_X(z, z) = \frac{2z}{1 + z^2}$$

attains every nonreal complex number for $|z| < 1$. Indeed, the inverse equation has two reciprocal roots; neither is on the unit circle for a nonreal target, because the displayed quotient is real there wherever finite. Exactly one inverse root is therefore in the disk.

If $b_j \neq 0$ for another site, q_{b_j} is a nonconstant holomorphic function, since b_j is nonzero and traceless. Freeze all other variable pairs at zero, and divide the polynomial of $I + tB$ by $\prod_i (1 + z_i w_i)$. The result has the form $C + taq_X(z_k, w_k) + tq_{b_j}(z_j, w_j)$. The open image of q_{b_j} allows its variables to be chosen so that the target needed from q_X is nonreal. The preceding inverse construction then gives an interior zero, a contradiction. Once the other components vanish, an imaginary part of c would make $-(1 + tc)/(ta)$ nonreal and give the same contradiction. $\square$

6.3. Routing and factorization of visible outputs. Let $\Phi : M_{2^m}(\mathbb{C}) \rightarrow M_{2^n}(\mathbb{C})$ be HP, TP and full Choi–LY, and put $\phi = \Phi^\dagger$. An output site i is called *transversely visible* if at least one of $\phi(X_i), \phi(Y_i)$ is nonscalar. Denote the set of such sites by A .

Lemma 6.4 (Local routing). *For each $i \in A$ there is a unique input site $\pi(i)$ such that, writing $k = \pi(i)$,*

$$\begin{aligned}\phi(X_i) &= b_{ix}I + D_{i,11}X_k + D_{i,21}Y_k, \\ \phi(Y_i) &= b_{iy}I + D_{i,12}X_k + D_{i,22}Y_k, \\ \phi(Z_i) &= \lambda_i Z_k,\end{aligned}\tag{75}$$

where all coefficients are real and $D_i \neq 0$. No injectivity of Φ or D_i is required.

Proof. For a local $U \in \mathfrak{g}_i$, the identity $U^2 = \delta I$, $\delta \in \mathbb{R}$, gives $e^{tU} = c_\delta(t)I + s_\delta(t)U$, with real analytic scalar coefficients satisfying $c_\delta(0) = 1$, $s_\delta(0) = 0$, and $s'_\delta(0) = 1$. These exponentials are stability units. Since ϕ is unital, their images, divided by $c_\delta(t)$, give a two-sided stable pencil $I + r\phi(U)$ near zero. A zero image is excluded near I . Lemma 6.2 puts $\phi(U)$ in $\mathfrak{e}_m$. HP then implies

$$\phi(X_i), \phi(Y_i) \in \mathbb{R}I + \sum_k \text{span}_{\mathbb{R}}\{X_k, Y_k\}, \quad \phi(Z_i) \in \mathbb{R}I + \sum_k \mathbb{R}Z_k.$$

Every nonscalar transverse image has a hyperbolic component, so Lemma 6.3 confines it to one input site. If two transverse directions used different sites, their sum would have two nonzero hyperbolic components. Thus the whole transverse plane uses the same site whenever its nonscalar image is nonzero.

Choose a transverse U_i with nonzero traceless image on this site, and apply the argument to $U_i + i\varepsilon Z_i$ for a small nonzero real ε . The routed component remains hyperbolic. Lemma 6.3 eliminates all other longitudinal components, and its real-scalar conclusion eliminates the scalar coefficient of $\phi(Z_i)$. This proves (75). $\square$

Theorem 6.5 (Visible-output decomposition). *Let $\Phi : M_{2^m}(\mathbb{C}) \rightarrow M_{2^n}(\mathbb{C})$ be HP, TP and full Choi–LY. With the preceding notation, put $K = \pi(A)$ and $G_k = \pi^{-1}(k)$. There are HP, TP, full Choi–LY maps $\Xi_k : M_2(\mathbb{C}) \rightarrow M_{2^{|G_k|}}(\mathbb{C})$ such that, after ordering the physical output groups,*

$$\text{Tr}_{A^c} \circ \Phi = \left(\bigotimes_{k \in K} \Xi_k \right) \circ \text{Tr}_{K^c}.\tag{76}$$

The routing function is unique. If Φ is CP, every Ξ_k is CP. The trace on the right is the ordinary, unnormalized partial trace.

Proof. For each $i \in A$, choose a small real open set $\mathcal{O}_i \subset \mathfrak{g}_i$ about a transverse direction whose image is nonscalar. Shrink it so that every $U_i \in \mathcal{O}_i$, and the traceless part of $\phi(U_i)$, is hyperbolic. By (75), $\phi(U_i) \in \mathcal{E}_{\pi(i)}$. These open sets have complex span equal to the entire local traceless matrix space, even when D_i has rank one.

Fix choices $U_i \in \mathcal{O}_i$. For $S \subseteq A$ set

$$F_S(t_S) = \phi \left(\prod_{i \in S} (I + t_i U_i) \right).$$

We prove by induction that, for all sufficiently small real parameters,

$$F_S(t_S) = \prod_{k \in \pi(S)} F_{k,S}(t_{S \cap G_k}), \quad F_{k,S} \in \mathcal{E}_k, \quad F_{k,S}(0) = I, \quad (77)$$

where each factor is multiaffine in its own parameters. Distinct factors act on distinct sites. They and their inverses are stability units near zero. The empty and singleton cases follow from unitality and Lemma 6.4.

Suppose (77) holds, and add $i \notin S$, with route k . Put

$$\Delta(t_S) = \phi \left(U_i \prod_{j \in S} (I + t_j U_j) \right), \quad B(t_S) = F_S(t_S)^{-1} \Delta(t_S).$$

The new input product is tensor-LY. Its image is $F_S + t_i \Delta$, which is nonzero near the identity. Left multiplication by the stable inverse gives the two-sided stable pencil $I + t_i B(t_S)$. Lemma 6.2 applies to this pencil. At $t_S = 0$ its coefficient is $\phi(U_i)$, whose component on site k is hyperbolic. Continuity and Lemma 6.3 therefore imply

$$B(t_S) \in \mathcal{E}_k \quad (78)$$

on a sufficiently small parameter neighborhood.

Fix a parameter t_j belonging to a different group G_ℓ , and fix all the other parameters. Write $F_{\ell,S} = C_0 + t_j C_1$. The matrices C_0, C_1 are linearly independent for sufficiently small fixed values of the other parameters: at zero they are $I, \phi(U_j)$, with a nonzero hyperbolic traceless part in the second matrix. Remove the fixed other-site factors from $\Delta = F_S B$. An affine function of t_j is thereby written as

$$(C_0 + t_j C_1) \otimes R(t_j), \quad R(t_j) = F_{k,S} B(t_S).$$

Choose linear functionals α, β with

$$\alpha(C_0) = 1, \quad \alpha(C_1) = 0, \quad \beta(C_0) = 0, \quad \beta(C_1) = 1.$$

Applying them shows that both $R(t_j)$ and $t_j R(t_j)$ are affine. Hence R , and therefore B , is constant in t_j . Repeating this for every parameter outside G_k gives the update

$$F_{S \cup \{i\}} = \prod_{\ell \neq k} F_{\ell,S} (F_{k,S} + t_i F_{k,S} B(t_{S \cap G_k})).$$

The last factor lies in the real algebra $\mathcal{E}_k$, which is closed under multiplication. It is multiaffine, as is seen by setting every other group's parameters to zero in the original definition of $F_{S \cup \{i\}}$. This completes the induction; only finitely many neighborhood shrinkings are needed.

Setting the other groups' parameters to zero identifies each $F_{k,S}$ as the image of its own group product. Extracting coefficients in (77) consequently gives

$$\phi \left(\prod_{i \in S} U_i \right) = \prod_{k \in \pi(S)} \phi \left(\prod_{i \in S \cap G_k} U_i \right).$$

This is a complex multilinear identity. Its validity on a product of real open sets extends first to their real spans and then to their complex spans. Adjoining identity factors spans the entire matrix algebra on A . Thus the restriction of ϕ to that algebra is the tensor product of maps on the groups, each with image on its routed input site.

For an observable O on G_k , extract the one-site image of $\phi(O \otimes I)$ by normalized partial trace over the other input sites. This gives a unital HP map ψ_k , CP if ϕ is CP. Its adjoint Ξ_k is TP and has the same positivity properties. Equivalently, Ξ_k is obtained by preparing every other input maximally mixed, applying Φ , and tracing the other outputs. These operations have tensor-LY Choi arrays. Their contraction is nonzero because Ξ_k is TP, so Ξ_k is full Choi-LY. Taking adjoints proves (76). The adjoint of the partial trace in that formula tensors with identity, which also verifies its normalization. $\square$

Theorem 6.6 (Invertible static classification). *An invertible complex-linear HP, TP map $\Phi : M_{2^n}(\mathbb{C}) \rightarrow M_{2^n}(\mathbb{C})$ is full Choi-LY if and only if*

$$\Phi = (\Phi_1 \otimes \cdots \otimes \Phi_n) \circ \mathcal{P}, \quad (79)$$

where $\mathcal{P}$ permutes the physical input wires and the Φ_i are invertible one-qubit HP, TP, full Choi-LY maps. The map Φ is CP if and only if all its local factors are CP.

Proof. Apply the preceding results to $\phi = \Phi^\dagger$. The induced map on the real space of transverse fields modulo scalars is injective. Indeed, if $\phi(U) = bI$, unitality gives $\phi(U - bI) = 0$; injectivity forces $U = bI$, and a traceless transverse field is then zero. In particular every output is visible, each of its two-dimensional transverse planes maps injectively to its routed two-dimensional input plane, and two such planes cannot route to the same input plane. With n planes on each side, the routes form a permutation and all groups have one site.

Theorem 6.5 now gives a tensor product of local factors. Each factor is invertible: a nonzero kernel vector of one factor, tensored with suitable nonzero vectors for the other factors, would give a kernel vector of the full map. The same theorem proves inheritance of CP. Conversely, tensor products and physical wire permutations preserve full Choi-LY and all the stated properties. $\square$

Remark 6.7. The proof of Theorem 6.6 does not require a semigroup representation, complete positivity, or injectivity of an unproved local restriction. For completeness, its adjoint local longitudinal coefficient in (75) is positive: stability of $I + s\lambda_i Z$, obtained from $I + sZ$ for small $s > 0$, gives $|1 + s\lambda_i| \geq |1 - s\lambda_i|$, hence $\lambda_i \geq 0$; injectivity excludes zero.

Example 6.8 (Ordinary static preservation is weaker). Let $n \geq 2$, $d = 2^n$, $\tau = \text{diag}(3/4, 1/4)$, and $\sigma = \tau^{\otimes n}$. For

$$0 < a < \frac{1}{4^n + 1}, \quad \Phi_a(M) = aM + (1 - a) \text{Tr}(M)\sigma,$$

the map Φ_a is invertible and CPTP, and sends every density matrix to a matrix with a polynomial nonzero on the closed unit polydisk. Nevertheless it is neither a product of local channels up to a physical permutation nor full Choi-LY.

Indeed, it fixes σ and multiplies the trace-zero subspace by a , proving invertibility. On the closed polydisk,

$$|G_\sigma| \geq 2^{-n} = d^{-1}, \quad |G_\rho| \leq d$$

for every density matrix ρ . The second inequality follows by bounding the two monomial-vector norms by $\sqrt{d}$ and $\|\rho\|_{\text{op}}$ by one. Hence

$$|G_{\Phi_a(\rho)}| \geq (1 - a)d^{-1} - ad > 0.$$

On input $|0^n\rangle\langle 0^n|$, the occupation indicators at any two distinct output sites have covariance $a(1-a)/16 > 0$. The output is not product, whereas a product of local channels, with or without a wire permutation, sends product inputs to products.

One may also exhibit the Choi zero directly. Its polynomial is

$$a \prod_i (1 + z_i x_i)(1 + w_i y_i) + (1-a)G_\sigma(z, w) \prod_i (1 + x_i y_i).$$

Set all output variables except z_1, w_1 to zero, and set $z_1 = w_1 = t$, $x_1 = y_1 = v$. Define

$$p(t) = (3/4)^{n-1}(3/4 + t^2/4), \quad C(t) = \frac{1-a}{a}p(t), \quad C_0 = C(1).$$

Choose $v_0 = e^{i\theta}$ with $\cos \theta = -1/(1+C_0)$, $\pi/2 < \theta < \pi$. Then $(1+v_0)^2/(1+v_0^2) = -C_0$. For small $\varepsilon > 0$, put

$$t = 1 - \varepsilon, \quad v = (1 - \varepsilon)v_0, \quad q = -\frac{(1+tv)^2}{C(t)(1+v^2)} - 1.$$

As $\varepsilon \downarrow 0$, $q \rightarrow 0$, so all of $|t|, |v|, |q|$ are smaller than one for sufficiently small positive ε . Set $x_2 = y_2 = \sqrt{q}$ and all remaining reference variables to zero. The Choi polynomial becomes $a(1+tv)^2 + (1-a)p(t)(1+v^2)(1+q) = 0$, at an interior point. Lemma 6.1 turns this into an ancillary physical witness. The same example is the late-time channel $e^{t(\mathcal{R}_\sigma - \text{Id})} = e^{-t} \text{Id} + (1 - e^{-t})\mathcal{R}_\sigma$, where $\mathcal{R}_\sigma(M) = \text{Tr}(M)\sigma$. Its ordinary strict-preservation property for $t > \log(4^n + 1)$ therefore does not imply preservation at times approaching zero.

6.4. Saturated columns and reflected polynomial factors. For a polynomial p , let $\|p\|_2$ be the Euclidean norm of its coefficient vector. If p is multiaffine in a specified finite set S , define its full reflection by

$$p^*(z) = z^S \overline{p(1/\bar{z})}. \quad (80)$$

The set S is part of the definition, and may include variables absent from p . In contrast, $\bar{p}(w) = \overline{p(\bar{w})}$ below denotes coefficientwise conjugation without reflection.

Lemma 6.9 (Saturation and disjoint factors). *Let $F(z, x) = \sum_{b \in \{0,1\}^m} A_b(z)x^b$ be a stable multiaffine polynomial. Suppose $\|A_{e_i}\|_2 = \|A_0\|_2$ for every input site i . Then, over the rational function field in z ,*

$$F(z, x) = A_0(z) \prod_{i=1}^m \left(1 + x_i \frac{A_{e_i}(z)}{A_0(z)} \right). \quad (81)$$

Writing the quotients in coprime form q_i/p_i , there is a polynomial factorization

$$F(z, x) = C(z) \prod_{i=1}^m (p_i(z) + x_i q_i(z)) \quad (82)$$

in which the output-variable supports of the displayed factors are pairwise disjoint. All these factors are stable, and $\|p_i\|_2 = \|q_i\|_2$.

Proof. Setting all input variables except x_i to zero gives a stable affine polynomial $A_0 + x_i A_{e_i}$, whence $|A_{e_i}(z)| \leq |A_0(z)|$ in the output polydisk. The inequality extends continuously to the output torus. Parseval and norm equality imply equality of these moduli everywhere on that torus. At a torus point where $A_0 \neq 0$, boundary Hurwitz makes the input polynomial stable; its nonzero constant coefficient

excludes the zero alternative. Write this input polynomial as $P_i(x_{\neq i}) + x_i Q_i(x_{\neq i})$. The holomorphic ratio Q_i/P_i is contractive on the remaining input polydisk, and has modulus one at its origin. It is therefore constant by the maximum modulus principle. Applying this to every input gives (81) at these torus points. The zero set of the nonzero polynomial A_0 has torus measure zero. Clearing denominators and using uniqueness of Fourier coefficients proves the rational identity everywhere.

Each $p_i + x_i q_i$ is primitive over $\mathbb{C}[z]$ and irreducible over the rational function field, where it is linear in its own input variable. Gauss's lemma applied to the primitive part of F , considered as a polynomial in x , gives (82) with C polynomial. Degrees in any one output variable add under multiplication in this integral domain. Since every such degree of F is at most one, each output variable belongs to at most one displayed factor. Stability of F implies stability of every factor and, by setting $x_i = 0$, of p_i . Finally coefficient norms multiply on disjoint supports. Comparing A_0 with A_{e_i} gives the claimed local norm equality. $\square$

Lemma 6.10 (Reduced reflected pairs). *Let $p + xq$ be stable and multiaffine, where p, q are coprime, have equal coefficient norms, and their union of output supports is S . Then*

$$q = e^{i\theta} p^* \quad (83)$$

for some real θ . Conversely, for any stable multiaffine p on a specified set S , the polynomial $p + e^{i\theta} x p^$ is stable.*

Proof. Stable affine slices and equality of norms imply $|p| = |q|$ on the output torus. Multiplication by z^S gives $pp^* = qq^*$. Coprimality implies $p \mid q^*$; write $q^* = ph$, so $p^* = qh$. The constant coefficient of the stable polynomial p is nonzero. Thus p^* has degree one in every variable of S . If h depends on z_j , degree additivity makes q independent of z_j , and the support-union property makes p depend on z_j . Independence of q implies $z_j \mid q^* = ph$. Since p has nonzero constant term, $z_j \mid h$, and then $z_j \mid p^* = qh$. The latter says p is independent of z_j , a contradiction. Hence h is constant. Reflecting twice shows $|h| = 1$, proving (83).

For the converse, p^*/p is holomorphic on the polydisk and has modulus at most one. To justify this assertion even in the presence of boundary zeros, use $p_r(z) = p(rz)$, $r < 1$. The polynomial p_r has no closed-polydisk zero, and p_r^*/p_r has unit modulus on the torus. Successive maximum modulus principles make it contractive inside. Letting $r \uparrow 1$ gives the asserted bound. Hence $1 + e^{i\theta} x p^*/p \neq 0$ for $x \in \mathbb{D}$. $\square$

Reflection descriptions of rational inner functions are classical; see [13]. The elementary proof above specifies the full support set and the coprime multiaffine reduction used here.

6.5. All stable isometries and their coding obstruction.

Theorem 6.11 (Stable isometries). *Let $V : (\mathbb{C}^2)^{\otimes m} \rightarrow (\mathbb{C}^2)^{\otimes n}$ be an isometry. Its coefficient polynomial is stable if and only if, after a physical output permutation,*

$$V = |\psi\rangle \otimes V_1 \otimes \cdots \otimes V_m, \quad (84)$$

where ψ is a normalized stable pure state on an optional prepared block S_0 . The remaining nonempty, disjoint blocks S_i carry one-input isometries whose polynomials are

$$G_{V_i}(z, x_i) = \frac{p_i(z) + e^{i\theta_i} x_i p_i^*(z)}{\|p_i\|_2}, \quad \gcd(p_i, p_i^*) = 1, \quad [z^{S_i}] p_i(z)^2 = 0. \quad (85)$$

Each p_i is stable and multiaffine, and reflection uses the whole block S_i . The sets $S_0, S_1, \dots, S_m$ partition the physical outputs. In particular, an isometric channel $M \mapsto VMV^*$ is full Choi-LY exactly for these isometries.

Proof. Write $G_V = \sum_b A_b(z)x^b$. Isometry gives $\langle A_b, A_c \rangle = \delta_{bc}$, so Lemma 6.9 applies. In its factorization, disjointness makes both norms and inner products multiplicative. Comparing columns 0 and e_i yields $\|p_i\|_2 = \|q_i\|_2 > 0$ and $\langle p_i, q_i \rangle = 0$. A factor with no output variable could not satisfy the latter equality, so every logical block is nonempty. Normalize the common prepared factor and each pair. There is no residual normalization, because the all-zero column has norm one. Variables absent from the polynomial are fixed-zero prepared outputs.

Lemma 6.10 gives the reflected form. Furthermore,

$$\langle p_i, p_i^* \rangle = \overline{\sum_{T \subseteq S_i} (p_i)_T (p_i)_{S_i \setminus T}} = \overline{[z^{S_i}] p_i(z)^2},$$

so orthogonality is exactly the final condition in (85). Conversely that condition gives orthogonal columns, reflection preserves their norms, and Lemma 6.10 gives stability. Tensor products prove the converse for V . The isometric channel's Choi polynomial is $G_V(z, x)\overline{G_V(w, y)}$ in independent variables, proving the final equivalence. $\square$

Lemma 6.12 (Strict longitudinal bias). *If p is stable and multiaffine on a nonempty specified set S , and $\gcd(p, p^*) = 1$, then for every $j \in S$,*

$$\|p|_{z_j=0}\|_2^2 > \|\partial_j p\|_2^2. \quad (86)$$

Proof. Write $p = a + z_j b$. Stable affine slices imply $|b| \leq |a|$ on the remaining polydisk, hence $\|b\|_2 \leq \|a\|_2$. If $b = 0$, strictness follows from $a \neq 0$. Otherwise suppose equality holds. Torus saturation gives $aa^* = bb^*$, reflecting both polynomials over $R = S \setminus \{j\}$, while the full reflection is $p^* = b^* + z_j a^*$. Let $g = \gcd(a, b)$ and $h = a/g + z_j b/g$. This polynomial is primitive over $\mathbb{C}[z_R]$ and has positive degree in z_j . Over $\mathbb{C}(z_R)[z_j]$,

$$p^* = \frac{a^*}{b} (a + z_j b).$$

Thus h divides both p and p^* over the fraction field. Gauss's lemma gives the same divisibility in the polynomial ring, contradicting coprimality. This proof also covers empty R and variables absent from either slice. $\square$

Corollary 6.13 (Single-output erasure). *For a reduced logical block in Theorem 6.11, every physical output $j \in S_i$ satisfies*

$$V_i^* Z_j V_i = \delta_j Z_i, \quad \delta_j = \frac{\|p_i|_{z_j=0}\|_2^2 - \|\partial_j p_i\|_2^2}{\|p_i\|_2^2} > 0. \quad (87)$$

Erasure of any such output is not exactly correctable on the logical space. Every nontrivial stable isometric quantum code therefore has distance one.

Proof. Reflection complements occupied subsets, so the two diagonal expectations of Z_j have opposite signs. The cross term is zero because complementary subsets cancel:

$$\langle p_i, Z_j p_i^* \rangle = \overline{\sum_{T \subseteq S_i} (-1)^{\mathbf{1}_{j \in T}} (p_i)_T (p_i)_{S_i \setminus T}} = 0.$$

Lemma 6.12 now proves (87). For erasure of output j , the two Kraus operators may be taken as $E_a = \langle a|_j \otimes I$, $a = 0, 1$. The Knill–Laflamme criterion [14] requires $V^* E_a^* E_b V$ to be scalar for all a, b . Subtracting the diagonal cases requires $V^* Z_j V$ to be scalar, contrary to (87). Equivalently, this weight-one phase observable is not detectable by the code. $\square$

Remark 6.14. Outputs in the prepared block are exempt from the erasure conclusion. Nor is there a uniform positive lower bound on the individual bias: for $p(z_1, z_2) = 1 + rz_1$, $0 \leq r < 1$, reflected over $\{1, 2\}$, one has $p^* = z_2(r + z_1)$, coprime orthogonal columns, and $\delta_1 = (1 - r^2)/(1 + r^2) \rightarrow 0$. The conclusion concerns exact correction only.

Example 6.15 (Coherent information invisible in one-site marginals). The repetition isometry $V|0\rangle = |00\rangle$, $V|1\rangle = |11\rangle$ has polynomial $1 + xz_1z_2$. The square channel $\Phi(M) = V \text{Tr}_2(M) V^*$ is full Choi–LY, with polynomial

$$(1 + x_1z_1z_2)(1 + y_1w_1w_2)(1 + x_2y_2).$$

All its single-output transverse adjoints vanish, although $\Phi^\dagger(X_1X_2) = X_1 \otimes I_2$. Thus Theorem 6.5 cannot identify its omitted outputs as a classical or input-independent sector.

Proposition 6.16 (No stable Stinespring representation in general). *There is a two-qubit CPTP full Choi–LY map with no Stinespring isometry whose coefficient tensor is LY, for any finite qubit environment.*

Proof. Let Δ be complete computational-basis dephasing on two qubits, let P swap the qubits, and set $\Phi = \frac{1}{2}\Delta + \frac{1}{2}\text{Ad}_P \circ \Delta$. This is CPTP. With $r_j = z_j w_j$ and $s_j = x_j y_j$, its Choi polynomial is

$$\begin{aligned} K(r, s) &= \frac{1}{2}(1 + r_1s_1)(1 + r_2s_2) + \frac{1}{2}(1 + r_1s_2)(1 + r_2s_1) \\ &= 1 + \frac{1}{2}(r_1 + r_2)(s_1 + s_2) + r_1r_2s_1s_2. \end{aligned} \tag{88}$$

For fixed $s_1, s_2 \in \mathbb{D}$, this is the symmetric multiaffine polarization in r_1, r_2 of $(1 + rs_1)(1 + rs_2)$. The Grace–Walsh–Szegő coincidence theorem on the convex circular domain $\mathbb{D}$ implies its nonvanishing for $r_1, r_2 \in \mathbb{D}$; see [6]. Thus the full Choi tensor is stable.

However, $\Phi^\dagger(Z_1) = \Phi^\dagger(Z_2) = (Z_1 + Z_2)/2$. If a stable Stinespring isometry existed, apply Theorem 6.11 to its physical and environmental output qubits together. Every one-site observable on a physical output would pull back either to a scalar or to an operator on one logical input, because its output belongs to one disjoint encoding block or to the prepared block. Tracing the environment merely inserts identities at its sites. The displayed two-input longitudinal observable is incompatible with this conclusion. $\square$

6.6. All Schur channels.

Lemma 6.17 (A scalar equal-diagonal criterion). *For a nonzero PSD matrix $H = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$, its tensor polynomial is stable if and only if $a > 0$ and $a \geq d$. For arbitrary complex β, γ , the polynomial $1 + \beta x + \gamma y + xy$ is stable if and only if $\gamma = \bar{\beta}$ and $|\beta| \leq 1$.*

Proof. A stable polynomial must have nonzero constant coefficient, so $a > 0$. The necessary boundary root-modulus inequality is

$$|a + \bar{b}\zeta|^2 - |b + d\zeta|^2 = (a - d)(a + d + 2\Re(\bar{b}\zeta)) \geq 0, \quad |\zeta| = 1.$$

Its average gives $a \geq d$. Conversely, PSD gives $|b|^2 \leq ad$, and $a \geq d$ gives $|b| \leq a$. If $|b| < a$, the affine constant factor $a + \bar{b}z$ is nonzero on the closed disk, and the displayed inequality and the maximum modulus principle bound the other coefficient divided by this one by one. This proves stability. If $|b| = a$, PSD and $a \geq d$ force $d = a$, and the polynomial factors as $a(1 + (\bar{b}/a)z)(1 + (b/a)w)$, also stable.

For the second assertion, boundary root-modulus inequalities give $|1 + \beta\zeta| \geq |\gamma + \zeta|$, and the analogous inequality with β, γ exchanged. Squaring and averaging gives $|\beta| = |\gamma|$. The remaining nonnegative linear trigonometric polynomial has mean zero, so vanishes identically, yielding $\beta = \bar{\gamma}$. Setting one variable to zero gives $|\beta| \leq 1$. The first assertion applied to the resulting equal-diagonal PSD matrix proves the converse. $\square$

Theorem 6.18 (Stable correlation matrices and Schur channels). *Let $C \succeq 0$ be a 2^m -by- 2^m matrix with $C_{aa} = 1$. Its tensor polynomial is stable if and only if*

$$C = \bigotimes_{i=1}^m \begin{pmatrix} 1 & \eta_i \\ \bar{\eta}_i & 1 \end{pmatrix}, \quad |\eta_i| \leq 1. \quad (89)$$

Equivalently, a CPTP map fixing every computational-basis state is full Choi-LY if and only if it is a tensor product of one-qubit phase/dephasing channels. Parameters $\eta_i = 0$ are allowed.

Proof. Separate one physical site and write $C = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix}$. For arbitrary $z \in \mathbb{D}^{m-1}$, compress the other sites against the product vector $v(z) = \sum_a z^a |a\rangle$. The two-by-two compression is PSD. Its tensor polynomial is an interior specialization of G_C , hence stable, and its constant coefficient is strictly positive. By Lemma 6.17,

$$\langle v(z), (A - D)v(z) \rangle \geq 0.$$

At any fixed positive interior coordinate moduli, the angular average of this expression is $\sum_a (A_{aa} - D_{aa})|z|^{2a} = 0$. Continuity and nonnegativity force pointwise equality on every such torus. Varying the moduli proves the polynomial identity $\langle v(z), (A - D)v(z) \rangle = 0$. Independence of the monomials $\bar{z}^a z^b$ gives $A = D$ entry-wise.

Writing u for the remaining independent row and column variables, the full polynomial has the form

$$a(u)(1 + xy) + b(u)x + c(u)y.$$

The specialization $a(u)$ is nonzero throughout the polydisk. Lemma 6.17 gives $c(u)/a(u) = \bar{b(u)}/a(u)$ and $|b(u)/a(u)| \leq 1$ for every complex u there. Both ratios are holomorphic on this connected domain, so they are constant. This factors off one matrix in (89). The remaining factor A is PSD, has diagonal one, and is tensor-LY. Induction proves the assertion. The converse follows from the one-site criterion and multiplication on disjoint variables.

For a Schur multiplier $\mathcal{S}_C(M) = C \circ M$, CP and TP are equivalent to $C \succeq 0$ and diagonal one. Its full Choi polynomial is $G_C(r, s)$, where $r_i = z_i u_i$ and $s_i = w_i v_i$. Each disk value is attainable by these products, so full Choi stability is exactly

tensor stability of C . Finally, any CPTP map fixing all computational-basis states has diagonal Kraus operators: each Kraus image of a basis vector must lie in the one-dimensional range of the corresponding pure output. It is therefore a Schur map, proving the equivalent formulation. $\square$

6.7. Channels with pure computational-basis outputs.

Theorem 6.19 (Pure-basis-output classification). *Let $\Phi : M_{2^m}(\mathbb{C}) \rightarrow M_{2^n}(\mathbb{C})$ be CPTP, and suppose every computational-basis output $\Phi(|a\rangle\langle a|)$ is pure. The outputs need not be orthogonal or distinct. Then Φ is full Choi-LY if and only if it has the following form.*

Partition the inputs as $E \sqcup K = [m]$, and the outputs into an optional prepared block S_0 and nonempty disjoint blocks S_i , $i \in K$. Prepare a normalized stable pure state ψ on S_0 . On each S_i take a reduced stable two-column matrix V_i with normalized, linearly independent columns v_{i0}, v_{i1} . Reduced means that their amplitude polynomials are coprime. Equivalently,

$$G_{V_i}(z, x_i) = \frac{p_i(z) + e^{i\theta_i} x_i p_i^*(z)}{\|p_i\|_2}, \quad \gcd(p_i, p_i^*) = 1, \quad (90)$$

where p_i is stable and multiaffine and reflection uses the full block. Put $\mu_i = \langle v_{i0}, v_{i1} \rangle$, and choose $|\eta_i| \leq 1$ subject to

$$\mu_i \eta_i = 0. \quad (91)$$

With

$$C_i = \begin{pmatrix} 1 & \eta_i \\ \bar{\eta}_i & 1 \end{pmatrix}, \quad \mathcal{B}_i(M) = V_i(C_i \circ M)V_i^*,$$

the channel, after output permutation, is exactly

$$\Phi(\rho) = |\psi\rangle\langle\psi| \otimes \left(\bigotimes_{i \in K} \mathcal{B}_i \right) (\text{Tr}_E \rho). \quad (92)$$

Empty K is allowed. If $\mu_i = 0$, the block is a stable isometric encoding preceded by arbitrary local phase/dephasing. If $\mu_i \neq 0$, it measures the input in the computational basis and prepares the corresponding nonorthogonal reflected pure state.

Proof. Assume first that Φ is full Choi-LY. Choose normalized output vectors ψ_a , with amplitude polynomials p_a . Complete input dephasing is full Choi-LY; its Choi polynomial is a product of factors $1 + x_i y_i u_i v_i$. After composing with this map and grouping each input row/column pair into its product, we obtain the stable polynomial

$$F(z, w, s) = \sum_a A_a(z, w) s^a, \quad A_a(z, w) = p_a(z) \bar{p}_a(w). \quad (93)$$

Grouping is an equivalence of disk stability because disk-variable products cover the disk. Each A_a has coefficient norm one, being the polynomial of a normalized rank-one projection. Lemma 6.9 gives, for every nonempty subset $T \subseteq [m]$,

$$A_T A_0^{|T|-1} = \prod_{i \in T} A_{e_i}. \quad (94)$$

Substitute (93) into this identity. The rational function

$$R_T(z) = \frac{p_T(z) p_0(z)^{|T|-1}}{\prod_{i \in T} p_{e_i}(z)}$$

satisfies $R_T(z)\overline{R_T(\bar{w})} = 1$ in independent variables. Fixing a generic w shows that it is a constant ξ_T of modulus one. We may therefore align the output column phases so that the polynomial

$$G(z, x) = p_0(z) \prod_i \left(1 + x_i \frac{p_{e_i}(z)}{p_0(z)} \right) \quad (95)$$

has exactly those phase-aligned columns. More explicitly, its coefficient at x^T is $\xi_T^{-1} p_T$; hence it is a multiaffine polynomial, not merely a rational expression. The stable constant slice $A_0 = p_0 \bar{p}_0$ makes p_0 zero-free in the output polydisk. The singleton modulus domination for (93), specialized to $w = \bar{z}$, gives $|p_{e_i}/p_0| \leq 1$. Thus (95) is stable. It defines a matrix V with normalized columns; no orthogonality is being asserted.

Apply Lemma 6.9 to G , whose column norms are all one, and then Lemma 6.10 to its primitive factors. We obtain a stable prepared factor and disjoint reduced reflected pairs with equal local norms. A dependent primitive pair $q_i = c_i p_i$ must have both polynomials constant, so has empty output support; conversely every nonempty primitive pair has independent columns. Assign the empty factors to E and the others to K . Their equal norms imply $|c_i| = 1$. Replacing each erased input variable by $c_i^{-1} x_i$ is a disk-preserving phase change of columns, and makes the two erased-input columns identical. Normalize the prepared factor and all retained pairs. The total normalization is one because the all-zero full column has norm one. All aligned output vectors now have the form

$$v_a = \psi \otimes \bigotimes_{i \in K} v_{i, a_i}, \quad (96)$$

independent of a_E . Common prepared factors and variables absent from G belong to S_0 . This proves all the polynomial and support assertions about the pairs in the theorem.

Choose Kraus operators K_ℓ of Φ . Since the positive sum of their output projections on input a has rank one, every $K_\ell|a\rangle$ is proportional to v_a . Writing $K_\ell|a\rangle = c_{\ell a} v_a$ gives

$$\Phi(|a\rangle\langle b|) = C_{ab} |v_a\rangle\langle v_b|, \quad C_{ab} = \sum_\ell c_{\ell a} \bar{c}_{\ell b}, \quad (97)$$

where $C \succeq 0$ and $C_{aa} = 1$. Original trace preservation is exactly

$$C_{ab} \langle v_b, v_a \rangle = \delta_{ab}. \quad (98)$$

In particular, if $a_K = b_K$, the aligned vectors in (96) are identical, so

$$C_{ab} = \delta_{ab} \quad \text{whenever } a_K = b_K. \quad (99)$$

For each retained block, its independent normalized columns have positive definite Gram matrix

$$H_i = V_i^* V_i = \begin{pmatrix} 1 & \mu_i \\ \bar{\mu}_i & 1 \end{pmatrix}, \quad |\mu_i| < 1.$$

Both V_i^* and

$$H_i^{-1} = \frac{1}{1 - |\mu_i|^2} \begin{pmatrix} 1 & -\mu_i \\ -\bar{\mu}_i & 1 \end{pmatrix}$$

are tensor-LY, the latter by Lemma 6.17. Consequently $L_i = H_i^{-1}V_i^*$ is tensor-LY and satisfies $L_i V_i = I$. Contraction cannot give zero because of this identity. The rectangular matrix $L = \langle \psi | \otimes \bigotimes_{i \in K} L_i$ is stable and obeys $L v_a = |a_K\rangle$.

The CP map $\Psi(M) = L\Phi(M)L^*$ is full Choi-LY or zero, and satisfies

$$\Psi(|a\rangle\langle b|) = C_{ab}|a_K\rangle\langle b_K|.$$

Its trace is $C_{ab}\delta_{a_K, b_K} = \delta_{ab}$ by (99). Thus Ψ is TP and nonzero. The compression itself need not be TP on arbitrary output matrices; this argument proves TP only for its composition with Φ , which is all that is required.

The Choi polynomial of Ψ is

$$\sum_{a,b} C_{ab} z^{a_K} w^{b_K} u^a v^b.$$

For $i \in K$, replace $z_i u_i, w_i v_i$ by r_i, s_i ; for $i \in E$, put $r_i = u_i, s_i = v_i$. The variables r_i, s_i range independently over the disk, so stability of this Choi polynomial is precisely tensor stability of the entire correlation matrix C . Theorem 6.18 gives $C = \bigotimes_i \begin{pmatrix} 1 & \eta_i \\ \bar{\eta}_i & 1 \end{pmatrix}$, with $|\eta_i| \leq 1$. Taking strings differing only at an erased input in (99) forces $\eta_i = 0$ there. Taking strings differing at a retained input in (98) gives $\eta_i \bar{\mu}_i = 0$, equivalently (91). Combining these tensor-product coefficients with (96) proves (92); the erased coefficient is exactly δ_{a_E, b_E} , hence ordinary partial trace.

Conversely, start from the displayed data. Each C_i is a PSD correlation matrix, so its Schur map is CP. Following it by $M \mapsto V_i M V_i^*$ preserves CP. Normalization of the columns gives the correct diagonal traces, while (91) makes the off-diagonal traces zero. Thus each $\mathcal{B}_i$ is TP and has pure basis outputs. Its two component Choi tensors are stable by Lemma 6.10 and Theorem 6.18; their composition is nonzero because it is TP. Stable pure preparation, partial trace, tensor products and physical permutations prove the converse. The descriptions of the two cases follow directly from $\mu_i \eta_i = 0$. $\square$

Theorems 6.5, 6.18, and 6.19 concern different singular strata. In particular, none asserts a classification of general channels with mixed computational-basis outputs. Proposition 6.16 prevents extending the isometry theorem to that class by an assumption of a stable Stinespring representation.

6.8. Pauli-diagonal maps without invertibility.

Lemma 6.20 (Factorization on slit domains). *Let $\Omega_1, \dots, \Omega_n$ be nonempty connected open subsets of $\mathbb{C}$ with $\mathbb{C} \setminus \Omega_j \subseteq \mathbb{R}$. A multiaffine polynomial nonvanishing on $\prod_j \Omega_j$ is a nonzero scalar times a product of one-variable affine polynomials, with constant factors permitted.*

Proof. Write $p(r_1, r') = a(r') + r_1 b(r')$. If $b = 0$, omit that coordinate. Otherwise, on a nonempty connected open neighborhood where $b \neq 0$, the holomorphic function $-a/b$ avoids Ω_1 and is therefore real-valued. It is constant, say $c \notin \Omega_1$. The polynomial identity principle gives $a + cb = 0$ everywhere, so $p = (r_1 - c)b$. The remaining factor is nonvanishing on the remaining domain. Induction proves the assertion. $\square$

Theorem 6.21 (All-rank Pauli-diagonal rigidity). *Suppose a complex-linear HP, TP map on n qubits is diagonal in the Pauli-string basis: $\Phi(\sigma_\alpha) = \lambda_\alpha \sigma_\alpha$, where*

$\sigma_\alpha = \bigotimes_j \sigma_{\alpha_j}$ and $\sigma_0 = I$. If Φ is full Choi-LY, then

$$\lambda_\alpha = \prod_j \lambda_{\alpha_j^{(j)}}, \quad \Phi = \bigotimes_j \Phi_j, \quad (100)$$

where $\alpha_j^{(j)}$ denotes the string with the stated Pauli at site j and identities elsewhere. Each local factor is HP, TP and full Choi-LY. If Φ is CP, all factors are CP, and the converse holds. Zero eigenvalues are allowed.

Proof. HP makes the λ_α real, and TP gives $\lambda_0 = 1$. Pauli orthogonality yields

$$J(\Phi) = 2^{-n} \sum_\alpha \lambda_\alpha \sigma_\alpha \otimes \sigma_\alpha^T.$$

The four one-site polynomials are $f_I = 1 + zw$, $f_X = z + w$, $f_Y = i(z - w)$, and $f_Z = 1 - zw$; the transpose contributes a minus sign only for Y .

At every site select a desired axis, independently of the others. For axis X , set the output variables to $z = w = 1$. Only the identity and X terms remain, and their normalized ratio is

$$r_X = \frac{x + y}{1 + xy}.$$

Its image on the bidisk is $\Omega_\perp = \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty))$. For a real $|r| \geq 1$, the inverse equation would require $y = (r - x)/(1 - rx)$, whereas

$$|r - x|^2 - |1 - rx|^2 = (r^2 - 1)(1 - |x|^2) \geq 0.$$

The endpoint equations also require a boundary variable. Conversely, setting $x = y = u$ gives an inverse quadratic with reciprocal roots. A unit-circle root would place r on the omitted rays. For every nonzero $r \in \Omega_\perp$, exactly one inverse root is therefore in the disk; zero is attained at $u = 0$.

For axis Y , set $z = i, w = -i$. The output factor is $f_Y = -2$, and the input transpose changes the sign again. The surviving ratio is $r_Y = i(x - y)/(1 + xy)$, with the same image $\Omega_\perp$, by replacing (x, y) with $(ix, -iy)$. For axis Z , set $z = s, w = -s, x = t, y = t$, all interior. The output specialization kills X and the input specialization kills Y . The ratio is

$$r_Z = \frac{1 + s^2}{1 - s^2} \frac{1 - t^2}{1 + t^2}.$$

Each factor ranges over the right half-plane, so their product ranges exactly over $\Omega_Z = \mathbb{C} \setminus (-\infty, 0]$. Indeed products have arguments strictly between $-\pi$ and π , and the principal square root realizes every point of this slit plane as such a product.

For axes $a_1, \dots, a_n$, divide out the nonzero identity factors after these substitutions. The resulting polynomial is

$$p_a(r) = \sum_{S \subseteq [n]} \lambda_{a_S} \prod_{j \in S} r_j, \quad p_a(0) = 1, \quad (101)$$

where a_S has the chosen axes on S . Boundary specialization at the X, Y sites gives stable or zero by Hurwitz. The zero alternative is impossible: the ratio maps have the nonempty open product image just computed, and the nonzero polynomial (101) cannot vanish identically there. The Z substitutions are interior. Thus p_a is nonvanishing on the corresponding product of slit domains.

Lemma 6.20 and $p_a(0) = 1$ give $p_a(r) = \prod_j (1 + \lambda_{a_j^{(j)}} r_j)$. The normalization at zero is algebraic and does not require zero to belong to Ω_Z . Comparing coefficients,

and varying the axes, proves (100) for every Pauli string. Local factors inherit the Choi and positivity properties by the marginal construction in the proof of Theorem 6.5. Tensor products prove the converse. $\square$

For CPTP local Pauli factors the exact one-site condition is

$$\lambda_z \geq \max\{|\lambda_x|, |\lambda_y|\},$$

or equivalently $\min(p_0, p_z) \geq \max(p_x, p_y)$ in the Pauli-error probabilities. This is the Pauli specialization of the local classification in Theorem 7.1; its sufficiency is also [19, Theorem 3]. Because the Pauli transform between error probabilities and eigenvalues is an invertible tensor-product transform, (100) says that the entire error distribution is a product distribution.

6.9. Classical channels and the purity boundary. Consider the channel that measures m input bits in the computational basis and prepares a diagonal distribution $P(b | a)$ on n output bits. Its full Choi polynomial reduces to

$$K(r, s) = \sum_{a,b} P(b | a) r^b s^a = \sum_a A_a(r) s^a, \quad K(1, s) = \prod_i (1 + s_i). \quad (102)$$

The coefficients are nonnegative. As above, reducing each row/column pair to its product is an equivalence of disk stability. Define the collision purity of column a by $\mathcal{C}(a) = \sum_b P(b | a)^2 = \|A_a\|_2^2$.

Theorem 6.22 (The constant-purity boundary). *Every stable stochastic kernel (102) satisfies $\mathcal{C}(a) \leq \mathcal{C}(0)$ for every input. Equality at every singleton input e_i holds if and only if, for a partition $S_0, S_1, \dots, S_m$ of the output variables, the kernel is*

$$K(r, s) = C(r_{S_0}) \prod_{i=1}^m (p_i(r_{S_i}) + s_i p_i^*(r_{S_i})). \quad (103)$$

Here C and the p_i are stable, multiaffine, have nonnegative coefficients, satisfy $C(1) = p_i(1) = 1$, and $\gcd(p_i, p_i^*) = 1$; reflection is over the full indicated block. Empty blocks S_i are allowed and contribute $1 + s_i$. Every column of such a kernel has the same collision purity.

Proof. Fix the output variables in the polydisk. For any input subset T , set its variables equal to t and the other input variables to zero. The resulting univariate polynomial is nonvanishing for $|t| < 1$, has constant coefficient A_0 , and top coefficient A_T in degree $|T|$. If that top coefficient is nonzero, the product of root moduli is at least one, whence $|A_T| \leq |A_0|$. If it is zero the inequality is immediate. Continuity and torus Parseval prove the purity inequality.

Under singleton equality, Lemma 6.9 gives disjoint primitive factors $C \prod_i (p_i + s_i q_i)$. The stable constant slice makes the constant coefficients of C, p_i nonzero. Fix their phases to make these constants positive. Each coefficient of each factor can be isolated as a coefficient of K by selecting the constant monomial in all the other factors; disjoint supports therefore make all coefficients of C, p_i, q_i nonnegative. Lemma 6.10 gives $q_i = e^{i\theta_i} p_i^*$, and nonnegativity forces the phase to be one. Stochasticity at output variables one gives $p_i(1) = q_i(1) > 0$. Normalizing these common values and the prepared factor gives (103), with no scalar remainder because every column has mass one. Variables absent from K are fixed-zero prepared outputs.

Conversely, Lemma 6.10 makes each factor stable. The displayed normalization makes the nonnegative coefficient kernel stochastic. Reflection preserves coefficient norms and disjoint supports make them multiply, so all columns have equal collision purity. $\square$

Corollary 6.23 (Deterministic classical channels). *A deterministic stochastic kernel is full Choi–LY if and only if every output bit is identically zero or copies one input bit. An input may be copied to any disjoint block of outputs or discarded.*

Proof. Write $P(b \mid a) = \mathbf{1}_{b=f(a)}$. All column purities equal one. The rational product identity in Lemma 6.9 therefore gives

$$r^{f(a)} = r^{f(0) + \sum_{i:a_i=1} (f(e_i) - f(0))}.$$

Stable evaluation at the full origin requires $f(0) = 0$. Each output coordinate is consequently a sum of its singleton Boolean values. Since it must still be a bit when all inputs are one, at most one singleton value can be one. This proves necessity. Conversely the kernel of such a map is $\prod_i (1 + s_i \prod_{j \in S_i} r_j)$, with disjoint copied-output blocks, and is manifestly stable. $\square$

Theorem 6.24 (One classical output). *A stochastic kernel from m input bits to one output bit is full Choi–LY if and only if*

$$P(1 \mid a) = \alpha + \sum_i \lambda_i a_i, \quad \alpha \geq 0, \quad \lambda_i \geq 0, \quad 2\alpha + \sum_i \lambda_i \leq 1. \quad (104)$$

Thus it chooses to copy input i with probability λ_i , to output zero with probability $1 - 2\alpha - \sum_i \lambda_i$, or to output a fresh fair bit with probability 2α .

Proof. Write $K(r, s) = A(s) + rB(s)$. Stochasticity gives $A + B = \prod_i (1 + s_i)$. Put $H = (A - B)/(A + B)$, so

$$K = \frac{\prod_i (1 + s_i)}{2} ((1 + r) + (1 - r)H(s)).$$

The quotient $(1 + r)/(1 - r)$ ranges over the open right half-plane; thus stability is exactly $\Re H(s) \geq 0$ on the input polydisk. Set $u_i = (1 - s_i)/(1 + s_i)$. Then H becomes a real multiaffine polynomial in the right-half-plane variables u_i , because every $s^a / \prod_i (1 + s_i)$ is a product of factors $(1 \pm u_i)/2$.

A real polynomial with nonnegative real part on a product of right half-planes has the form $c + \sum_i \lambda_i u_i$, with $c, \lambda_i \geq 0$. To see this, restrict to $u_i = v_i t$, $v_i > 0$, $\Re t > 0$. A real one-variable polynomial of degree at least two has a leading term with negative real part along some interior right-half-plane ray at large radius. Every such restriction therefore has degree at most one. Varying v in its open positive orthant forces all higher homogeneous parts to vanish. Positive real evaluations near zero, or with one variable tending to infinity, show that the remaining coefficients are nonnegative.

Evaluating the transformed polynomial identity at $u_i = (-1)^{a_i}$ gives $1 - 2P(1 \mid a) = c + \sum_i \lambda_i (-1)^{a_i}$. This is polynomial evaluation, including at vertices corresponding to poles of the original change of variables. Consequently (104) holds with $\alpha = (1 - c - \sum_i \lambda_i)/2$. The original probability at $a = 0$ makes $\alpha \geq 0$, and $c \geq 0$ gives the last inequality. Conversely these conditions define the stated stochastic mixture and a positive-real polynomial H , proving stability. $\square$

Every single-output marginal of a multibit classical full Choi–LY channel therefore has the form (104). Marginalization sets the other output variables to one; boundary Hurwitz applies, and stochastic normalization excludes zero. This does not determine the joint output correlations.

Proposition 6.25 (Nonconvexity among reset channels). *For $n \geq 3$, the distribution $\nu_\theta = (1 - \theta)\delta_{0^n} + \theta \text{Unif}(\{0, 1\}^n)$, $0 \leq \theta \leq 1$, has a stable polynomial precisely when*

$$0 \leq \theta \leq \frac{1}{1 + \cos^n(\pi/n)} \quad \text{or} \quad \theta = 1. \quad (105)$$

For $n = 1, 2$, all $\theta \in [0, 1]$ are allowed. Consequently the set of full Choi–LY channels is not convex, even among classical reset channels.

Proof. The distribution polynomial is $f_\theta(r) = 1 - \theta + 2^{-n}\theta \prod_j (1 + r_j)$. The endpoints $\theta = 0, 1$ are stable. For $0 < \theta < 1$, an interior zero means that a product of n points in the open disk centered at one with radius one equals $-2^n(1 - \theta)/\theta$. Write these nonzero points as $\rho_j e^{i\alpha_j}$, where $|\alpha_j| < \pi/2$ and $0 < \rho_j < 2 \cos \alpha_j$. A negative real product requires $\sum_j \alpha_j$ to be an odd multiple of π , which is impossible for $n = 1, 2$. For $n \geq 3$, concavity of $\log \cos$ implies

$$\prod_j \rho_j < 2^n \prod_j \cos \alpha_j \leq [2 \cos(\pi/n)]^n.$$

An odd multiple of larger absolute value can only reduce the bound. Every smaller positive product magnitude is attained by setting all arguments to π/n and choosing equal moduli. Therefore an interior zero occurs exactly when $2^n(1 - \theta)/\theta < [2 \cos(\pi/n)]^n$, the complement of (105) away from the separately treated endpoint. Equality requires boundary points and is stable for the open-disk definition. Resetting to this distribution multiplies its polynomial by the stable input trace factor $\prod_i (1 + s_i)$, so the same criterion applies to the channel. $\square$

7. THE COMPLETE ONE-QUBIT STATIC CLASSIFICATION

We give exact parameter criteria for a single qubit, including every singular stratum. The stability classification first requires only Hermiticity preservation and trace preservation. Complete positivity is then imposed by explicit matrix or scalar inequalities.

Write a complex-linear Hermiticity-preserving, trace-preserving map in Bloch coordinates as

$$\Phi(I) = I + t \cdot \sigma, \quad \Phi(\sigma_j) = \sum_i T_{ij} \sigma_i, \quad t \in \mathbb{R}^3, \quad T \in \mathbb{R}^{3 \times 3}, \quad (106)$$

where $\sigma = (X, Y, Z)$. On trace-one Hermitian matrices its action is $r \mapsto t + Tr$. Invertibility means complex-linear invertibility on $M_2(\mathbb{C})$, equivalently $\det T \neq 0$. Full Choi stability always refers to all four variables of $F_{J(\Phi)}$.

Theorem 7.1 (All one-qubit HP/TP maps). *The map (106) has a full Lee–Yang Choi kernel if and only if it belongs to exactly one of the following two classes.*

(1) *There is a nonzero real 2×2 matrix D such that*

$$T = \begin{pmatrix} D & 0 \\ 0 & \lambda \end{pmatrix}, \quad \lambda > 0, \quad t = (\tau, 0), \quad \tau \in \mathbb{R}^2, \quad \|\tau\| < 1, \quad (107)$$

and, for every $n \in \mathbb{R}^2$ with $\|n\| = 1$,

$$\|D^\top n\|^2 \leq m_\lambda(\tau \cdot n), \quad m_\lambda(r) = \min_{-1 \leq x \leq 1} ((x+r)^2 + \lambda^2(1-x^2)). \quad (108)$$

(2) The transverse linear action vanishes and

$$T = \text{diag}(0, 0, \lambda), \quad \lambda \geq 0, \quad t_z \geq 0, \quad \|t_\perp\| \leq 1. \quad (109)$$

For $|r| < 1$, the minimum in (108) is

$$m_\lambda(r) = \begin{cases} \lambda^2(1-r^2/(1-\lambda^2)), & 0 < \lambda < 1, |r| \leq 1-\lambda^2, \\ (1-|r|)^2, & 0 < \lambda < 1, |r| > 1-\lambda^2, \\ (1-|r|)^2, & \lambda \geq 1. \end{cases} \quad (110)$$

The complex-linear rank is $2 + \text{rank } D$ in the first class and $1 + \mathbf{1}_{\{\lambda > 0\}}$ in the second. In particular the first class is invertible exactly when D is invertible.

7.1. Boundary kernels and the proof of the classification.

Lemma 7.2 (Trace-one scalar kernels). *A Hermitian trace-one matrix with transverse Bloch norm R and longitudinal coordinate h has a stable coefficient polynomial if and only if $R \leq 1$ and $h \geq 0$. For $0 \leq R \leq 1$, the function*

$$K_R(z, w) = \frac{1 + zw + R(z + w)}{1 - zw} \quad (111)$$

has strictly positive real part on $\mathbb{D}^2$, and the infimum of that real part is zero.

Proof. A rotation about Z makes the transverse component $(R, 0)$ and only rotates the disk variables. The coefficient polynomial becomes

$$f(z, w) = \frac{1 + h + R(z + w) + (1 - h)zw}{2} = \frac{1 - zw}{2}(K_R(z, w) + h).$$

For $a = |z| < 1$, $b = |w| < 1$ and $0 \leq R \leq 1$, multiplication by $|1 - zw|^2$ shows that the real part of the numerator of K_R is at least

$$1 - a^2b^2 - R(a(1 - b^2) + b(1 - a^2)) \geq (1 - ab)(1 - a)(1 - b) > 0.$$

This proves sufficiency. Setting $z = w = is$ and letting $s \uparrow 1$ shows that the infimum of $\Re K_R$ is zero.

For necessity, stability gives the boundary root-modulus inequality obtained by treating f as affine in w . At $z = e^{i\theta}$ its squared difference between the constant and linear coefficients is $h(1 + R \cos \theta)$. Its average is h , so $h \geq 0$. If $h > 0$, the same inequality for all θ gives $R \leq 1$. If $h = 0$ and $R > 1$, setting $w = 0$ gives the interior zero $z = -1/R$. $\square$

Lemma 7.3 (The local block form). *Every map in Theorem 7.1 whose full Choi kernel is stable satisfies*

$$T = D \oplus \lambda, \quad \lambda \geq 0. \quad (112)$$

If $D \neq 0$, then $t_z = 0$ and $\lambda > 0$.

Proof. Let $\phi = \Phi^\dagger$. It is unital and Hermiticity preserving. Lemma 6.1 makes it a preserver of complex matrix-LY polynomials, with a possible zero output. Near the identity that zero alternative is excluded by continuity. Apply Lemma 6.2 to

the two-sided stable curves $\phi(e^{sX})$, $\phi(e^{sY})$, and $\phi(e^{siZ})$. Hermiticity preservation, and its anti-Hermitian counterpart, give

$$\begin{aligned}\phi(X) &= t_x I + D_{11}X + D_{12}Y, \\ \phi(Y) &= t_y I + D_{21}X + D_{22}Y, \\ \phi(Z) &= t_z I + \lambda Z.\end{aligned}\tag{113}$$

The stable matrix $I + sZ$, $s > 0$ small, maps to $(1 + st_z)I + s\lambda Z$. Its root-modulus inequality is

$$|1 + st_z + s\lambda|^2 - |1 + st_z - s\lambda|^2 = 4s\lambda(1 + st_z) \geq 0,$$

so $\lambda \geq 0$.

If $D \neq 0$, choose a real transverse field A such that the transverse component of $\phi(A)$ is nonzero. For small nonzero real ε , both $A + i\varepsilon Z$ and the traceless part of its image are hyperbolic. The input pencil $I + s(A + i\varepsilon Z)$ is stable for both signs of sufficiently small s : its exponentials are stability units by Lemma 6.2, and the identity $(A + i\varepsilon Z)^2 \in \mathbb{R}I$ rewrites these exponentials as scalar multiples of the pencil near the identity. Its image is a stable pencil whose scalar part is $t_A + i\varepsilon t_z$. Lemma 6.3 forces this scalar to be real, hence $t_z = 0$.

Finally specialize the two input Choi variables to $u = e^{i\theta}$, $v = -e^{-i\theta}$. The input matrix is $Z + in \cdot \sigma_\perp$, with n ranging over the real unit circle. The output matrix is $\lambda Z + i(Dn) \cdot \sigma_\perp$. Hurwitz's theorem makes its polynomial stable or zero. If $\lambda = 0$, choose n with $Dn \neq 0$. The polynomial is then nonzero but its constant coefficient is zero, contradicting stability. This proves $\lambda > 0$. $\square$

Proof of Theorem 7.1. By Lemma 7.3, it remains to treat the two possible transverse blocks. Suppose first that $D = 0$, and rotate about Z so that $t_\perp = (R, 0)$. Direct expansion of the Choi kernel gives

$$F_{J(\Phi)} = \frac{(1 + uv)(1 - zw)}{2} \left(K_R(z, w) + t_z + \lambda \frac{1 - uv}{1 + uv} \right). \tag{114}$$

The last quotient ranges over the right half-plane. Thus $R \leq 1$, $t_z \geq 0$, and $\lambda \geq 0$ imply strict nonvanishing by Lemma 7.2. Conversely, specialization $u = v = 0$ gives a trace-one Hermitian kernel with transverse norm R and longitudinal coordinate $t_z + \lambda$, so $R \leq 1$. If $\lambda = 0$, that same lemma gives $t_z \geq 0$. If $\lambda > 0$ and $t_z < 0$, choose z, w with $\Re(K_R + t_z) < 0$. A value of $(1 - uv)/(1 + uv)$ in the right half-plane then cancels the parenthesis in (114), a contradiction. This proves the second class completely, without positivity.

Now suppose $D \neq 0$. Lemma 7.3 gives $\lambda > 0$ and $t_z = 0$. Set

$$\begin{aligned}A_0 &= (1 + \lambda)/2, & B_0 &= (1 - \lambda)/2, & \zeta &= (t_x - it_y)/2, \\ a &= (D_{11} + D_{22} + i(D_{12} - D_{21}))/2, & b &= (D_{11} - D_{22} + i(D_{12} + D_{21}))/2.\end{aligned}\tag{115}$$

The full polynomial has the reciprocal decomposition

$$F_{J(\Phi)} = p + vp^\sharp, \quad p = g + uh, \quad g = A_0 + \zeta w + \bar{\zeta}z + B_0zw, \quad h = \bar{b}w + \bar{a}z, \tag{116}$$

where $p^\sharp(z, w, u) = zwu \overline{p(1/\bar{z}, 1/\bar{w}, 1/\bar{u})}$. This is an algebraic identity with no positivity assumption.

If the full polynomial is stable, its specializations p and g are stable, since their constant coefficient is $A_0 > 0$. Lemma 7.2 applied to g gives $R = \|t_\perp\| \leq 1$.

The root test in u and continuity give $|h| \leq |g|$ on the output two-torus. Write $z = e^{i(\theta+\varphi)}$, $w = e^{i(\theta-\varphi)}$ and $n = (\cos \varphi, -\sin \varphi)$. Direct expansion yields

$$|h|^2 = \|D^\top n\|^2, \quad |g|^2 = (\cos \theta + t_\perp \cdot n)^2 + \lambda^2 \sin^2 \theta. \quad (117)$$

Minimization in $x = \cos \theta$ proves (108). If $R = 1$, rotate so that $t_\perp = e_1$. The endpoint $x = -1$ gives $\|D^\top n\| \leq 1 - n_1$ for every unit n . At $n = e_1$ the first row of D vanishes. Setting $n = (\cos \alpha, \sin \alpha)$ then gives $|\sin \alpha| \|D^\top e_2\| \leq 1 - \cos \alpha$. Division by $|\sin \alpha|$ and passage to $\alpha = 0$ kills the second row. Since $D \neq 0$, necessarily $R < 1$.

Conversely, assume (107) and (108). The polynomial g has no zero on the closed bidisk. Indeed its constant coefficient in w , namely $A_0 + \bar{\zeta}z$, has no zero on the closed disk, and at $|z| = 1$ the squared root-modulus difference is

$$|A_0 + \bar{\zeta}z|^2 - |\zeta + B_0 z|^2 = \lambda(1 + 2\Re(\bar{\zeta}z)) > 0.$$

Here $2|\zeta| = R < 1$. The maximum modulus principle in z proves the asserted closed-bidisk nonvanishing. Equations (108) and (117), followed by the maximum modulus principle in each variable, imply $|h/g| \leq 1$ on the bidisk. Hence $p = g + uh$ is stable.

For any stable multiaffine p , one has $|p^\sharp/p| \leq 1$ in the open polydisk. To include boundary zeros, apply the maximum modulus principle to $p_r(z, w, u) = p(rz, rw, ru)$, $r < 1$, which has no zeros on the closed polydisk. On its distinguished boundary $|p_r^\sharp| = |p_r|$, so $|p_r^\sharp/p_r| \leq 1$ inside. Let $r \uparrow 1$. This proves the assertion, and hence $p + vp^\sharp \neq 0$ when $|v| < 1$. Sufficiency follows.

The formula (110) follows by minimizing the quadratic $(1 - \lambda^2)x^2 + 2rx + r^2 + \lambda^2$: for $\lambda < 1$ its critical point is $-r/(1 - \lambda^2)$, and otherwise its minimum on the interval is at an endpoint. Finally the Pauli transfer matrix is $\begin{pmatrix} 1 & 0 \\ t & T \end{pmatrix}$, whose rank is $1 + \text{rank } T$. This gives all rank assertions. $\square$

7.2. Complete positivity and finite inequalities.

Theorem 7.4 (The CPTP subfamilies). *In the first class of Theorem 7.1, complete positivity is equivalent to*

$$J = \begin{pmatrix} A_0 & 0 & \zeta & a \\ 0 & B_0 & \bar{b} & \zeta \\ \bar{\zeta} & b & B_0 & 0 \\ \bar{a} & \bar{\zeta} & 0 & A_0 \end{pmatrix} \succeq 0, \quad (118)$$

with coefficients (115). For a positive trace-preserving map of this form with $0 < \lambda < 1$, the stability criterion (108) is equivalent to the finite matrix inequality

$$\frac{DD^\top}{\lambda^2} + \frac{\tau\tau^\top}{1 - \lambda^2} \preceq I_2. \quad (119)$$

At $\lambda = 1$, positivity forces $\tau = 0$, and the stability criterion is $\|D\|_{\text{op}} \leq 1$.

For the singular rank-three subclass, input and output rotations about Z put $D = \text{diag}(d, 0)$ with $d > 0$ and $\tau = (p, q)$. The complete CPTP and full-Choi-LY conditions are the two scalar inequalities

$$q^2 + (d + \sqrt{p^2 + \lambda^2})^2 \leq 1, \quad \frac{d^2}{\lambda^2} + \frac{p^2}{1 - \lambda^2 - q^2} \leq 1, \quad d, \lambda > 0. \quad (120)$$

The first inequality makes the denominator in the second strictly positive.

In the transverse-erasure class, the complete conditions are

$$t_z, \lambda \geq 0, \quad \|t_\perp\|^2 + (t_z + \lambda)^2 \leq 1. \quad (121)$$

Proof. The matrix in (118) is the Choi matrix in the order (00, 01, 10, 11), with each pair ordered as (output, input). Its partial trace over the output equals the input identity. Choi's criterion gives the first assertion.

For the finite inequality, let $k = 1 - \lambda^2 > 0$ and fix a transverse unit vector n . Reversing n if necessary, set $r = \tau \cdot n \geq 0$ and $s^2 = \|D^\top n\|^2$. The desired scalar form of (119) is

$$Q = \lambda^2 - s^2 - \lambda^2 r^2 / k \geq 0. \quad (122)$$

If $r \leq k$, this is exactly the first branch of (110). If $r > k$, put $a = k/r \in (0, 1)$. Positivity means that the affine image of the Bloch unit ball lies in the unit ball. Its support in the unit output direction $(an, \sqrt{1 - a^2})$ therefore satisfies

$$ar + \sqrt{a^2 s^2 + \lambda^2 (1 - a^2)} \leq 1.$$

Squaring the nonnegative sides gives

$$0 \leq k - 2ar + a^2(r^2 - s^2 + \lambda^2) = \frac{k^2}{r^2} Q.$$

Thus positivity itself implies (122) in the clipped region. Conversely (122) bounds the unconstrained minimum over all real x , hence implies (108) on $[-1, 1]$. Varying n proves (119). Positivity at the input Bloch points $\pm e_z$ gives $\|\tau\|^2 + \lambda^2 \leq 1$, proving both $\lambda \leq 1$ and the stated endpoint assertion.

Suppose now D has rank one. Real rotations in the transverse input and output planes put it in the form $\text{diag}(d, 0)$, $d > 0$; these are implemented by rotations about Z and preserve both properties under consideration. In these coordinates the Choi matrix satisfies $2J = I + H$, where

$$H = pX \otimes I + qY \otimes I + dX \otimes X + \lambda Z \otimes Z.$$

Pauli multiplication gives

$$H^2 = (p^2 + q^2 + d^2 + \lambda^2)I + 2pdI \otimes X - 2d\lambda Y \otimes Y.$$

The last two Pauli matrices anticommute. The eigenvalues of H^2 are $q^2 + (\sqrt{p^2 + \lambda^2} \pm d)^2$, each twice. Also $\text{Tr } H = \text{Tr } H^3 = 0$, by Pauli orthogonality. Newton's identities make the characteristic polynomial of H even, so its spectrum is symmetric about zero. Its least eigenvalue is consequently $-\sqrt{q^2 + (\sqrt{p^2 + \lambda^2} + d)^2}$. This proves the first inequality in (120) as the exact CP condition. Since $d, \lambda > 0$, it also implies $\lambda < 1$ and $1 - \lambda^2 - q^2 > 0$. Taking the Schur complement of the lower-right entry in (119) now gives precisely the second inequality in (120).

Finally, a transverse-erasure map measures in the Z basis and prepares

$$\rho_\pm = \frac{1}{2}(I + t_x X + t_y Y + (t_z \pm \lambda)Z). \quad (123)$$

Its Choi matrix is block diagonal on the input and is positive exactly when both prepared matrices are positive, equivalently $\|t_\perp\|^2 + (t_z \pm \lambda)^2 \leq 1$. Since $t_z, \lambda \geq 0$, the two inequalities reduce to (121). $\square$

Remark 7.5. Full Choi stability does not require both states in (123) to be individually Lee-Yang: the quantity $t_z - \lambda$ may be negative. The choice $t_\perp = 0$, $t_z = \lambda = 1/3$ is a rank-two CPTP example with positive longitudinal drift. A rank-three example retaining transverse information is $t = 0$, $D = \text{diag}(1/5, 0)$, $\lambda = 2/5$.

At rank one, the maps are precisely resets $M \mapsto \text{Tr}(M)\rho$; their CPTP full-Choi-LY condition is $t_z \geq 0$ and $\|t\| \leq 1$. These singular examples explain why the zero longitudinal drift conclusion for an invertible map cannot be extended without a rank hypothesis.

8. INDEPENDENTLY COVARIANT CHANNELS BETWEEN FINITE SPINS

Let U_k denote the spin- $k/2$ representation on $\mathcal{H}_k = \mathbb{C}^{k+1}$, now allowing $k = 0$. For input capacities $\kappa = (\kappa_1, \dots, \kappa_n)$ and output capacities $\ell = (\ell_1, \dots, \ell_n)$, a complex-linear map $\Phi : \text{End}(\bigotimes_i \mathcal{H}_{\kappa_i}) \rightarrow \text{End}(\bigotimes_i \mathcal{H}_{\ell_i})$ is independently covariant if

$$\Phi(U_\kappa(g)MU_\kappa(g)^*) = U_\ell(g)\Phi(M)U_\ell(g)^*, \quad U_k(g) = \bigotimes_i U_{k_i}(g_i), \quad (124)$$

for every $g = (g_1, \dots, g_n) \in SU(2)^n$. The input and output sites in this definition are matched in advance.

For a capacity vector k , write $b_a^k = \prod_i \binom{k_i}{a_i}$. Use the Choi convention $J(\Phi) = \sum_{a,b} \Phi(|a\rangle\langle b|) \otimes |a\rangle\langle b|$. Its coherent polynomial is

$$F_{J(\Phi)}(z, w, u, v) = \sum_{c,e,a,b} \sqrt{b_c^\ell b_e^\ell b_a^\kappa b_b^\kappa} [\Phi(|a\rangle\langle b|)]_{ce} z^c w^e u^a v^b. \quad (125)$$

Full coherent Choi stability means stability in all four groups of variables. For one matched pair of sites put

$$K_o = 1 + zu, \quad K_i = 1 + uv, \quad A = (1 + zu)(1 + wv), \quad B = K_o K_i. \quad (126)$$

Theorem 8.1 (Covariant root classification). *Suppose Φ is complex linear, trace preserving, and independently covariant as in (124). Its full coherent Choi kernel is Lee–Yang if and only if $\Phi = \bigotimes_i \Phi_i$, where each local factor is specified by an integer $0 \leq d_i \leq \min(\kappa_i, \ell_i)$ and an unordered list $s_{i1}, \dots, s_{id_i} \geq 0$ as follows. With $q_i(r) = \prod_{a=1}^{d_i} (r + s_{ia})$, its local kernel is*

$$F_{J(\Phi_i)} = \frac{K_{o,i}^{\ell_i - d_i} K_{i,i}^{\kappa_i - d_i} \prod_{a=1}^{d_i} (A_i + s_{ia} B_i)}{(\ell_i + 1) \int_0^1 q_i(x) dx}. \quad (127)$$

Every such map is completely positive, although neither positivity nor Hermiticity preservation was assumed. Its complex-linear rank is

$$\text{rank } \Phi = \prod_i (d_i + 1)^2. \quad (128)$$

It is injective exactly when $d_i = \kappa_i$ at every site, surjective exactly when $d_i = \ell_i$ at every site, and invertible exactly when $d_i = \kappa_i = \ell_i$ at every site.

We prove the theorem together with a factorization through an intermediate spin representation. The representation theory of covariant maps and the general covariant completely positive simplex are established in [1]; coherent descriptions and multipliers are also studied in [4]. We use the multiplicity-free commutant description, and derive the additional root restriction directly from stability.

Lemma 8.2 (Range of the coherent ratio). *The image of A/B on $\mathbb{D}^4$ is precisely $\Omega = \mathbb{C} \setminus (-\infty, 0]$.*

Proof. For $a \in \mathbb{D}$, the disk automorphisms

$$f_a(x) = \frac{x - a}{1 - \bar{a}x}, \quad h_a(y) = \frac{y + \bar{a}}{1 + ay}$$

satisfy

$$1 + f_a(x)h_a(y) = \frac{(1 - |a|^2)(1 + xy)}{(1 - \bar{a}x)(1 + ay)}.$$

Apply f_a to z, v and h_a to u, w . The ratio A/B is unchanged. Choosing $a = z$ makes the new z zero, and reduces the ratio to $(1 + w'v')/(1 + u'v')$. It is nonzero. If it equalled $-t$ for some $t > 0$, then $1+t = -v'(w' + tu')$, whereas $|v'|(|w'| + t|u'|) < 1+t$. This proves inclusion in Ω . Conversely, setting $u = z$ and $v = w = -z$ gives

$$\frac{A}{B} = \left(\frac{1 + z^2}{1 - z^2} \right)^2.$$

The successive maps $z \mapsto z^2$, $x \mapsto (1+x)/(1-x)$, and $y \mapsto y^2$ map onto the disk, the right half-plane, and Ω , respectively. This proves surjectivity. $\square$

Lemma 8.3 (Polynomial factorization on a product of slit planes). *If a nonzero polynomial $Q \in \mathbb{C}[r_1, \dots, r_n]$ has no zero on Ω^n , then*

$$Q(r) = c \prod_i q_i(r_i), \quad c \neq 0, \quad (129)$$

where each q_i is monic and all its roots belong to $(-\infty, 0]$.

Proof. Write $Q(x, y) = \sum_{k=0}^d a_k(y)x^k$, with $a_d \neq 0$. On any connected open patch of Ω^{n-1} where $a_d \neq 0$, every root of the monic polynomial $Q(x, y)/a_d(y)$ is real and nonpositive. Its coefficients a_k/a_d are therefore real-valued holomorphic functions, and hence constant on the patch. Polynomial identity gives $Q(x, y) = a_d(y)p(x)$ everywhere, for a fixed monic polynomial p with all roots in $(-\infty, 0]$. Since p is nonzero on Ω , the polynomial a_d has no zero on Ω^{n-1} . Induction proves the claim, skipping variables of degree zero. This reasoning includes repeated roots and slices on which the degree might initially drop. $\square$

For $0 \leq r \leq k$, let $R_{k \rightarrow r}$ be partial trace from a Dicke-encoded k -qubit input to r qubits, identified with $\text{End}(\mathcal{H}_r)$. The reduced operator is supported on the symmetric sector: a vector antisymmetric in a retained pair is orthogonal to the fully symmetric input sector. Define the adjoint map

$$C_{r \rightarrow k}(N) = W_k^* (W_r N W_r^* \otimes I_2^{\otimes(k-r)}) W_k. \quad (130)$$

These maps are completely positive and covariant; $R_{k \rightarrow r}$ is trace preserving. At $r = 0$ they are the ordinary trace and scalar multiplication of I_{k+1} , respectively.

Lemma 8.4 (Coherent compression kernels). *For $0 \leq r \leq \min(\kappa, \ell)$, the map $T_r^{\kappa, \ell} = C_{r \rightarrow \ell} R_{\kappa \rightarrow r}$ has coherent Choi kernel*

$$A^r K_o^{\ell-r} K_i^{\kappa-r}. \quad (131)$$

These $\min(\kappa, \ell) + 1$ maps form a basis of all complex-linear $SU(2)$ -covariant maps between the two matrix spaces. Their trace effects are

$$(T_r^{\kappa, \ell})^*(I_{\ell+1}) = \frac{\ell+1}{r+1} I_{\kappa+1}. \quad (132)$$

Proof. The Dicke coefficients give, for $c = a + k$ and $e = b + k$,

$$[C_{r \rightarrow \ell}(|a\rangle\langle b|)]_{c,e} = \frac{\sqrt{\binom{r}{a}\binom{r}{b}\binom{\ell-r}{k}}}{\sqrt{\binom{\ell}{c}\binom{\ell}{e}}}, \quad (133)$$

and zero if the two shifts do not agree. In particular,

$$F_{C_{r \rightarrow \ell}(N)}(z, w) = (1 + zw)^{\ell-r} F_N(z, w). \quad (134)$$

Adjunction gives the coefficients of $R_{\kappa \rightarrow r}$, with ℓ replaced by κ . If its input indices are $a + m, b + m$, the two square-root binomial weights from the output and input of the Choi kernel cancel exactly the two denominators in these formulas. What remains is

$$\binom{r}{a} \binom{r}{b} \binom{\ell-r}{k} \binom{\kappa-r}{m},$$

the coefficient in the expansion of (131).

Covariance is equivalent to the Choi matrix commuting with $U_\ell \otimes \overline{U_\kappa}$. The Clebsch–Gordan decomposition is multiplicity free, with spins $|\ell - \kappa|/2, |\ell - \kappa|/2 + 1, \dots, (\ell + \kappa)/2$. Its commutant therefore has dimension $\min(\kappa, \ell) + 1$. The displayed covariant kernels are independent, since division by $K_o^\ell K_i^\kappa$ gives the successive powers of A/B , whose range is open by Lemma 8.2. They consequently span the entire complex commutant, without a Hermiticity assumption.

Finally $C_{r \rightarrow \ell}(I_{r+1}) = I_{\ell+1}$: the projection onto the symmetric sector of the first r qubits acts identically on the fully symmetric ℓ -qubit sector. Covariance and irreducibility imply that $C_{r \rightarrow \ell}^*(I_{\ell+1})$ is a scalar multiple of I_{r+1} . Taking traces identifies this scalar as $(\ell + 1)/(r + 1)$. Composing with the trace-preserving $R_{\kappa \rightarrow r}$ proves (132). $\square$

Proof of Theorem 8.1. For the independent product group, the commutant is the tensor product of the local commutants in Lemma 8.4. There is therefore a unique polynomial Q of separate degrees at most $\min(\kappa_i, \ell_i)$ such that

$$F_{J(\Phi)} = \prod_i K_{o,i}^{\ell_i} K_{i,i}^{\kappa_i} Q\left(\frac{A_1}{B_1}, \dots, \frac{A_n}{B_n}\right). \quad (135)$$

The separate degree bounds make this a polynomial. The prefactor never vanishes in the polydisk. The variables of different sites are independent, so Lemma 8.2 shows that full coherent stability is equivalent to nonvanishing of Q on Ω^n . Lemma 8.3 gives $Q = c \prod_i q_i$, where $q_i(r) = \prod_{a=1}^{d_i} (r + s_{ia})$ and $s_{ia} \geq 0$. The kernel thus factors, up to one nonzero scalar, into the numerators of (127).

For one factor write $q(r) = \sum_{a=0}^d q_a r^a$ and set $\Psi_{\kappa,\ell,q} = \sum_{a=0}^d q_a T_a^{\kappa,\ell}$. Every coefficient q_a is nonnegative, so this map is completely positive. By (132) its trace effect is

$$\Psi_{\kappa,\ell,q}^*(I_{\ell+1}) = c_\ell(q) I_{\kappa+1}, \quad c_\ell(q) = (\ell + 1) \sum_{a=0}^d \frac{q_a}{a+1} = (\ell + 1) \int_0^1 q(x) dx > 0. \quad (136)$$

In the global factorization, trace preservation forces $c \prod_i c_{\ell_i}(q_i) = 1$. Thus the initial scalar is positive real and the map is the tensor product of the individually normalized completely positive maps in (127). Conversely these maps are covariant

and trace preserving by construction, and their kernels are stable because q_i has no root in Ω . This proves the classification and automatic complete positivity.

To prove the rank assertion, first consider the square case $d = \kappa = \ell$. The map $\Psi_{d,d,q}$ is also the Dicke compression

$$\Psi_{d,d,q}(M) = W_d^* \left[\left(\bigotimes_{a=1}^d \mathcal{T}_{s_a} \right) (W_d M W_d^*) \right] W_d, \quad \mathcal{T}_s(N) = N + s \operatorname{Tr}(N) I_2, \quad (137)$$

where the tensor product in brackets acts on the d qubit matrix factors. Its kernel is $\prod_a (A + s_a B)$ by diagonalization of the binary variables, which verifies the identity with $\Psi_{d,d,q}$. As an operator on Hilbert–Schmidt space, $\mathcal{T}_s$ is positive definite: its eigenvalues are 1 on traceless matrices and $1 + 2s$ on the scalar space. The embedding $M \mapsto W_d M W_d^*$ is a Hilbert–Schmidt isometry. Its compression of the positive definite tensor product is therefore positive definite and invertible. Define

$$\Lambda_{d,q} = \frac{\Psi_{d,d,q}}{c_d(q)}, \quad \Lambda_{0,1} = \operatorname{Id}_{\mathbb{C}}. \quad (138)$$

The multiplication identity (134) gives $C_{d \rightarrow \ell} C_{a \rightarrow d} = C_{a \rightarrow \ell}$, and adjunction gives $R_{d \rightarrow a} R_{\kappa \rightarrow d} = R_{\kappa \rightarrow a}$. Hence

$$\Psi_{\kappa,\ell,q} = C_{d \rightarrow \ell} \Psi_{d,d,q} R_{\kappa \rightarrow d}. \quad (139)$$

The map $C_{d \rightarrow \ell}$ is injective by (134); its adjoint $R_{\ell \rightarrow d}$ is therefore surjective, and the same holds for $R_{\kappa \rightarrow d}$. The middle map in (139) is invertible on a space of dimension $(d+1)^2$. This proves local rank $(d+1)^2$. Ranks multiply under tensor products, which gives (128) and all assertions about injectivity, surjectivity, and invertibility. $\square$

Corollary 8.5 (Trace, invertible core, and symmetric cloning). *Each local map in Theorem 8.1 has the exact factorization*

$$\Phi_i = \mathcal{C}_{d_i \rightarrow \ell_i} \circ \Lambda_{d_i, q_i} \circ R_{\kappa_i \rightarrow d_i}, \quad \mathcal{C}_{d \rightarrow \ell} = \frac{d+1}{\ell+1} C_{d \rightarrow \ell}. \quad (140)$$

All three maps are trace preserving and completely positive, and the middle map is an invertible coherent Lee–Yang map. For equal input and output capacities the local factors are also unital.

Proof. The first two assertions follow from (139), (136), and $c_d(q)/c_\ell(q) = (d+1)/(\ell+1)$. For equal capacities, covariance makes $\Phi(I)$ scalar, and trace preservation makes this scalar one. $\square$

The normalized map $\mathcal{C}_{d \rightarrow \ell}$ is the universal symmetric cloner of [18, Eq. (3.3)]; its dimension-ratio normalization and concatenation law are classical. Here stability forces the entire factorization (140). The integer d specifies an intermediate spin dimension and the exact algebraic rank; it is not an assertion of noiseless recovery or an operational quantum communication capacity. If a specified covariance relation permutes the matched output sites, the theorem applies after undoing that permutation; an invertible map then matches equal capacities.

Example 8.6 (The spin-one spectral region). For $\kappa = \ell = 2$, let λ_1, λ_2 be the multipliers of a trace-preserving covariant map on the dipole and quadrupole subspaces. The exact coherent Choi condition is

$$0 \leq \lambda_2 \leq \lambda_1, \quad 3\lambda_1 \leq 2 + \lambda_2, \quad 3\lambda_1^2 + \lambda_2^2 \geq 4\lambda_2. \quad (141)$$

Indeed, in the degree-two case write $q(r) = r^2 + ar + b$. Its roots are real and nonpositive exactly when $a, b \geq 0$ and $a^2 \geq 4b$. The maps with kernels A^2 , AB , and B^2 have scalar multipliers 1, 3/2, 3, respectively, dipole multipliers 1, 1, 0, and quadrupole multipliers 1, 0, 0. For the middle map, partial trace annihilates the quadrupole $\text{diag}(1, -2, 1)$, while the difference of the two extreme projectors is sent to S^z ; covariance determines its action on each irreducible operator space. Therefore

$$\lambda_1 = \frac{1+a}{1+3a/2+3b}, \quad \lambda_2 = \frac{1}{1+3a/2+3b}.$$

Solving for a, b gives (141) when $\lambda_2 > 0$. Degree one gives $\lambda_2 = 0$ and $0 < \lambda_1 \leq 2/3$, and degree zero gives the origin, proving all endpoint cases. In particular, the ordinary qutrit depolarizing map with $\lambda_1 = \lambda_2 = t$ fails coherent Choi stability for every $0 < t < 1$, although both endpoints are stable.

Remark 8.7. Theorem 8.1 assumes independent local covariance, not only covariance under a common rotation. Its proof uses the static coherent Choi kernel. The argument does not apply the invertible qubit classification to a polarized higher-spin map: a Dicke lift is generally singular on the full qubit matrix space. No unrestricted noncovariant higher-spin static classification is asserted here.

9. QUANTITATIVE RIGIDITY AND CORRELATED TESTS

The finite-test rigidity of Theorem 4.3 is an exact statement. Its quantitative behaviour depends on the allowed inputs. We prove that products of two GHZ states detect precisely the failure of a diagonal Hamiltonian to have additive energies. The approximate-additivity theorem of Kalton and Roberts then gives a dimension-independent comparison with distance from one-site dynamics. In contrast, even all product Lee–Yang inputs admit no such comparison.

Throughout this section, a density matrix acts on $\mathcal{H}_n = (\mathbb{C}^2)^{\otimes n}$, and

$$F_X(z, w) = \sum_{x, y \in \{0, 1\}^n} X_{xy} z^x w^y,$$

$$\mathcal{Y}_n = \{\rho \geq 0 : \text{Tr } \rho = 1, F_\rho \neq 0 \text{ on } \mathbb{D}^{2n}\}.$$

The condition is zero-freeness, not merely that F_ρ is a nonzero polynomial. We use the unhalved trace norm $\|\cdot\|_1$ and diamond norm $\|\cdot\|_\diamond$, and write $\text{dist}_1(X, \mathcal{Y}_n) = \inf_{\rho \in \mathcal{Y}_n} \|X - \rho\|_1$. Let LocHP_n be the real vector space of maps

$$\mathcal{M} = \sum_{i=1}^n \mathcal{M}_i \otimes \text{Id}_{\{1, \dots, n\} \setminus \{i\}},$$

where every $\mathcal{M}_i$ is complex linear and Hermiticity preserving. Let LocGKLS_n be the cone obtained by requiring each $\mathcal{M}_i$ to be a one-qubit GKLS generator. In particular, local Hamiltonian generators belong to both classes.

9.1. The diagonal quotient norm. Identify a subset $S \subseteq [n]$ with the computational basis vector whose occupied sites are exactly S . For a real function h on $2^{[n]}$, put

$$H = \text{diag}(h(S) : S \subseteq [n]), \quad \mathcal{L}_H(X) = -i[H, X],$$

$$\text{osc } h = \max_S h(S) - \min_S h(S).$$

Proposition 9.1. *For every diagonal Hermitian H ,*

$$\begin{aligned} d(H) &:= \text{dist}_\diamond(\mathcal{L}_H, \text{LocHP}_n) \\ &= \text{dist}_\diamond(\mathcal{L}_H, \text{LocGKLS}_n) \\ &= \min_{b \in \mathbb{R}^n} \text{osc} \left(h(S) - \sum_{i \in S} b_i \right). \end{aligned} \quad (142)$$

If $h(S) = h([n] \setminus S)$ for all S , then $d(H) = \text{osc } h = \|\mathcal{L}_H\|_\diamond$.

Proof. Fix $\mathcal{M} \in \text{LocHP}_n$ and set $E_{xy} = |x\rangle\langle y|$. The coefficient of E_{xy} in $\mathcal{M}(E_{xy})$ equals

$$\langle x | \mathcal{M}(E_{xy}) | y \rangle = \sum_i \langle x_i | \mathcal{M}_i(E_{x_i y_i}) | y_i \rangle.$$

If $x_i = y_i$, the summand is real. The summands for $(x_i, y_i) = (0, 1)$ and $(1, 0)$ are conjugates. Thus, for a vector $b \in \mathbb{R}^n$ independent of x, y , its imaginary part is $b \cdot (y - x)$. Since $\|E_{xy}\|_1 = 1$ and every matrix entry is bounded by the trace norm,

$$\|\mathcal{L}_H - \mathcal{M}\|_\diamond \geq \max_{x, y} |(h(x) - b \cdot x) - (h(y) - b \cdot y)| = \text{osc}(h - b \cdot x).$$

Here and below $h(x)$ is the subset notation written in occupation coordinates.

For a Hermitian matrix D one has

$$\| -i[D, \cdot] \|_\diamond = \lambda_{\max}(D) - \lambda_{\min}(D). \quad (143)$$

Indeed, subtract the midpoint of the spectral interval from D . The trace-norm ideal inequality, also after adjoining an identity ancilla, gives the upper bound $2\|D\|_{\text{op}}$ for the resulting commutator. The matrix unit between eigenvectors at the two spectral endpoints gives the reverse bound. Apply this identity to $D = H - H_b$, where $H_b = \sum_i b_i |1\rangle\langle 1|_i$. The local Hamiltonian generator $\mathcal{L}_{H_b}$ attains the corresponding oscillation. Because $\text{LocGKLS}_n \subseteq \text{LocHP}_n$, the two distance infima agree with the infimum of these oscillations. It is a minimum: the objective is continuous and

$$\text{osc}(h - b \cdot x) \geq \text{osc}(b \cdot x) - \text{osc } h = \|b\|_{\ell^1} - \text{osc } h.$$

Suppose now that h is invariant under complementation. Replacing x by $1 - x$ shows that $\text{osc}(h - b \cdot x) = \text{osc}(h + b \cdot x)$. The oscillation is a convex function of its argument, so

$$\text{osc } h \leq \frac{1}{2} \text{osc}(h - b \cdot x) + \frac{1}{2} \text{osc}(h + b \cdot x) = \text{osc}(h - b \cdot x).$$

Taking $b = 0$ proves the last assertion. $\square$

9.2. Two correlated blocks. For disjoint nonempty $A, B \subseteq [n]$, define

$$\begin{aligned} |\psi_{A,B}\rangle &= \frac{|0_A\rangle + |1_A\rangle}{\sqrt{2}} \otimes \frac{|0_B\rangle + |1_B\rangle}{\sqrt{2}} \otimes |0_{[n] \setminus (A \cup B)}\rangle, \\ \rho_{A,B} &= |\psi_{A,B}\rangle\langle\psi_{A,B}|. \end{aligned} \quad (144)$$

The pure-state coefficient polynomial is $\frac{1}{2}(1 + \prod_{i \in A} z_i)(1 + \prod_{i \in B} z_i)$. It is zero-free on $\mathbb{D}^n$, and hence $\rho_{A,B} \in \mathcal{Y}_n$. Set

$$\begin{aligned} \Delta_{A,B}(H) &= h(A \cup B) - h(A) - h(B) + h(\emptyset), \\ \Delta(H) &= \max_{A \cap B = \emptyset} |\Delta_{A,B}(H)|. \end{aligned}$$

Theorem 9.2 (Correlated diagonal rigidity). *Let $n \geq 2$ and let H be a diagonal Hermitian Hamiltonian. For every pair of disjoint nonempty subsets A, B the following limit exists:*

$$\lim_{t \downarrow 0} \frac{\text{dist}_1(e^{-itH} \rho_{A,B} e^{itH}, \mathcal{Y}_n)}{t} = \frac{|\Delta_{A,B}(H)|}{2}. \quad (145)$$

If $\delta(H)$ denotes the maximum of these limits, then

$$\delta(H) = \frac{\Delta(H)}{2}, \quad \delta(H) \leq d(H) \leq 4K \delta(H), \quad (146)$$

where K is any universal constant in the real Kalton–Roberts approximate-additivity theorem. In particular, one may take $K = 45$.

We first establish the polynomial and boundary facts used in the proof.

Lemma 9.3 (Block compression). *Let $f(u_1, \dots, u_k, v)$ be multiaffine and zero-free on the open unit polydisk. If $a(v)$ and $d(v)$ are its constant and full-degree coefficients in the u variables, then $a(v) + d(v)s$ is zero-free on the open unit polydisk in (s, v) .*

Consequently, if $V : \mathbb{C}^2 \otimes \mathbb{C}^2 \rightarrow \mathcal{H}_n$ is the isometry

$$V|a, b\rangle = |a_A, b_B, 0_{[n] \setminus (A \cup B)}\rangle,$$

then $F_{V^* \sigma V}$ is zero-free on $\mathbb{D}^4$ for every $\sigma \in \mathcal{Y}_n$.

Proof. Fix the remaining variables v in their polydisk. The polynomial $g(u) = f(u, \dots, u, v)$ has no zero in $\mathbb{D}$, and $a(v) = g(0) \neq 0$. If $\deg g = k$, all its roots have modulus at least one, and the product-of-roots identity gives $|d(v)| \leq |a(v)|$. If $\deg g < k$, then $d(v) = 0$. Thus $a(v) + d(v)s \neq 0$ for $|s| < 1$. This proves the first assertion; it is the unit-disk form of the Asano coefficient contraction [3].

For the second assertion, first set all row and column variables outside $A \cup B$ to zero. Apply the first assertion successively to the row variables in A , the row variables in B , and the corresponding two groups of column variables. The resulting coefficient polynomial is exactly $F_{V^* \sigma V}$. No normalization or zero-polynomial exception occurs in this argument. $\square$

The compression in Lemma 9.3 is completely positive and trace nonincreasing, but need not preserve trace. We use its unnormalized output. For every Hermitian X , $\|V^* X V\|_1 \leq \|X\|_1$, as follows either from trace-norm duality or from the positive and negative parts of X .

Lemma 9.4 (A two-qubit boundary contact). *Write $\rho_0 = |++\rangle\langle++|$, where $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, and put*

$$\begin{aligned} u &= |00\rangle + |10\rangle + |01\rangle + |11\rangle, \\ v &= |00\rangle - |10\rangle - |01\rangle + |11\rangle, \quad \ell(X) = \Im\langle v|X|u\rangle. \end{aligned}$$

If $t_j \downarrow 0$, X_j is Hermitian, F_{X_j} is zero-free on $\mathbb{D}^4$, and $\|X_j - \rho_0\|_1 = O(t_j)$, then $\ell(X_j) = o(t_j)$. Moreover,

$$|\ell(X)| \leq 2\|X\|_1 \quad \text{for every Hermitian } X. \quad (147)$$

Proof. The Hermitian observable representing ℓ is

$$R = \frac{|u\rangle\langle v| - |v\rangle\langle u|}{2i}.$$

The vectors u, v are orthogonal and both have norm two. On their span R has eigenvalues 2, -2 , and it vanishes on the orthogonal complement. Thus $\|R\|_{\text{op}} = 2$, proving (147).

Consider

$$g_j(s) = F_{X_j}(s-1, s-1, 1, 1) = a_{2,j}s^2 + a_{1,j}s + a_{0,j}.$$

For $X_j = \rho_0$ this polynomial is s^2 . Continuity of its coefficients gives $a_{2,j} = 1 + O(t_j)$ and $a_{1,j}, a_{0,j} = O(t_j)$. The column variables have been specialized to boundary points, so we justify their use. For fixed j , the polynomials $F_{X_j}(s-1, s-1, r, r)$, with $0 < r < 1$, are zero-free on $|s-1| < 1$. Hurwitz's theorem shows that their limit as $r \uparrow 1$ is zero-free there or identically zero. For all sufficiently large j , $a_{2,j} \neq 0$, which excludes the latter alternative.

Let $r_{1,j}, r_{2,j}$ be the two roots of g_j , counted with multiplicity. Vieta's formulas give their sum and product as $O(t_j)$; the quadratic formula then gives $|r_{k,j}| = O(\sqrt{t_j})$. Because $|r_{k,j} - 1| \geq 1$,

$$\Re r_{k,j} \leq \frac{1}{2}|r_{k,j}|^2 = O(t_j).$$

Their real parts have sum $O(t_j)$, so the same real parts are bounded below by $-O(t_j)$. Consequently $\Re r_{k,j} = O(t_j)$ and $\Im r_{k,j} = O(\sqrt{t_j})$. It follows that

$$\Im(r_{1,j}r_{2,j}) = O(t_j^{3/2}), \quad \Im a_{0,j} = O(t_j^{3/2}) + O(t_j^2) = o(t_j).$$

Finally $a_{0,j} = F_{X_j}(-1, -1, 1, 1) = \langle v | X_j | u \rangle$. $\square$

Proof of Theorem 9.2. Fix A, B , let V be as in Lemma 9.3, and abbreviate $\Delta = \Delta_{A,B}(H)$ and $\rho(t) = e^{-itH}\rho_{A,B}e^{itH}$. The image of V is invariant under H , and $V^*\rho(t)V = e^{-itD}\rho_0e^{itD}$, where the logical diagonal Hamiltonian D has energies $h(\emptyset), h(A), h(B), h(A \cup B)$. Since $\rho_0 = |u\rangle\langle u|/4$ and $\langle v | u \rangle = 0$, differentiation gives

$$\ell(V^*\rho(t)V) = -\Delta t + O(t^2). \quad (148)$$

Let $t_j \downarrow 0$ be arbitrary, and choose $\sigma_j \in \mathcal{Y}_n$ with

$$\|\rho(t_j) - \sigma_j\|_1 \leq \text{dist}_1(\rho(t_j), \mathcal{Y}_n) + t_j^2.$$

Because $\rho_{A,B} \in \mathcal{Y}_n$ and $\rho(t_j) = \rho_{A,B} + O(t_j)$, both the distance on the right and $\|\sigma_j - \rho_{A,B}\|_1$ are $O(t_j)$. Lemma 9.3 shows that $X_j = V^*\sigma_j V$ has a zero-free coefficient polynomial. It is Hermitian and $X_j = \rho_0 + O(t_j)$, so Lemma 9.4 applies without rescaling its trace. Equations (147) and (148) imply

$$2\|\rho(t_j) - \sigma_j\|_1 \geq |\ell(V^*\rho(t_j)V - X_j)| = |\Delta|t_j + o(t_j).$$

As the sequence was arbitrary, this proves the lower bound $|\Delta|/2$ for the liminf in (145).

For a matching upper bound, subtract $(\Delta/4)Z_1Z_2$ from D , where $Z = \text{diag}(1, -1)$. The remaining logical energy is affine in the two occupation bits: its mixed difference is zero. Its evolution of the logical vector $|++\rangle$, embedded by V , is therefore a product of two GHZ states with unit-modulus relative phases and a vacuum on the remaining sites. This comparison state belongs to $\mathcal{Y}_n$. Since all four logical configurations have equal probability and the two values of Z_1Z_2 occur equally often, the inner product between the actual and comparison pure vectors is $\cos(t\Delta/4)$. Their trace distance is

$$2|\sin(t\Delta/4)| \leq t|\Delta|/2 \quad (t \geq 0).$$

This proves (145). Empty subsets have mixed difference zero, and the family of pairs is finite, so $\delta(H) = \Delta(H)/2$.

We now invoke the precise inherited input. The Kalton–Roberts theorem [12, Theorem 4.1] supplies a universal $K \leq 45$ with the following property: if a real set function f on a set algebra satisfies $f(\emptyset) = 0$ and

$$|f(A \cup B) - f(A) - f(B)| \leq \eta \quad (A \cap B = \emptyset),$$

then there is an additive real set function μ such that $\sup_S |f(S) - \mu(S)| \leq K\eta$. Apply it on $2^{[n]}$ to $f(S) = h(S) - h(\emptyset)$ and $\eta = \Delta(H)$. On this finite algebra, $\mu(S) = \sum_{i \in S} b_i$ for some real b_i , and hence

$$\text{osc}(h - b \cdot x) \leq 2K\Delta(H).$$

If $\Delta(H) = 0$, the same conclusion follows directly from finite additivity. Proposition 9.1 now yields $d(H) \leq 2K\Delta(H) = 4K\delta(H)$.

Finally choose a minimizer b in (142). The flow of H_b consists of independent diagonal one-qubit unitaries. It rotates each polynomial variable by a unit complex number and thus preserves $\mathcal{Y}_n$. The Duhamel formula and unitary trace-norm invariance imply, for every density matrix ρ and $t \geq 0$,

$$\|e^{t\mathcal{L}_H}\rho - e^{t\mathcal{L}_{H_b}}\rho\|_1 \leq t\|\mathcal{L}_H - \mathcal{L}_{H_b}\|_\diamond = t d(H).$$

Apply this estimate to each $\rho_{A,B}$, divide by t , and use (145). Taking the maximum gives $\delta(H) \leq d(H)$. $\square$

Remark 9.5. The universal constant in (146) is inherited from Kalton–Roberts. The bridge to that theorem is the exact root-contact identity (145). The probes may contain arbitrarily large blocks, their family is exponential in n , and the defect uses distance to the full set $\mathcal{Y}_n$. No efficient certification procedure or corresponding statement for non-diagonal Hamiltonians is asserted.

9.3. Product inputs do not give uniform robustness. Let $\mathcal{P}_n \subseteq \mathcal{Y}_n$ be the set of all product density matrices in $\mathcal{Y}_n$, including mixed products, and put

$$\delta_{\text{prod}}(\mathcal{L}) = \sup_{\rho \in \mathcal{P}_n} \limsup_{t \downarrow 0} \frac{\text{dist}_1(e^{t\mathcal{L}}\rho, \mathcal{Y}_n)}{t}.$$

We give two different mechanisms by which this defect can be small. For use with mixed states, recall the elementary estimate

$$\|e^{-itH}\rho e^{itH} - \rho\|_1 \leq 2|t|\sqrt{\text{Tr}(\rho H^2) - (\text{Tr}(\rho H))^2}. \quad (149)$$

To prove it, purify ρ to a unit vector Ψ and evolve the purification by $H \otimes I$. For a pure-state projection $P = |\Psi\rangle\langle\Psi|$, the two possibly nonzero eigenvalues of $-i[H \otimes I, P]$ are $\pm\sqrt{\text{Var}_\rho H}$. Its trace norm is therefore twice this square root. The variance is constant along the unitary orbit, so integration gives the same bound for purified states. Partial trace gives (149).

Proposition 9.6 (A nearest-neighbour obstruction). *On a path with $m \geq 1$ edges and $n = m + 1$ vertices, let*

$$H_m = \frac{1}{2m} \sum_{j=1}^m Z_j Z_{j+1}, \quad \mathcal{L}_m = -i[H_m, \cdot].$$

Then

$$\|\mathcal{L}_m\|_\diamond = d(H_m) = 1,$$

$$\text{dist}_1(e^{t\mathcal{L}_m}\rho, \mathcal{Y}_n) \leq \sqrt{\frac{3}{m}}|t| \quad (\rho \in \mathcal{P}_n, t \in \mathbb{R}).$$

For the equatorial pure product $|\psi_\theta\rangle = \bigotimes_{j=1}^n(|0\rangle + e^{i\theta_j}|1\rangle)/\sqrt{2}$ and its density ρ_θ , the sharper formula is

$$\|e^{t\mathcal{L}_m}\rho_\theta - \rho_\theta\|_1 = 2\sqrt{1 - \cos(t/(2m))^{2m}} \leq \frac{|t|}{\sqrt{m}}. \quad (150)$$

Proof. The diagonal energy is invariant under simultaneous complementation. Equal computational spins have energy $1/2$, alternating spins have energy $-1/2$, and these are the extrema. Thus Proposition 9.1 gives the two norm identities.

For a product density, the computational spins are independent signs. Write $W_j = Z_j Z_{j+1}$ for their edge products, regarded as random variables in this diagonal distribution. Disjoint edges have zero covariance, each variance is at most one, and each adjacent covariance has absolute value at most one by Cauchy-Schwarz. Hence

$$\text{Var}\left(\sum_{j=1}^m W_j\right) \leq m + 2(m-1) \leq 3m.$$

Equation (149) gives the stated mixed-state bound. The initial state itself belongs to $\mathcal{Y}_n$ and is an admissible comparison state in the distance.

For ψ_θ , the computational distribution is uniform. The root sign and the m edge products determine all vertex signs bijectively, so the edge products are independent uniform signs. It follows that

$$\langle\psi_\theta|e^{-itH_m}|\psi_\theta\rangle = \cos(t/(2m))^m.$$

The trace distance of unit pure states ψ, ϕ is $2\sqrt{1 - |\langle\psi, \phi\rangle|^2}$; this follows by restricting the difference of their projections to their two-dimensional span. It proves the equality in (150). Independence gives $\text{Var}_{\rho_\theta} H_m = 1/(4m)$, so (149) proves its last inequality. Finally, $\rho_\theta \in \mathcal{Y}_n$, since its coefficient polynomial is a product of linear factors with roots on the unit circle. $\square$

Lemma 9.7 (One-qubit physical states). *A one-qubit density matrix $\tau = \begin{pmatrix} a & \bar{b} \\ b & d \end{pmatrix}$ belongs to $\mathcal{Y}_1$ if and only if $a \geq d$.*

Proof. Its polynomial is $f(z, w) = a + bz + \bar{b}w + dzw$. Suppose $a \geq d$. Positivity gives $|b|^2 \leq ad \leq a^2$, and $a > 0$ because $a + d = 1$. The denominator of

$$q(z) = \frac{\bar{b} + dz}{a + bz}$$

has no zero in $\mathbb{D}$. For $|z| = 1$,

$$|a + bz|^2 - |\bar{b} + dz|^2 = (a - d)(a + d + 2\Re(bz)) \geq 0.$$

If $|b| < a$, the maximum principle gives $|q(z)| \leq 1$ in $\mathbb{D}$. The remaining case $|b| = a$ forces $a = d = |b|$, and cancellation makes q a constant of modulus one. Thus in either case $f(z, w) = (a + bz)(1 + wq(z))$ is nonzero for $|z|, |w| < 1$.

Conversely, suppose $a < d$. Then $|b| \leq \sqrt{ad} < (a + d)/2$, so the displayed boundary difference is strictly negative on the unit circle. By continuity, some z

inside the circle satisfies $|a + bz| < |\bar{b} + dz|$. The denominator in $w = -(a + bz)/(\bar{b} + dz)$ is nonzero and $|w| < 1$; this gives an interior zero of f . $\square$

Proposition 9.8 (An exponential product-input obstruction). *For $n \geq 2$, put $P_n = |1^n\rangle\langle 1^n|$ and $\mathcal{K}_n = -i[P_n, \cdot]$. Then*

$$\|\mathcal{K}_n\|_\diamond = 1, \quad d(P_n) = \frac{n-1}{n}. \quad (151)$$

For every $\rho \in \mathcal{P}_n$ and every real t ,

$$\text{dist}_1(e^{t\mathcal{K}_n}\rho, \mathcal{Y}_n) \leq \|e^{t\mathcal{K}_n}\rho - \rho\|_1 \leq 2^{1-n/2}|t|. \quad (152)$$

For equatorial pure products, the middle expression is exactly

$$4\sqrt{2^{-n}(1-2^{-n})}|\sin(t/2)|.$$

Proof. The commutator norm is one by (143). To evaluate (142), average b over coordinate permutations. Convexity of oscillation and permutation symmetry of $h(x) = \mathbf{1}_{\{x=1^n\}}$ show that this cannot increase the objective. We may therefore take $b_i = c$ for all i . The residual energies are

$$\{-ck : 0 \leq k \leq n-1\} \cup \{1-cn\}.$$

If $c \leq 1/n$, the last value minus the value at weight $n-1$ is $1-c \geq (n-1)/n$. If $c \geq 1/n$, the difference between weights zero and $n-1$ has absolute value $c(n-1) \geq (n-1)/n$. For $c = 1/n$, all residual energies lie in $[-(n-1)/n, 0]$ and both endpoints occur. This proves (151).

The coefficient polynomial of a product density is the product of the individual coefficient polynomials in disjoint variables. Thus it is zero-free precisely when all factors are zero-free. Lemma 9.7 implies that every occupied population is at most $1/2$. Consequently $p = \text{Tr}(P_n\rho) \leq 2^{-n}$. For a purification Ψ of ρ ,

$$\langle \Psi | (e^{-itP_n} \otimes I) | \Psi \rangle = 1 - p + pe^{-it}.$$

The pure-state trace-distance formula and partial-trace contractivity therefore give

$$\|e^{t\mathcal{K}_n}\rho - \rho\|_1 \leq 4\sqrt{p(1-p)}|\sin(t/2)| \leq 2\sqrt{p}|t|.$$

This proves (152), using the initial LY state as comparison. For an equatorial pure product $p = 2^{-n}$, and no partial trace is needed, giving the asserted equality. $\square$

Corollary 9.9. *There is no function $\omega : [0, \infty) \rightarrow [0, \infty)$ with $\omega(0) = 0$ and $\lim_{s \downarrow 0} \omega(s) = 0$, independent of n , such that*

$$\text{dist}_\diamond(\mathcal{L}, \text{LocGKLS}_n) \leq \omega(\delta_{\text{prod}}(\mathcal{L}))$$

for every Hamiltonian generator $\mathcal{L}$ of diamond norm one. This remains false if the Hamiltonian is required to be a sum of nearest-neighbour two-site terms on a path.

Proof. Proposition 9.6 gives a sequence with distance one and $0 \leq \delta_{\text{prod}}(\mathcal{L}_m) \leq \sqrt{3/m} \rightarrow 0$, contradicting the required behaviour of ω . Proposition 9.8 separately gives exponentially small product defect and distance tending to one. $\square$

The normalization of the path interactions is by their total strength; no positive lower bound on each edge coefficient is assumed. These examples concern a global diamond norm and do not obstruct estimation of individual interaction coefficients in other access models. Nor do they contradict Theorem 9.2: correlated inputs can detect what all product inputs detect only weakly. For instance, the GHZ state on

all n sites belongs to $\mathcal{Y}_n$ and has $\text{Tr}(P_n\rho) = 1/2$, so the product-population bound used above does not extend to arbitrary physical Lee–Yang states.

DISCLOSURE OF ARTIFICIAL INTELLIGENCE USE

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